

INTRODUCTION TO EARTHQUAKE ENGINEERING

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M Y S O R E

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Natural Disasters



Earthquake



Tornado, Cyclone



Floods



Fire

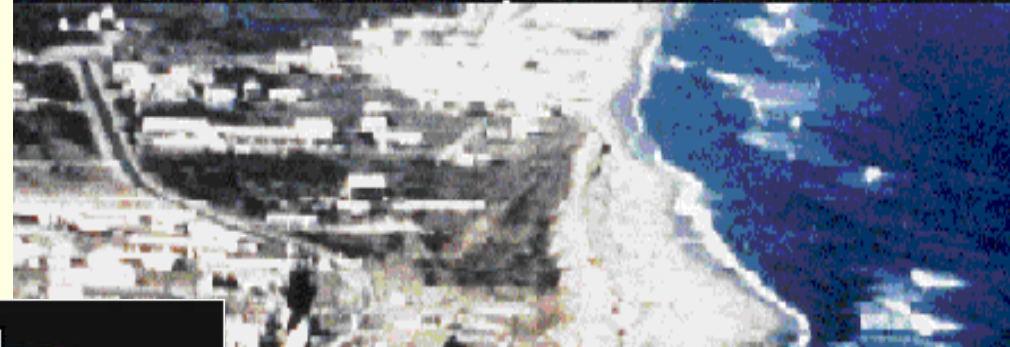
Natural Disasters



Volcano

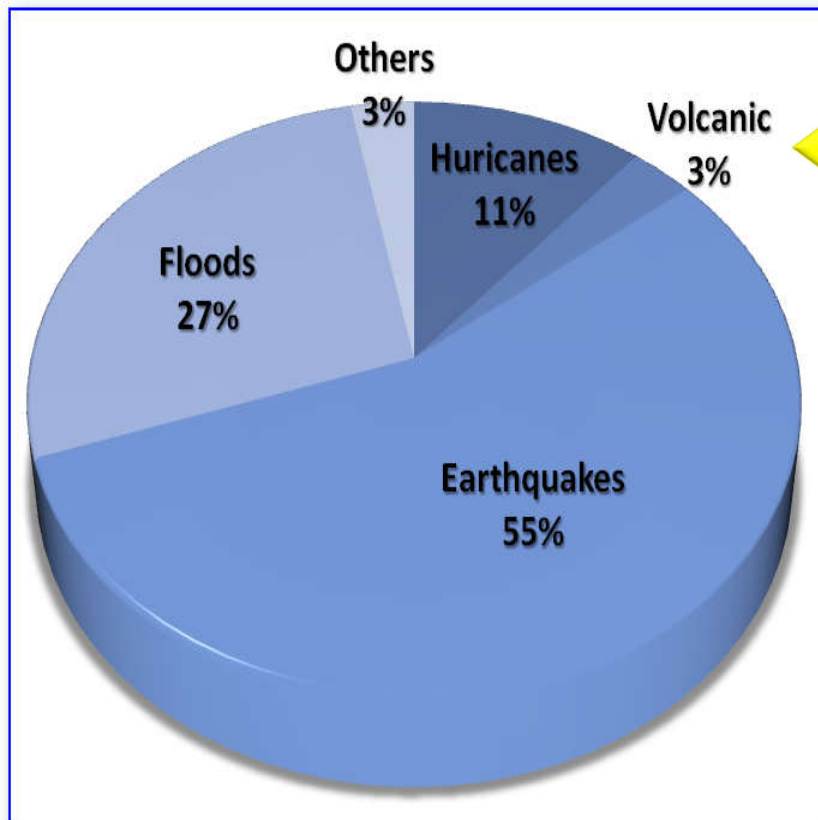


Coastal Storms and Tsunamis



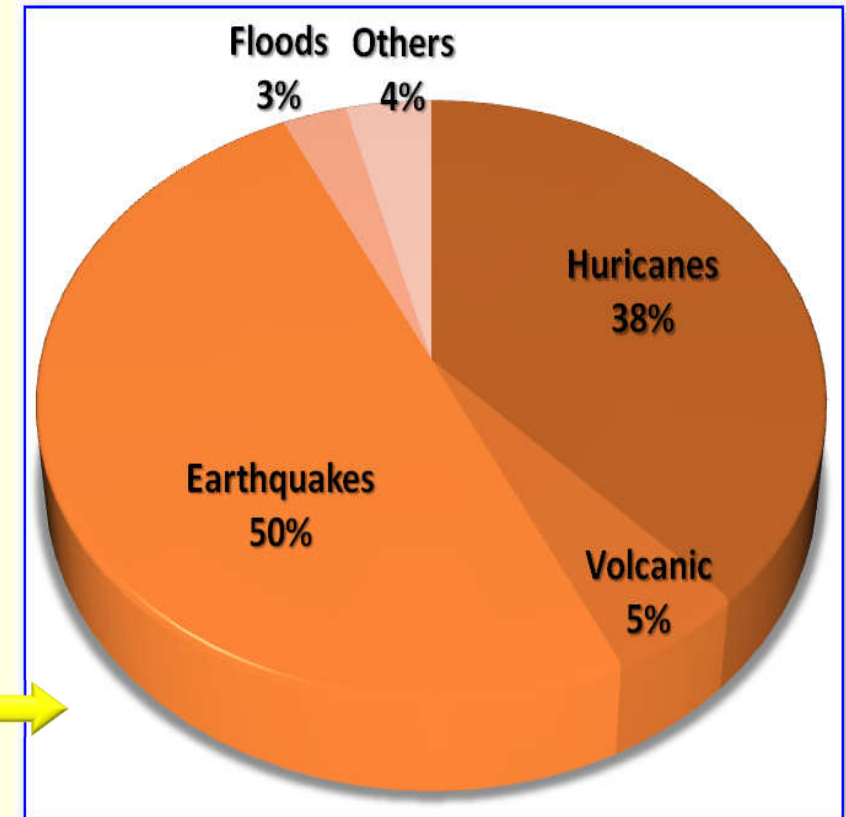
Hurricane

Damage during Natural Disasters

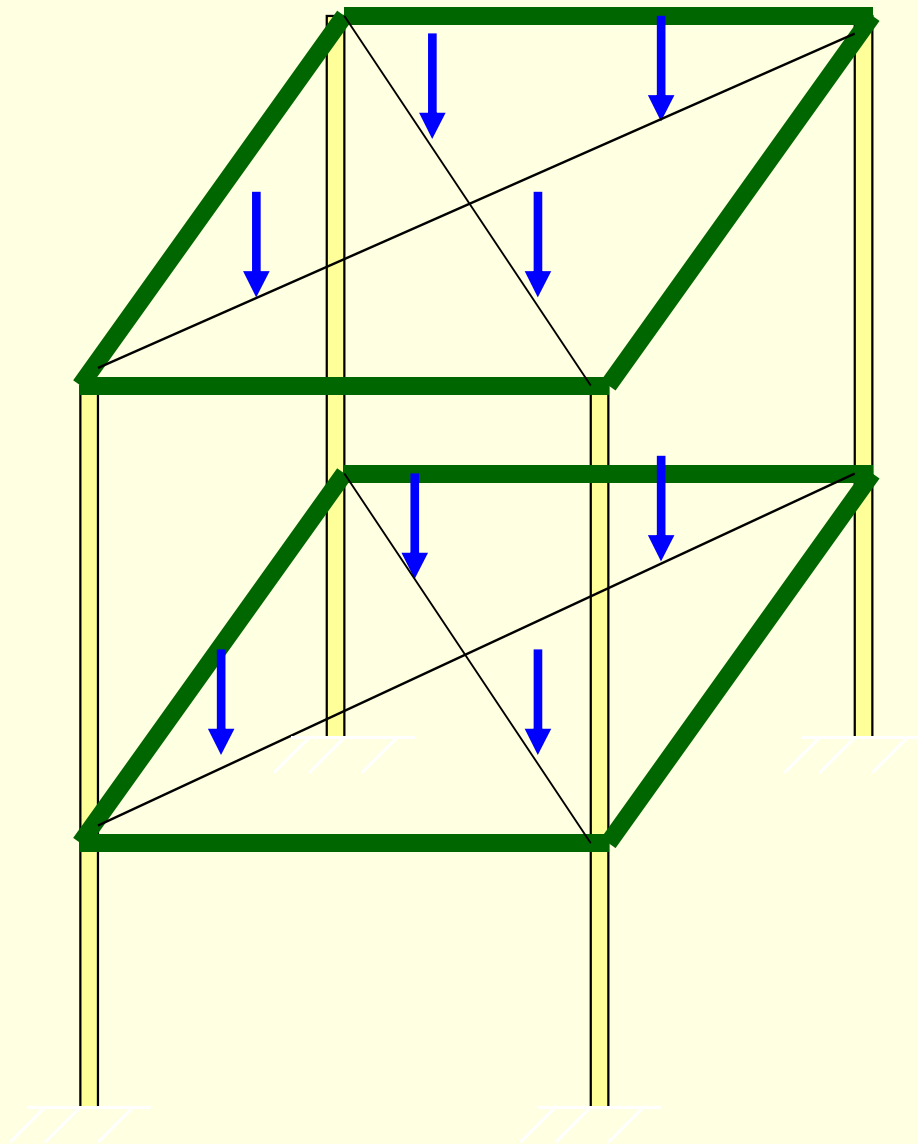


**Loss of built environment
from natural disasters**
(Source: Andrew and Robin, 2002)

Loss of life from natural disasters
(Source: Herath and Katayama, 1994)



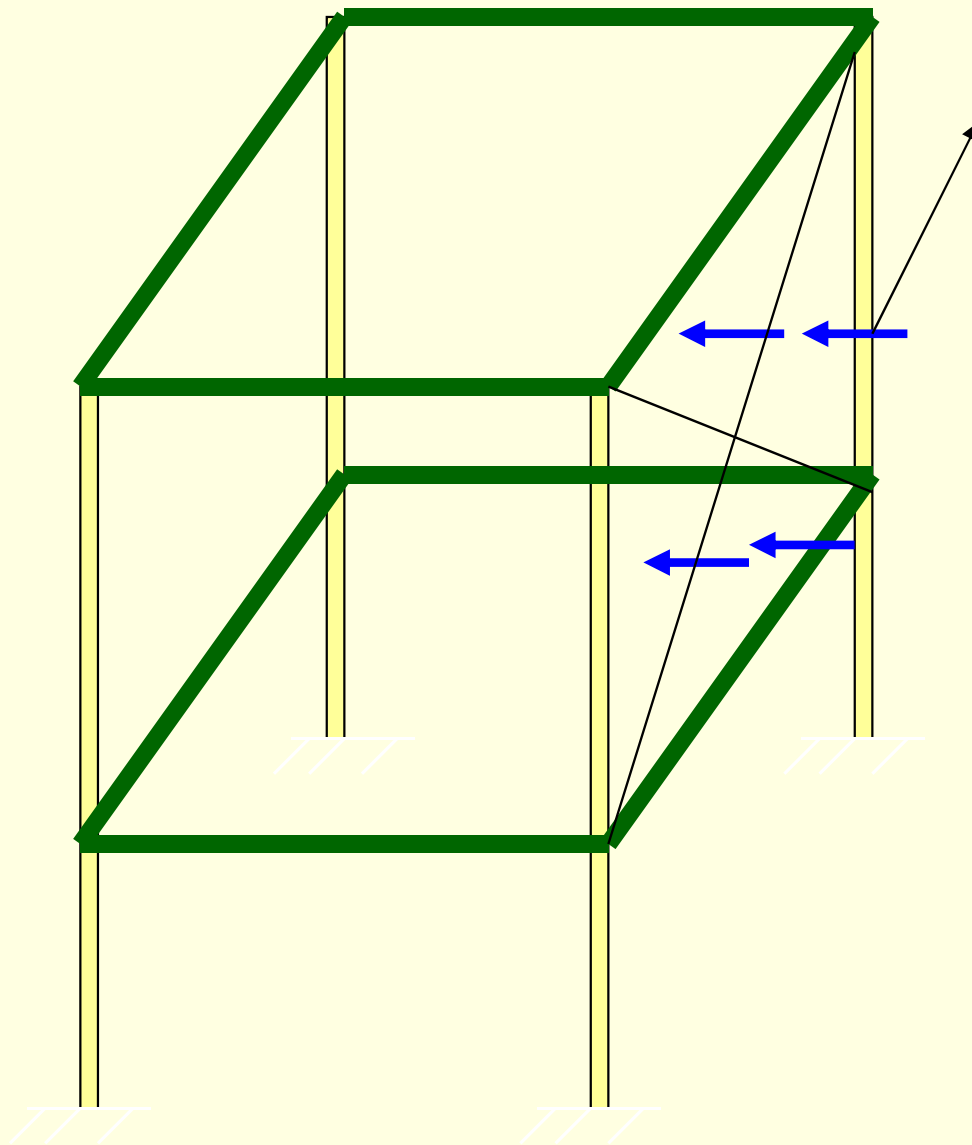
Dead & Live loads



Direction Of Dead
& Live loads

Depends on self
weight and
functional aspects
of building

Wind loads



Direction Of wind loads

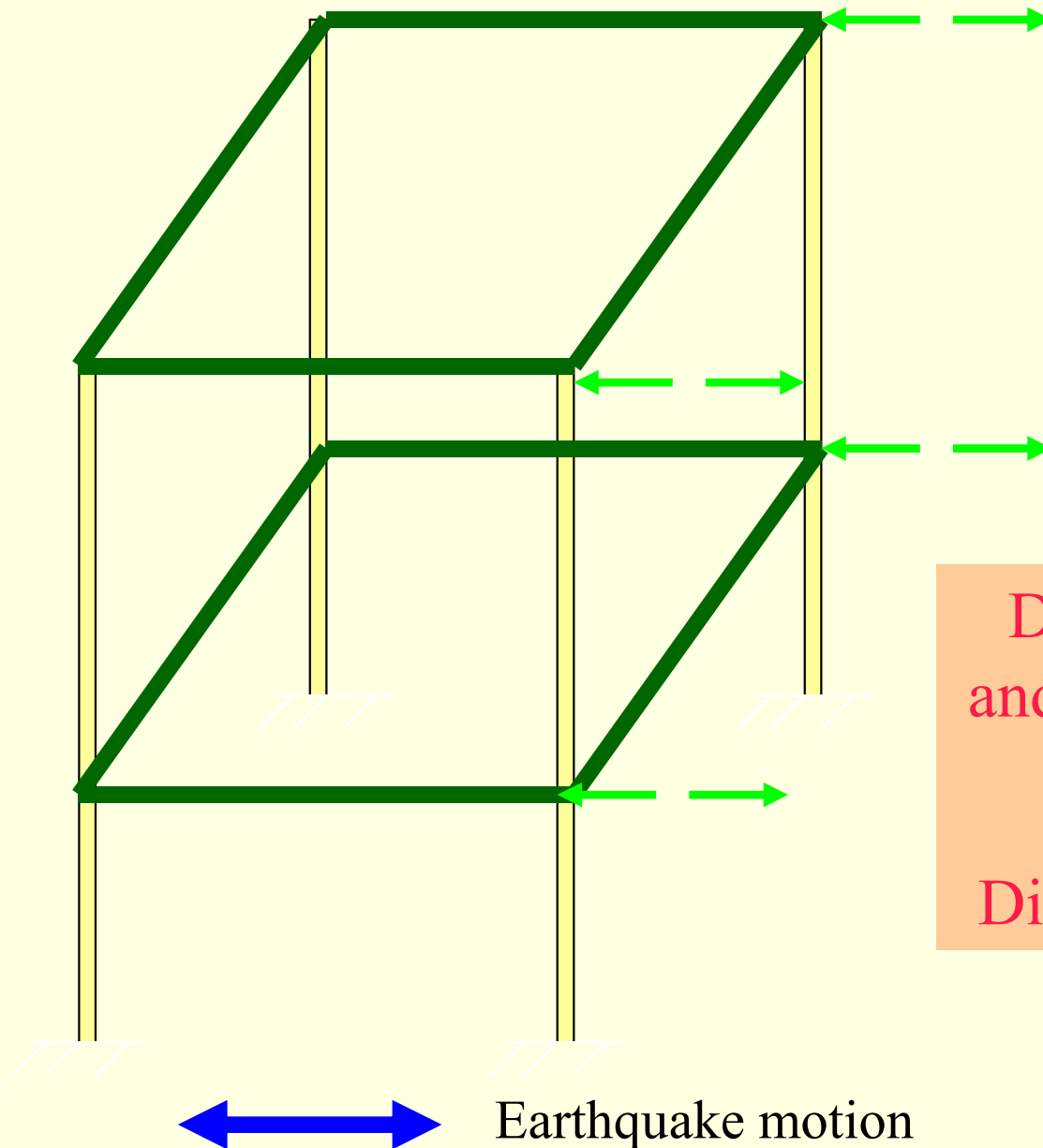
Depends on Wind
intensity and exposed
area of the building

&

Distribution is uniform
along the height

$$F = p * \text{area}$$

Seismic loads



Direction of seismic forces

$$F = \text{Mass} * \text{Acceleration}$$

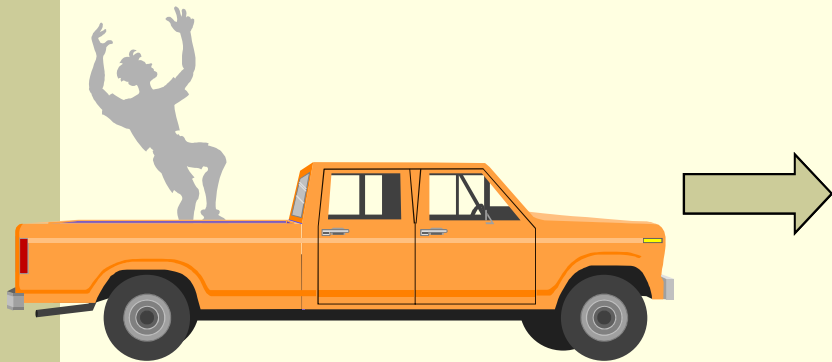
Depends on acceleration
and weight of the building

&

Distribution is not uniform

Effects of Earthquake

Inertia Force $F = m a$

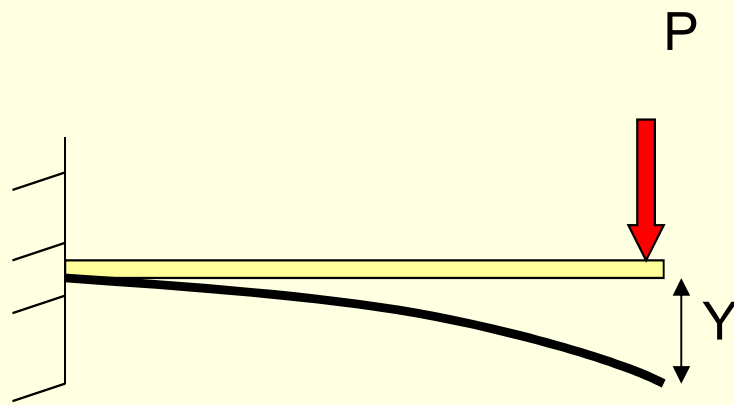


ACCELERATION

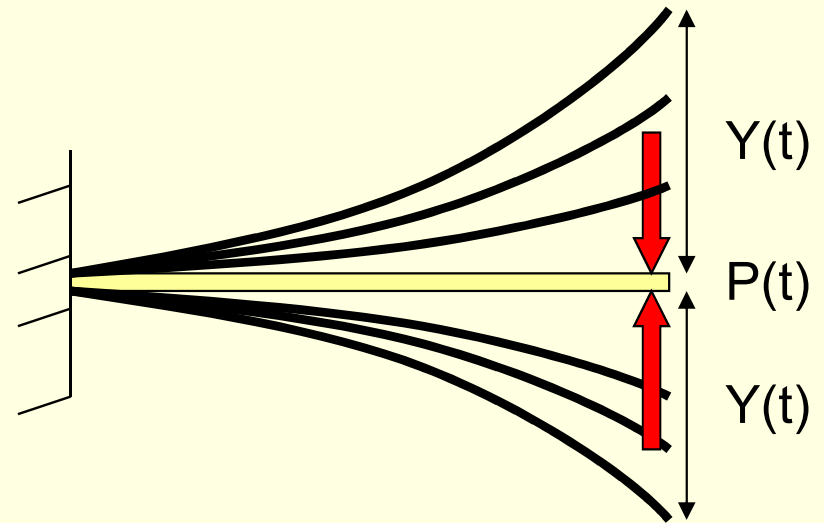


DECELERATION

Static Vs Dynamic Loading

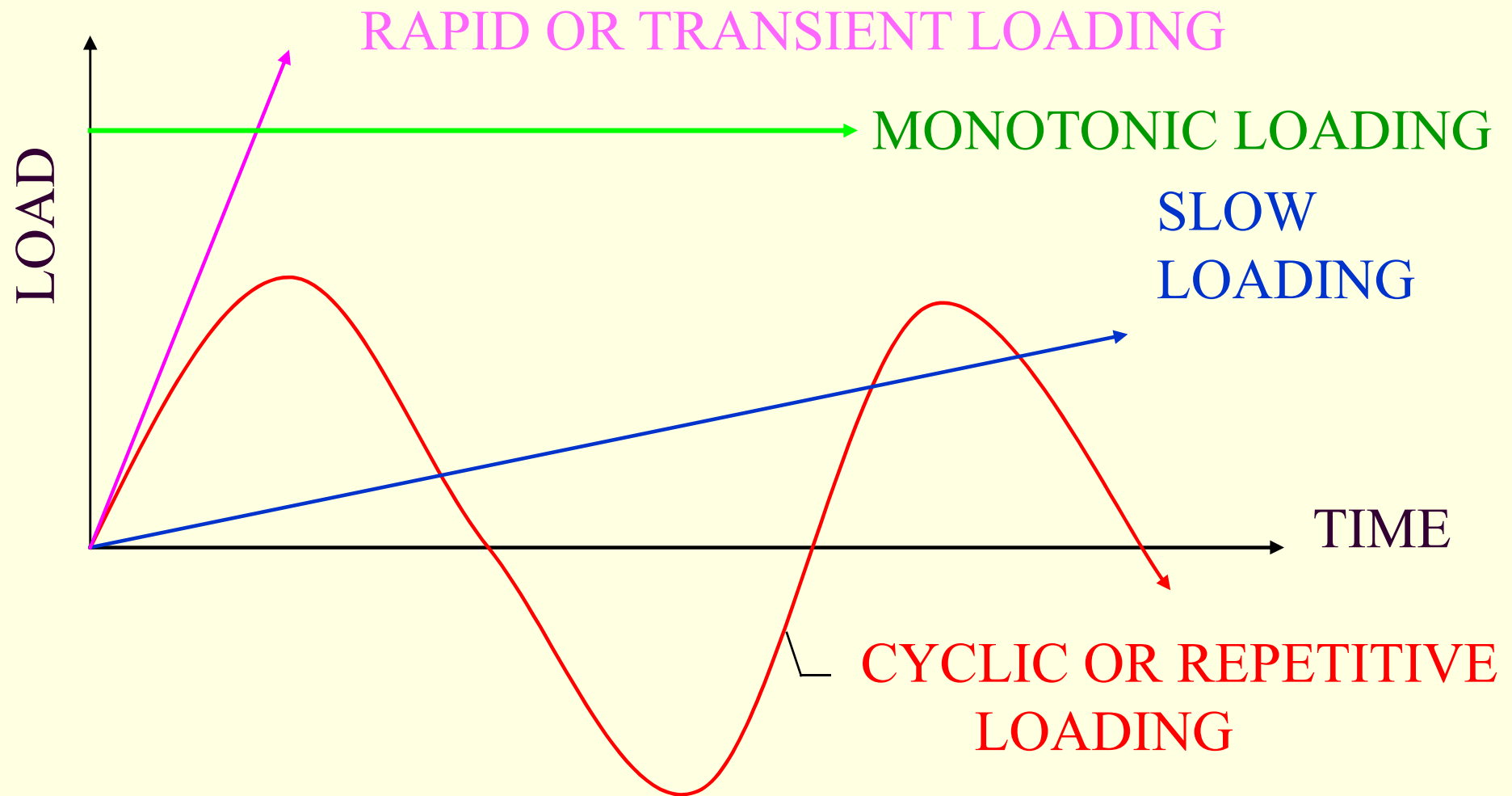


Static Loading

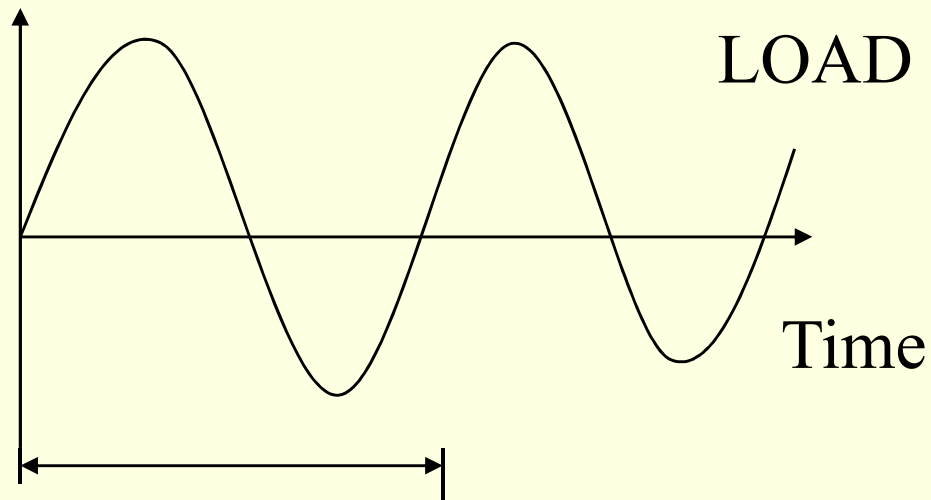


Dynamic Loading

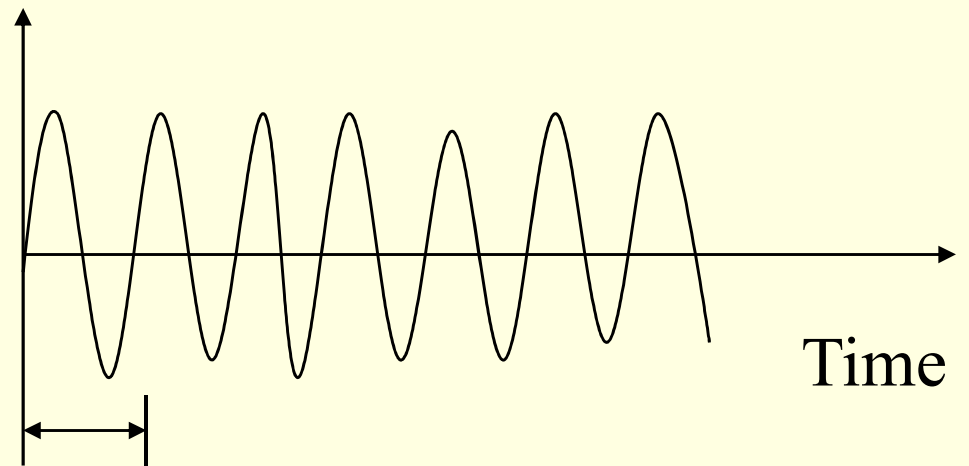
TYPES OF DYNAMIC LOADING



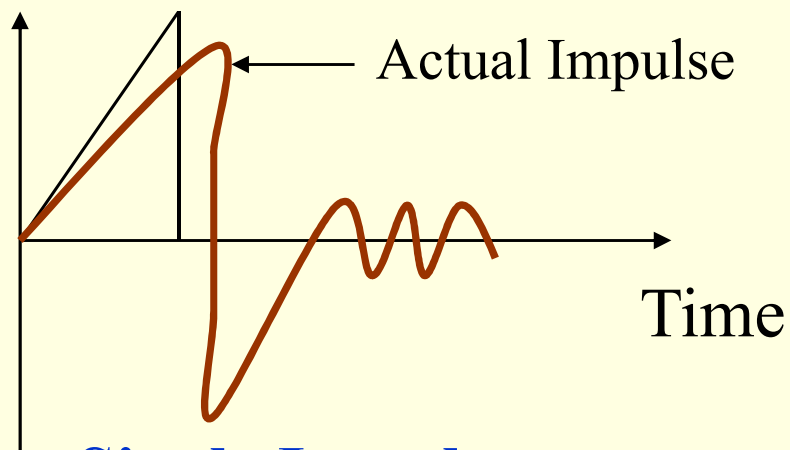
WHAT IS DYNAMIC FORCE ?



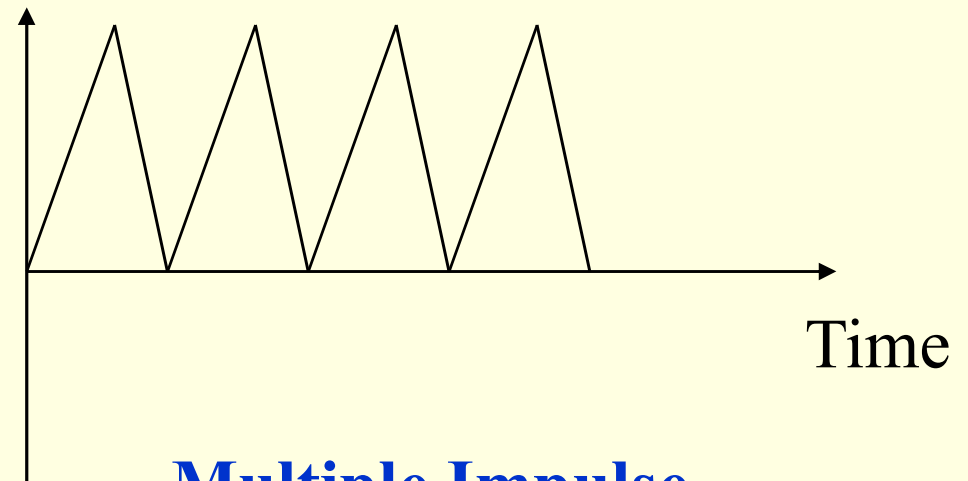
Large Period



Small Period

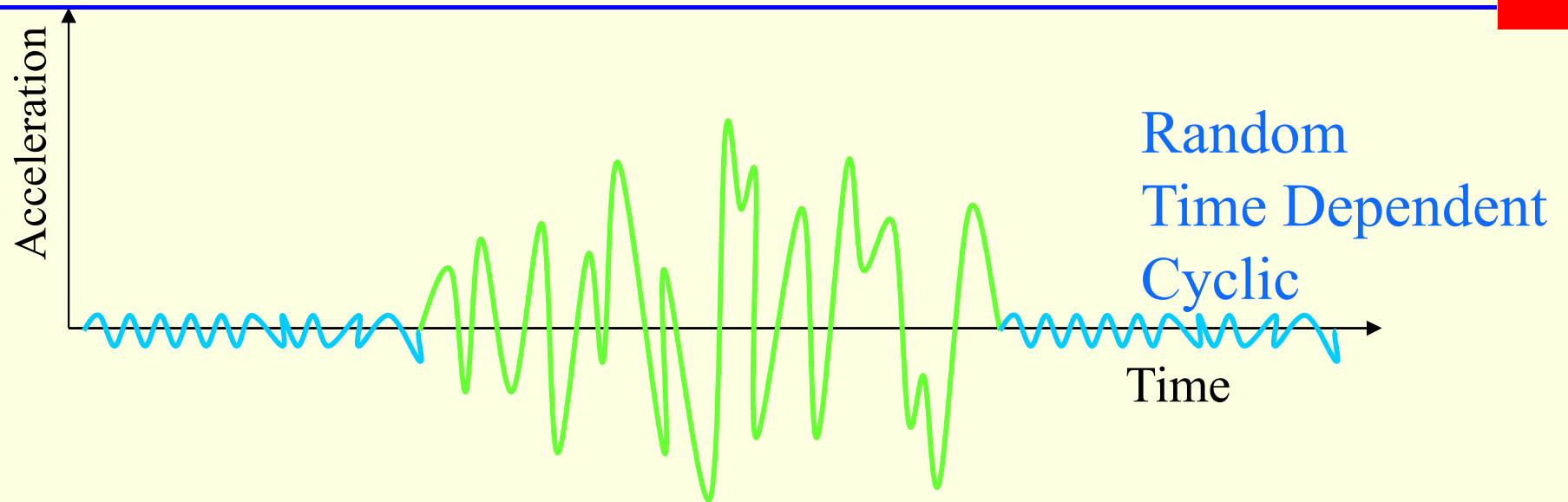


Single Impulse

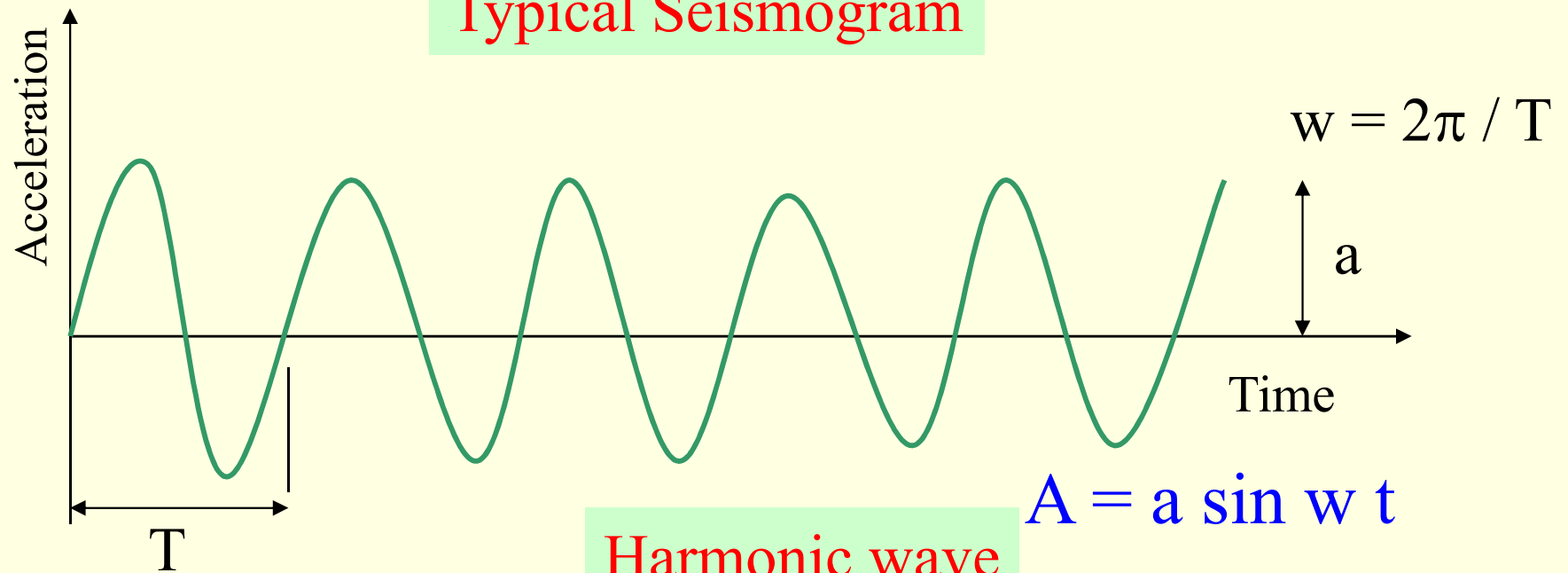


Multiple Impulse

Seismic and Harmonic Waves



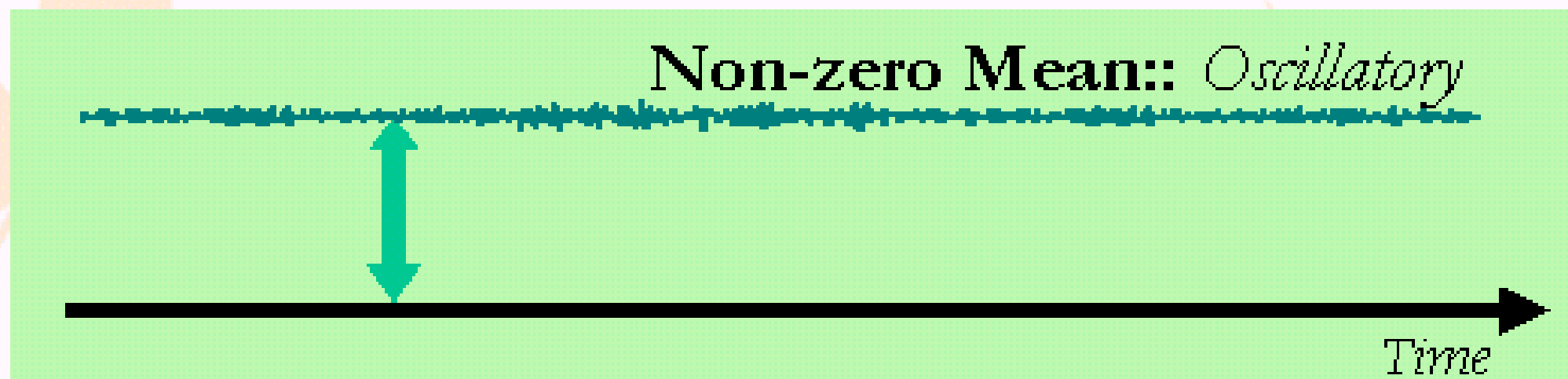
Typical Seismogram



Harmonic wave

Earthquake-Resistant Design

- Earthquake Force *versus* Wind Force



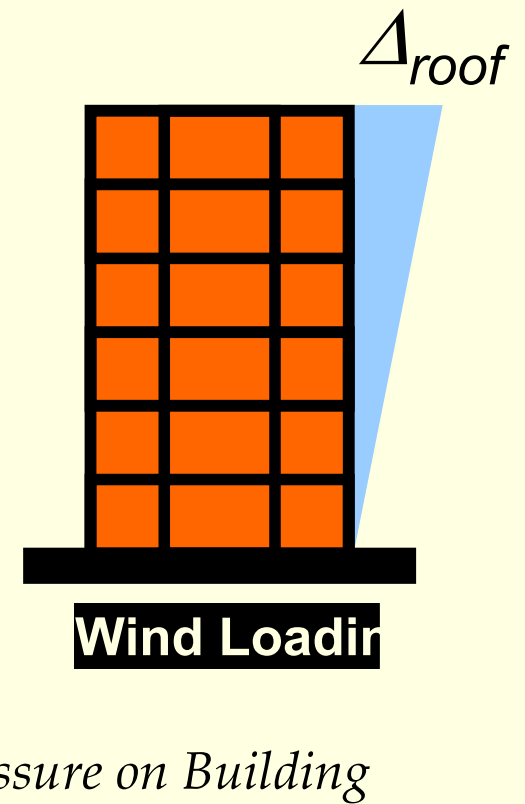
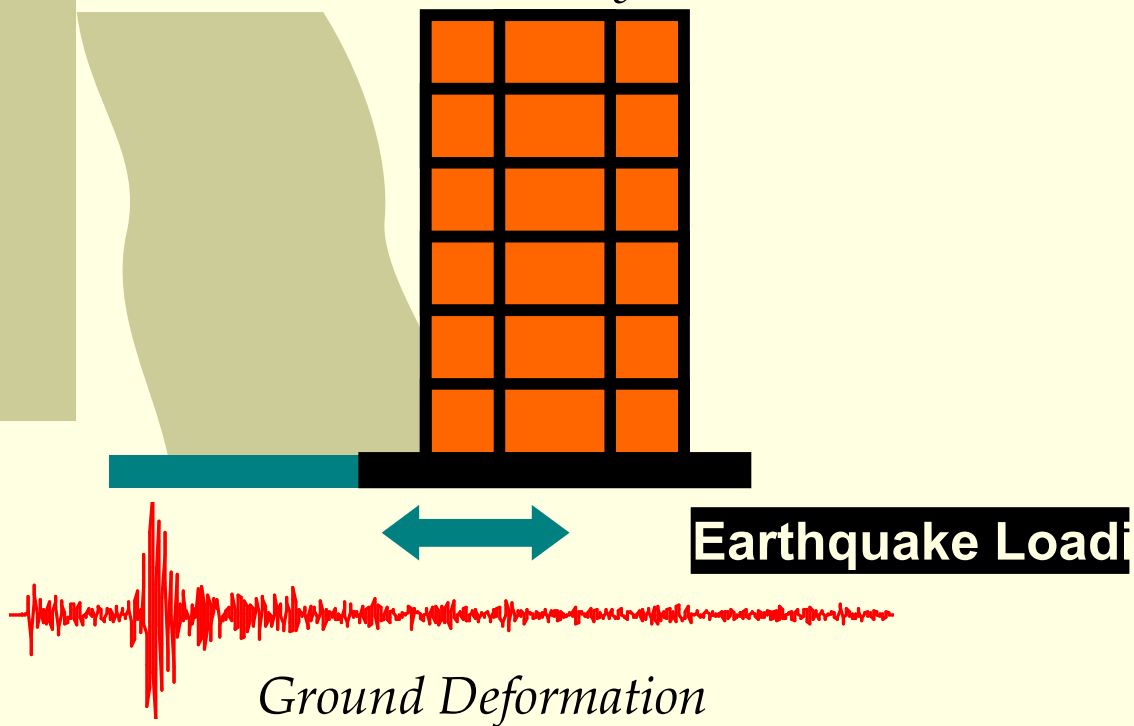
Nature of Loading

- Earthquake

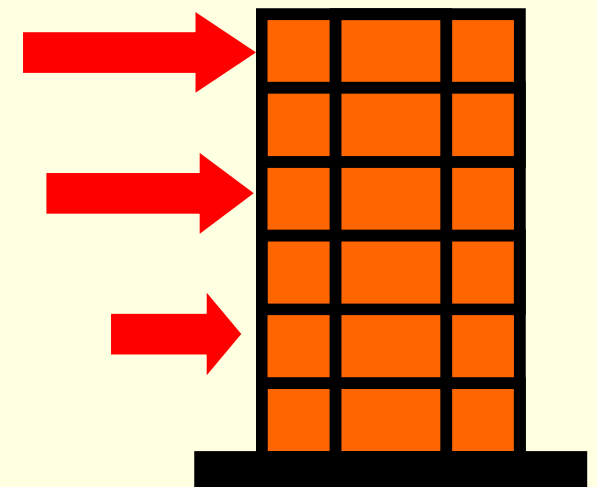
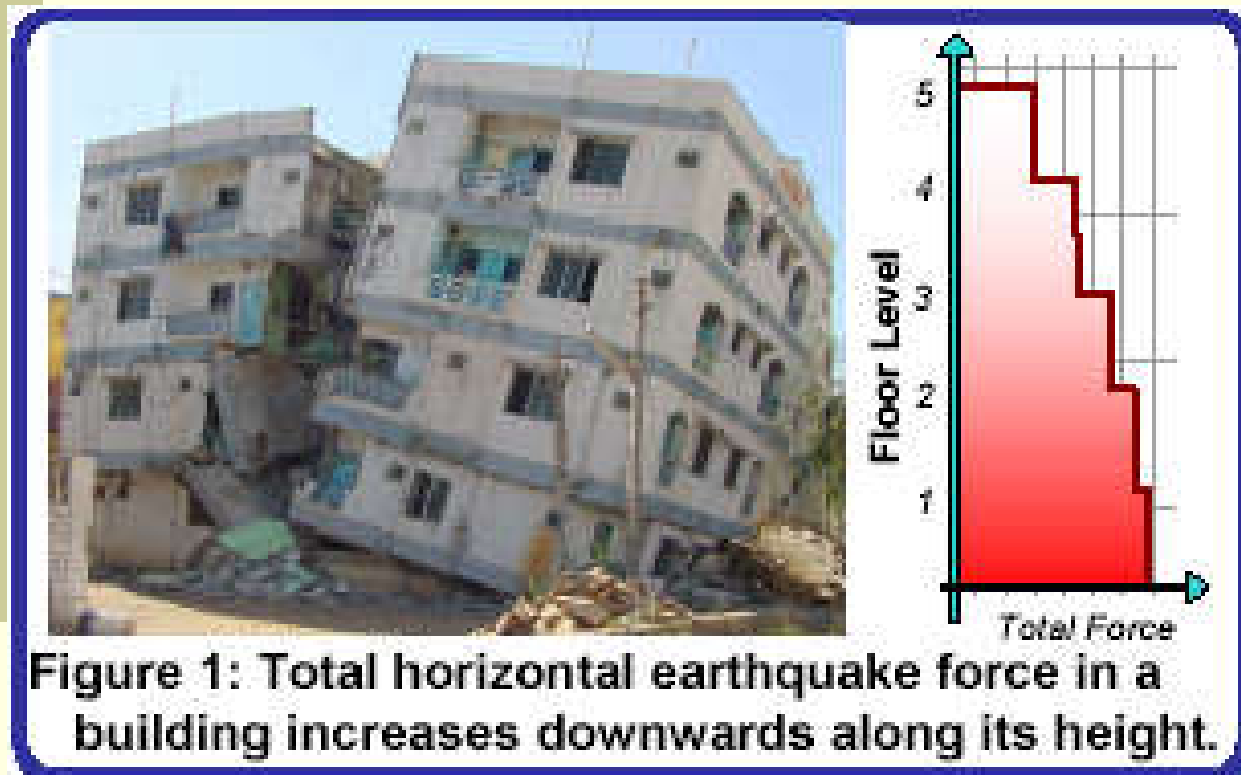
- *Random, Dynamic & Cyclic loading*

- Wind

- *Oscillatory & Monotonic loading*



Distribution of Lateral Forces Due to Earthquake



Wind force

1. Wind Force depends on exposed surface
2. Pressure from above the surface and in one direction
3. Estimated wind speed is used to find pressure
4. Less uncertainties
5. No stress reversal
6. Mostly structural problem
7. Pseudo-static analysis possible
8. Non-zero mean

Seismic Force

1. Earthquake force depends on Mass of Structure
2. Base motion from below the GL in both directions
3. Peak acceleration & frequency are used
4. Uncertainties are more
5. Reversal of stresses
6. Seismologist, geotechnical engineer and structural engineer problem
7. Pseudo-static analysis yields erroneous results
8. Zero mean

Effects of Earthquake

■ Primary Effects

- Ground Break, Fault formation

■ Secondary Effects

- Failure of R. C. Structures
- Failure of railway, highway & bridges
- Land slides and slope failure
- Liquefaction and Foundation Failure
- Failure of retaining walls
- Tsunami

Ground Break



Failure of Buildings and Loss of Human Life



The question...

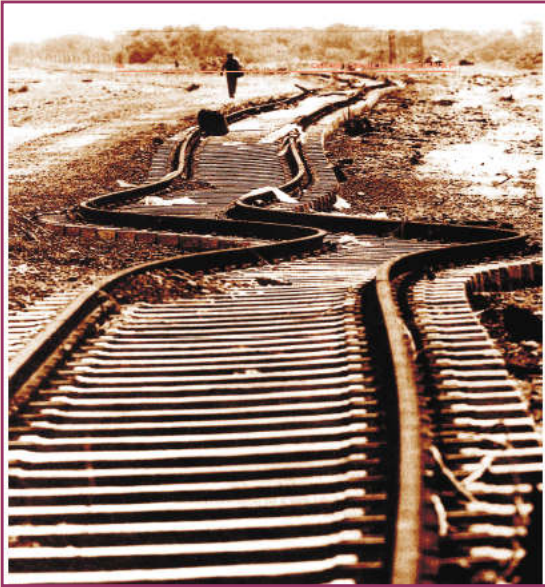


Is it only Low-Cost, and not Safety ??

Fallen Hanshin Expressway & loads of debris



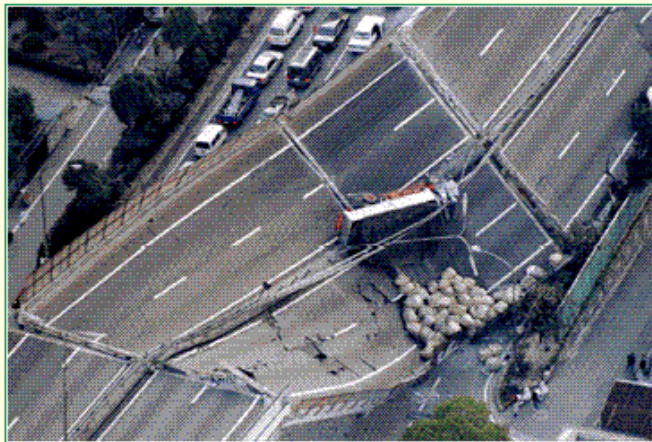
Failures after Earthquake



Railway track after
the Earthquake



Navalakki Port, Bhuj 2001



Failure of Express highway in Kobe



Ahmedabad, Bhuj 2001

Concrete Jungle





**Rapar Town after
Bhuj Earthquake
of 26th Jan 2001**

Japan during & one year later (11/03/2012)



Embankment failure of a fast track bullet train





**Hyogo Ken Nambu
Earthquake, Japan
January 17, 1995**

Failure of Dams



Failure and Tilting of Building Blocks due to Liquefaction



Is it Leaning
Tower
of Pisa ?

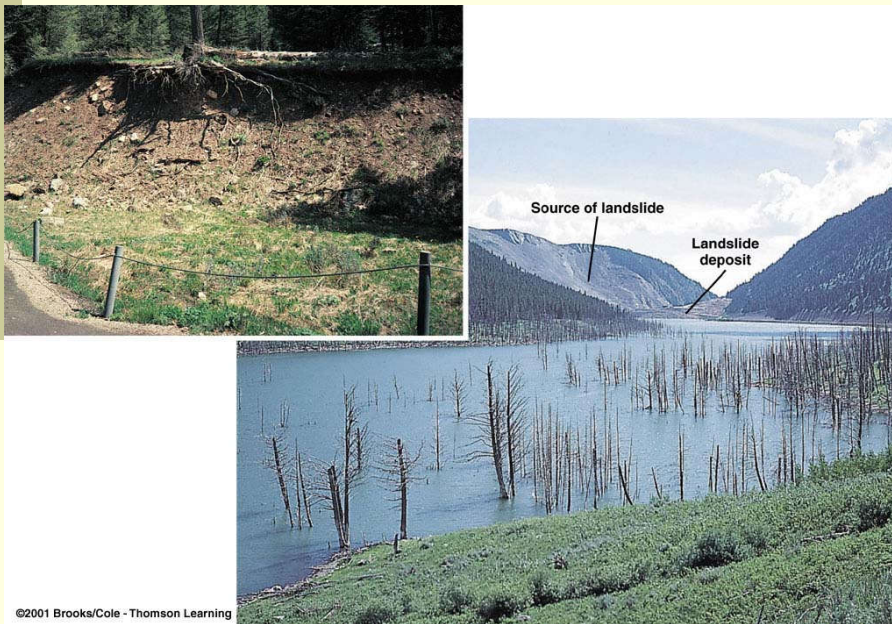


Short Circuiting & Leaking gas



The Economics and Societal Impacts of Earthquakes

- Building collapse
- Fire
- Tsunami
- Ground failure



Tsunami Effects

津波

Japanese word:

“Tsu” means
“harbor”

“Nami”
means
“wave”

English translation:
“Harbor wave”



Importance of Seismic Design

Region	Date	M	Death	Injured & Homeless
Kobe, Japan	17 th Jan 1995	7.2	5500	3 Lakh Homeless
Izmit, Turkey	17 th Aug 1999	7.8	18000	50000 Injured
Chi Chi, Taipei	21 st Sept 1999	7.3	2500	Thousands Injured
Gujarat, India	26 th Jan 2001	7.9	25000	12 Lakh Homeless
Seattle, USA	28 th Feb 2001	6.8	1	272
Kaman, Iran	26 th Dec 2003	6.6	20000	80000 Casualties in 1 Lakh Population

BIGGEST EARTHQUAKES RECORDED

SI No	Magnitude	Date	Place	Damage
1	9.5	22/05/1960	Chile	5000 deaths, 20 Lakh homeless
2	9.2	28/03/1964	Alaska	125 deaths, Tsunami
3	9.1	26/12/2004	Indonesia	2.26 Lakh killed, Tsunami
4	9.0	04/11/1952	Russia	0 death, Tsunami
5	9.0	11/03/2011	Japan	15000 deaths, Tsunami
6	8.8	27/02/2010	Chile	500 deaths, Tsunami
7	8.8	31/01/1906	Ecuador	1000 deaths
8	8.6 – 8.9	11/04/2012	Indonesia	0 death
9	8.7	04/02/1965	Alaska	0 death, Tsunami
10	8.6	28/03/2005	Indonesia	1300 deaths

Indian Earthquakes of the Past

Year of Occurrence	Place	Maximum	Intensity	Other Features
1618	Bombay	-	-	2000 lives lost
1720	Delhi	6.5	-	some lives lost
1737	Bengal	-	-	300,000 lives lost
1803	Mathura	6.5	-	The shock felt up to Calcutta.
1803	Kumaon	6.5	-	Killed 200-300 people.
1819	Kutchch	8.0	XI	Towns of Tera, Kathara & Mothala razed to ground.
1828	Srinagar	6.0	Intensity	1000 people killed.
1833	Bihar	7.7	X	Hundreds of people killed
1848	Mt.Abu	6.0	-	Few people killed
1869	Assam	7.5	-	Affected an area of 2,50,000 Sq. miles.
1885	Srinagar	7.0	-	Kamiarary area destroyed.
1897	Shillong	8.7	XII	Wide spread destruction in Shillong.
1905	HP	8.0	XI	Thousands of people killed.
1906	HP	7.0	-	Heavy damage.
1916	Nepal	7.5	-	All houses collapsed at Dharchulla.
1918	Assam	7.6	-	Heavy damage.
1930	Meghalaya	7.1	IX	Heavy damage in Dhubri.
1934	Bihar, Nepal	8.3	XI	Large number of border area people killed.
1935	Quetta (Pak)	7.5	IX	25,000 people killed
1941	Andaman	8.1	X	Very heavy damage.

Past earthquakes in India

Date	Event	Time	Magnitude	Max. Intensity	Deaths
16 June 1819	Cutch	11:00	8.3	IX	1,500
12 June 1897	Assam	16:25	8.7	XII	1,500
8 Feb. 1900	Coimbatore	03:11	6.0	VII	Nil
4 Apr. 1905	Kangra	06:10	8.0	X	19,000
15 Jan. 1934	Bihar-Nepal	14:13	8.3	X	11,000
15 Aug. 1950	Assam	19:39	8.6	X	1,530
21 Jul. 1956	Anjar	21:02	6.1	IX	115
10 Dec. 1967	Koyna	04:30	6.5	VIII	200
23 Mar. 1970	Bharuch	20:56	5.2	VII	30
21 Aug. 1988	Bihar-Nepal	04:39	6.6	IX	1,004
20 Oct. 1991	Uttarkashi	02:53	6.4	IX	768
30 Sep. 1993	Killari (Latur)	03:53	6.2	VIII	7,928
22 May 1997	Jabalpur	04:22	6.0	VIII	38
29 Mar. 1999	Chamoli	00:35	6.6	VIII	63
26 Jan. 2001	Bhuj	08:46	7.7	X	13,805

Past earthquakes in India

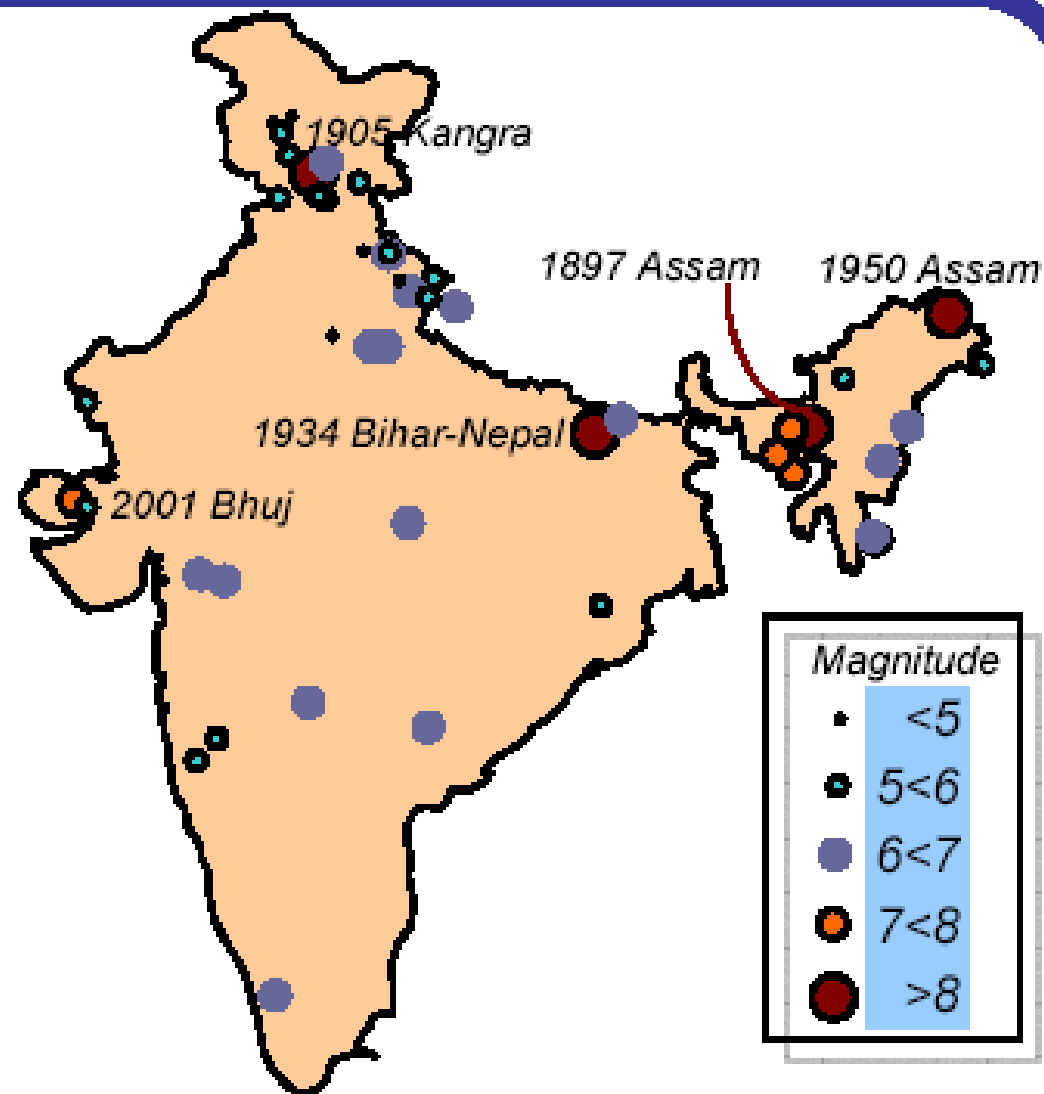
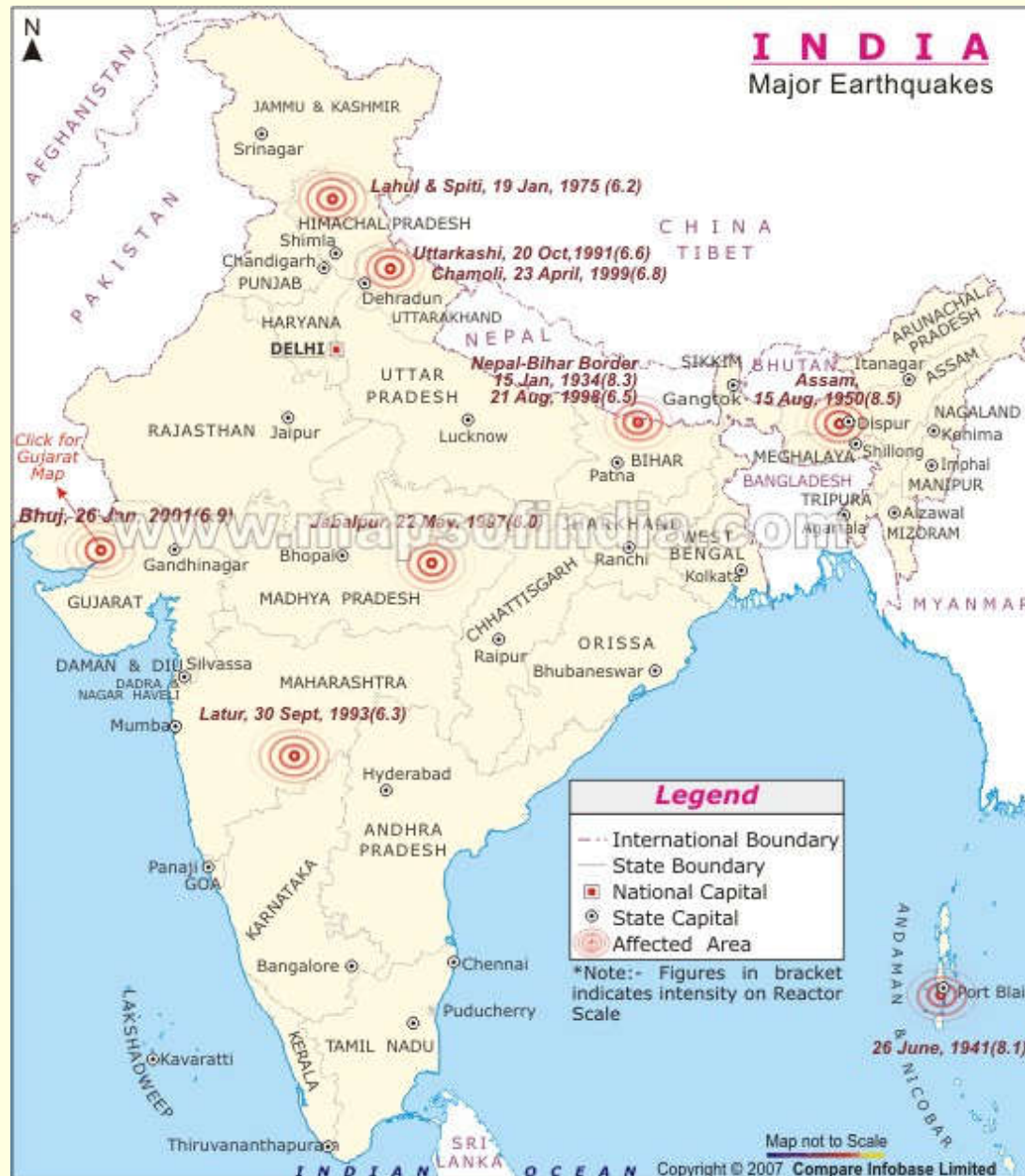


Figure 2: Some Past Earthquakes

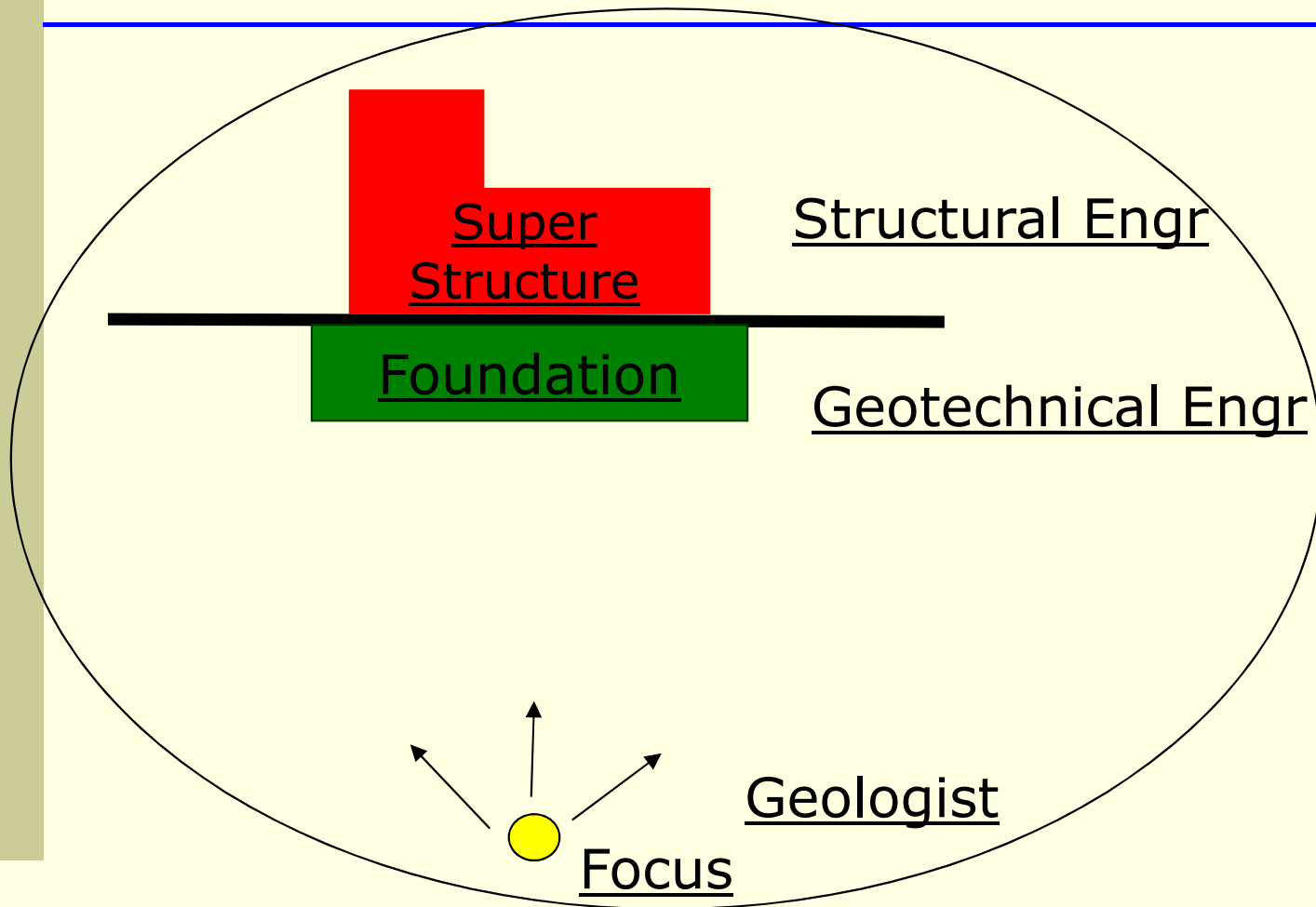
Recent Major Earthquakes in India



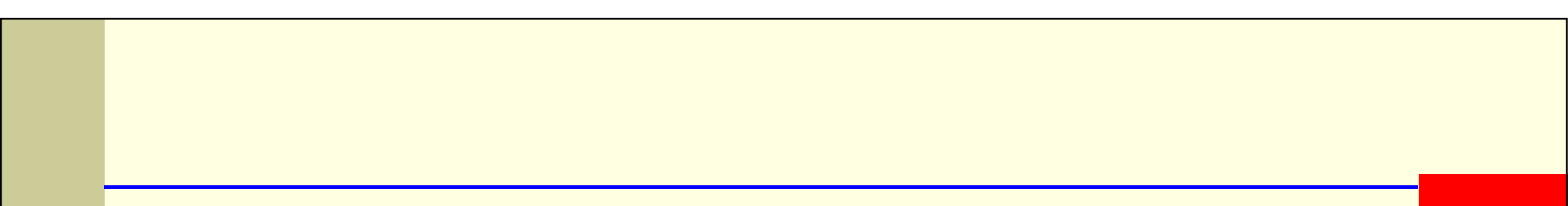
Sumatra Earthquake (2004)
Kashmir Earthquake (2005)
Sikkim Earthquake (2011)

Frequency of Earthquakes of Different Magnitudes

Frequency of Earthquakes of Different Magnitude Worldwide		
Magnitude	Number of Earthquakes per Year	Description
> 8.5	0.3	Great
8.0 - 8.4	1	
7.5 - 7.9	3	Major
7.0 - 7.4	15	
6.6 - 6.9	56	
6.0 - 6.5	210	Destructive
5.0 - 5.9	800	Damaging
4.0 - 4.9	6,200	Minor
3.0 - 3.9	49,000	
2.0 - 2.9	300,000	
0 - 1.9	700,000	

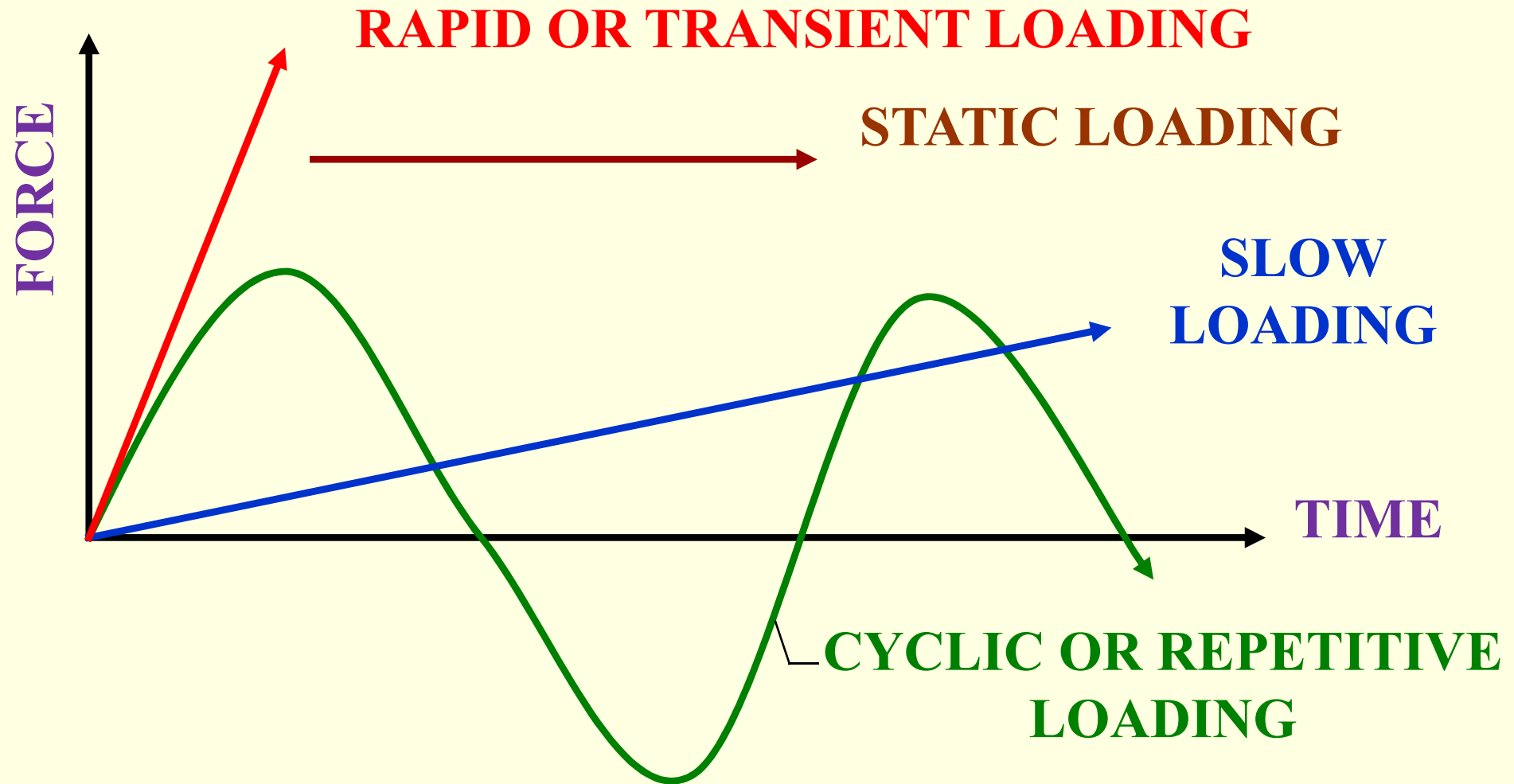


**Water Resource Engr, Mechanical Engr, Environmental Engr,
Electrical Engr, Sociologist + Good Manager, Earthquake
Engineering is Interdisciplinary**

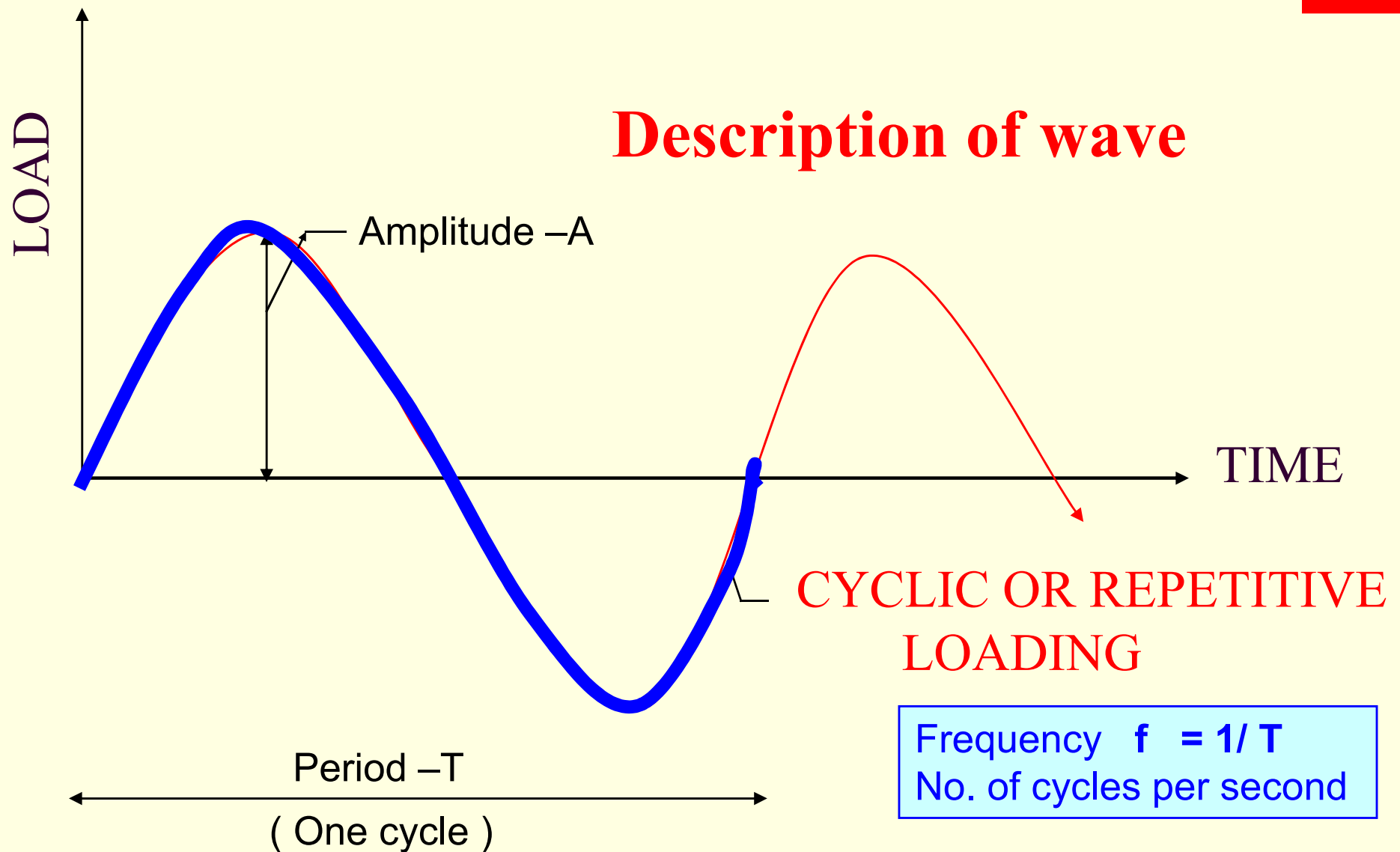


Characteristics of Seismic Ground Motion

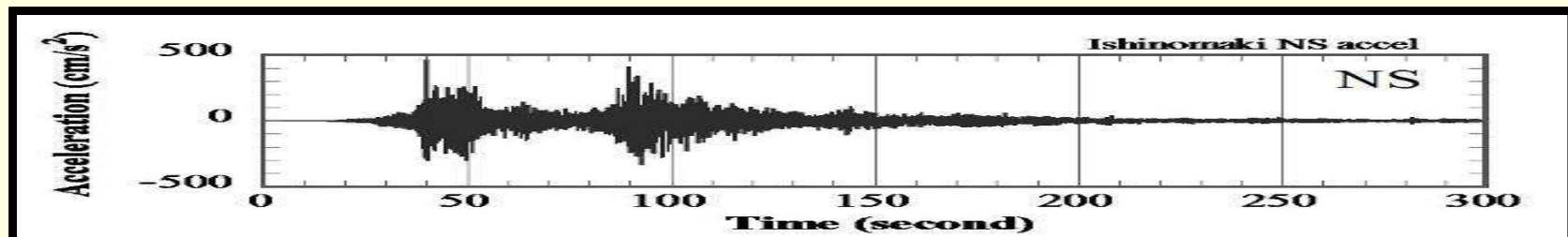
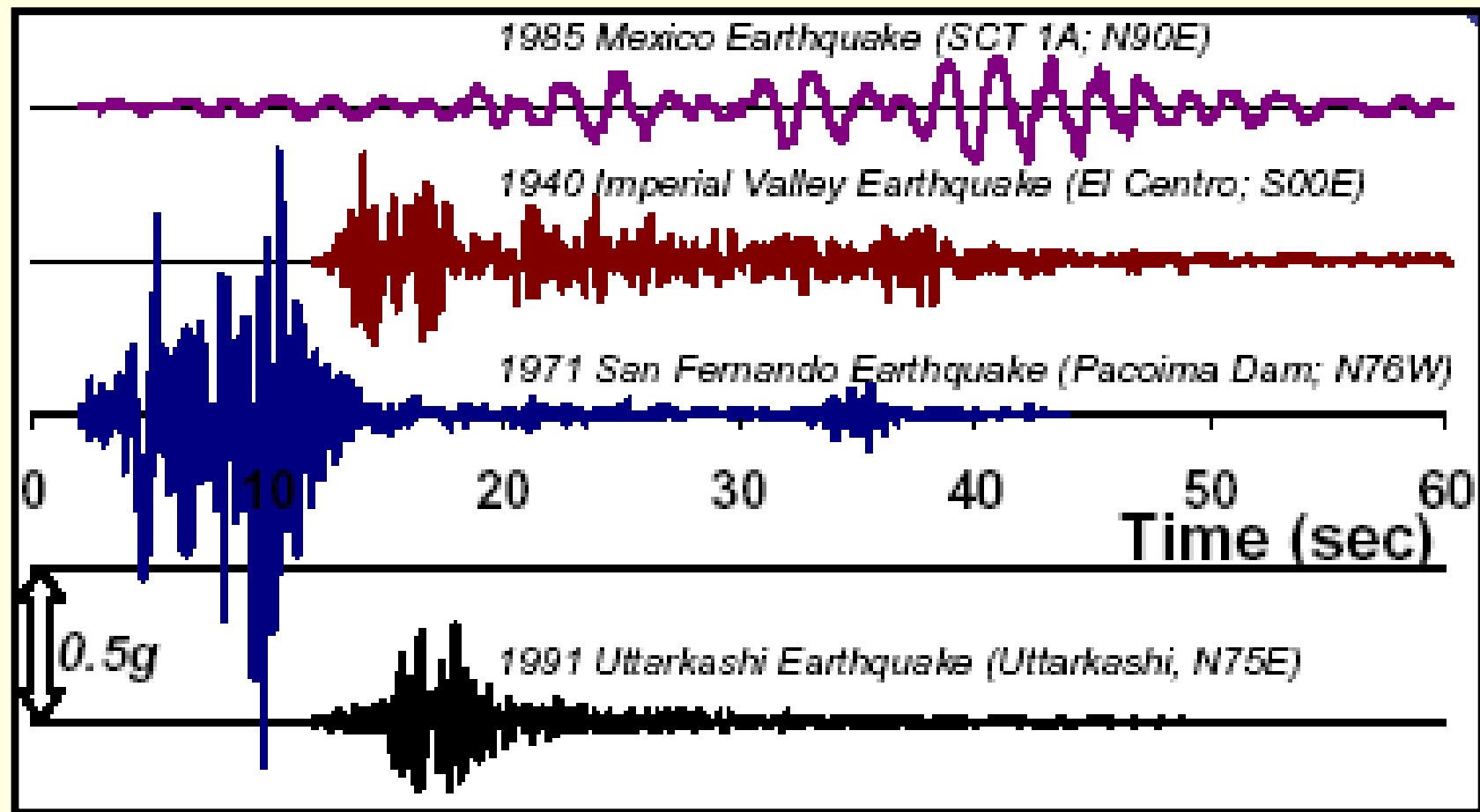
TYPES OF LOADING



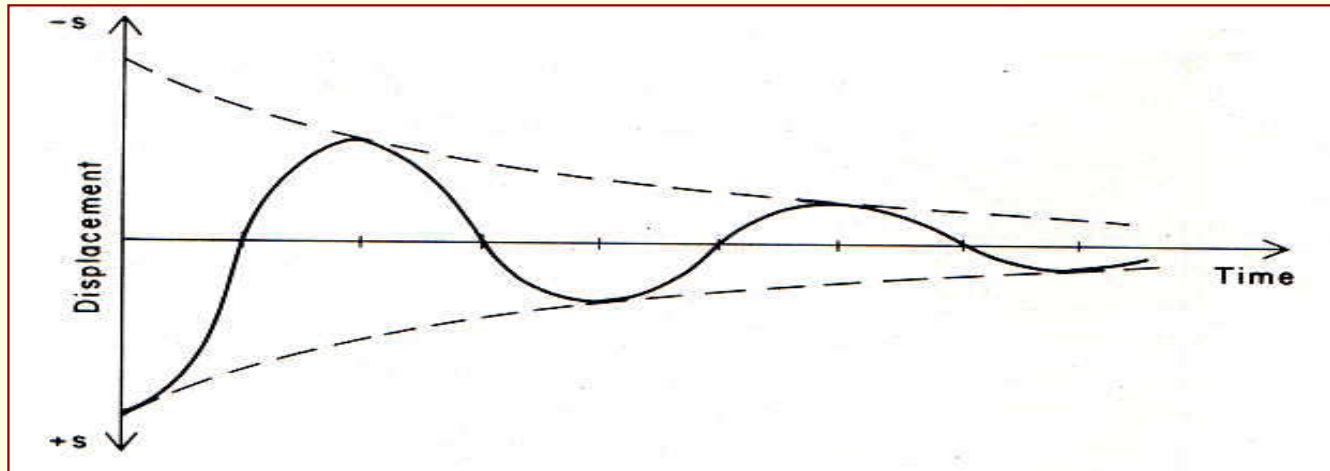
Description of wave



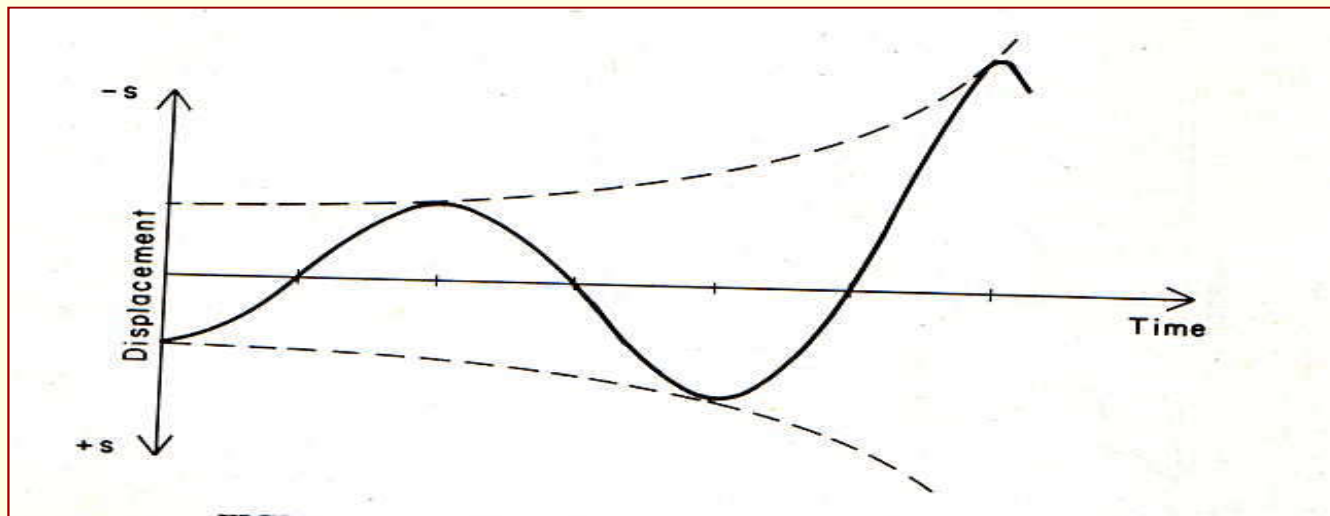
No two earthquake motions are similar



DAMPING AND RESONANCE



**Effect of
Damping**



**Effect of
Resonance**



Earthquake

Vibrations of the earth surface caused by waves originating from a source of disturbance in the earth mass

What are Earthquakes?

- The shaking or trembling caused by the sudden release of energy
- Usually associated with faulting or breaking of rocks
- Continuing adjustment of position results in aftershocks

Structure and Diameter of Earth

Diameter along Equator- 12740 km

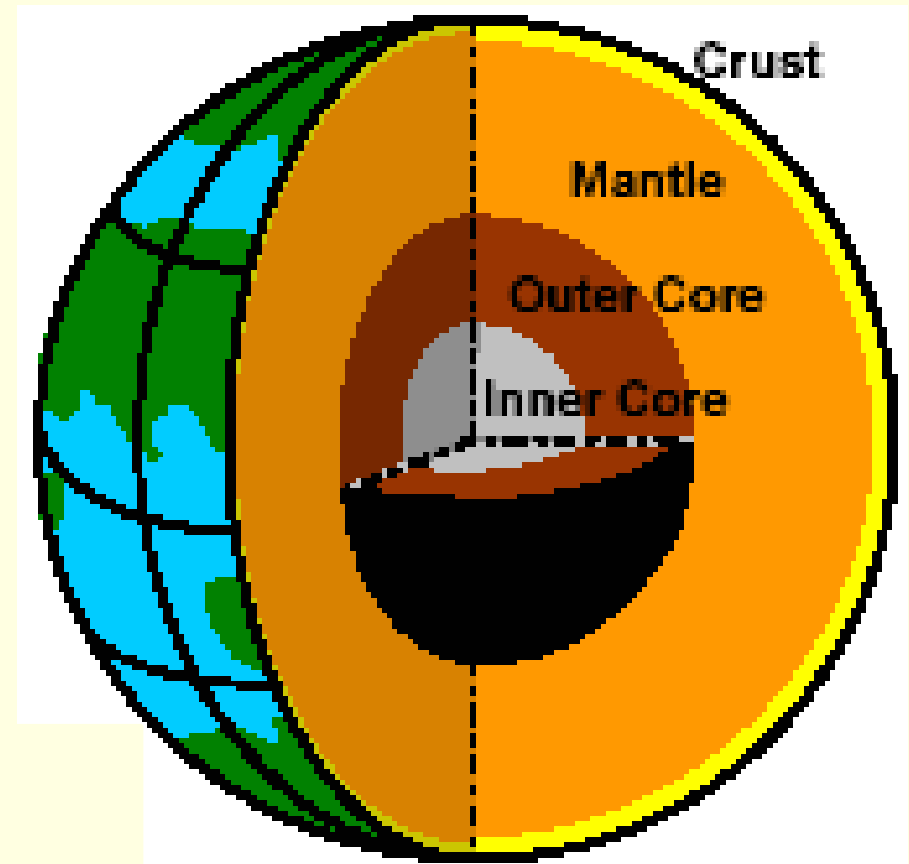
Polar Diameter _ 12700 km

The higher diameter along equator is caused by the higher centrifugal forces generated along the equator due to rotation of earth

Anatomy of Earth

Region	Radius (km)
Inner Core	1290
Outer Core	2200
Mantle	2900
Crust	5 to 70

- Temperature
- Pressure
- Density



Anatomy of Earth

Four Parts

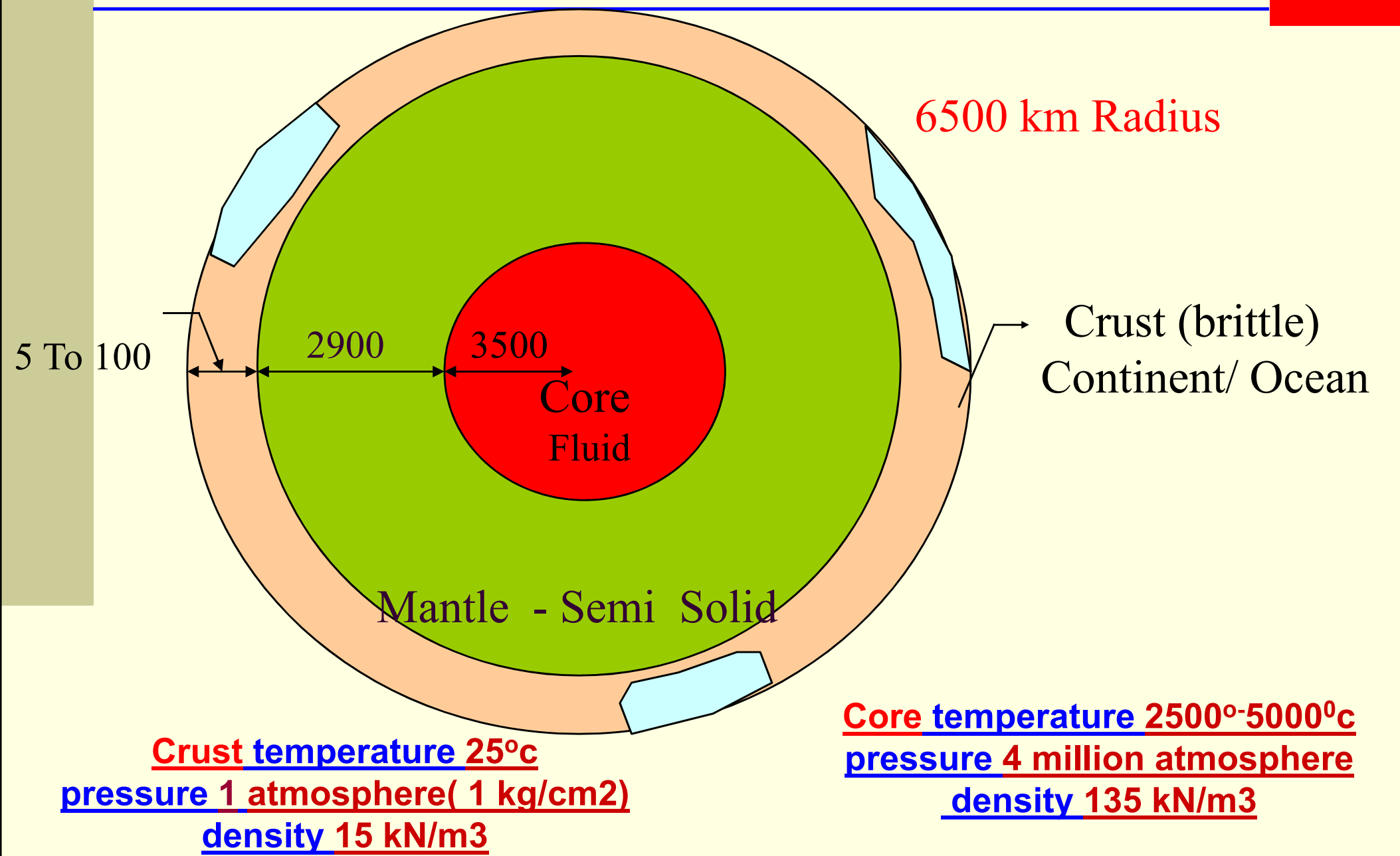
Earth Crust – 5- 40 km, density 15-20 kN/m³

Lithosphere – crust and upper part of mantle-
70-100 km – under Deep Ocean
and 100-150 km – under continent (properties same as crust)

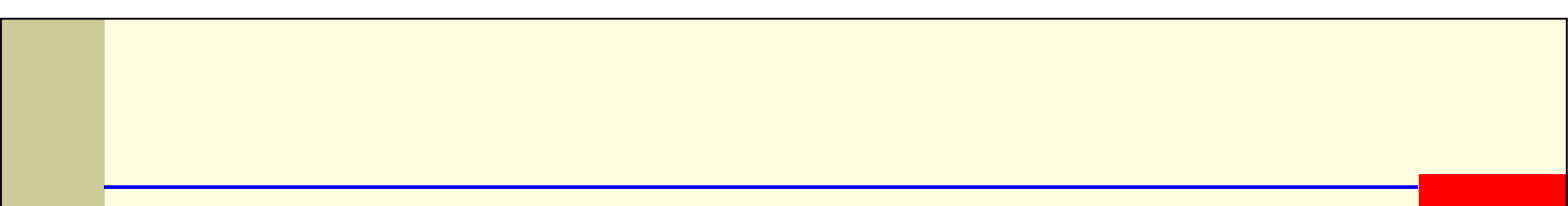
Asthenosphere – 150 km thick, has lower rigidity and partially molten
plays imp. Role in plate tectonics, density- 50-60 kN/m³

Barysphere – core (inner-1224 and Outer-2200) , density – 130 kN/m³
Composed of Iron and Nickel alloys with silica. Pressure 4×10^6 Atmosphere

Composition of Earth

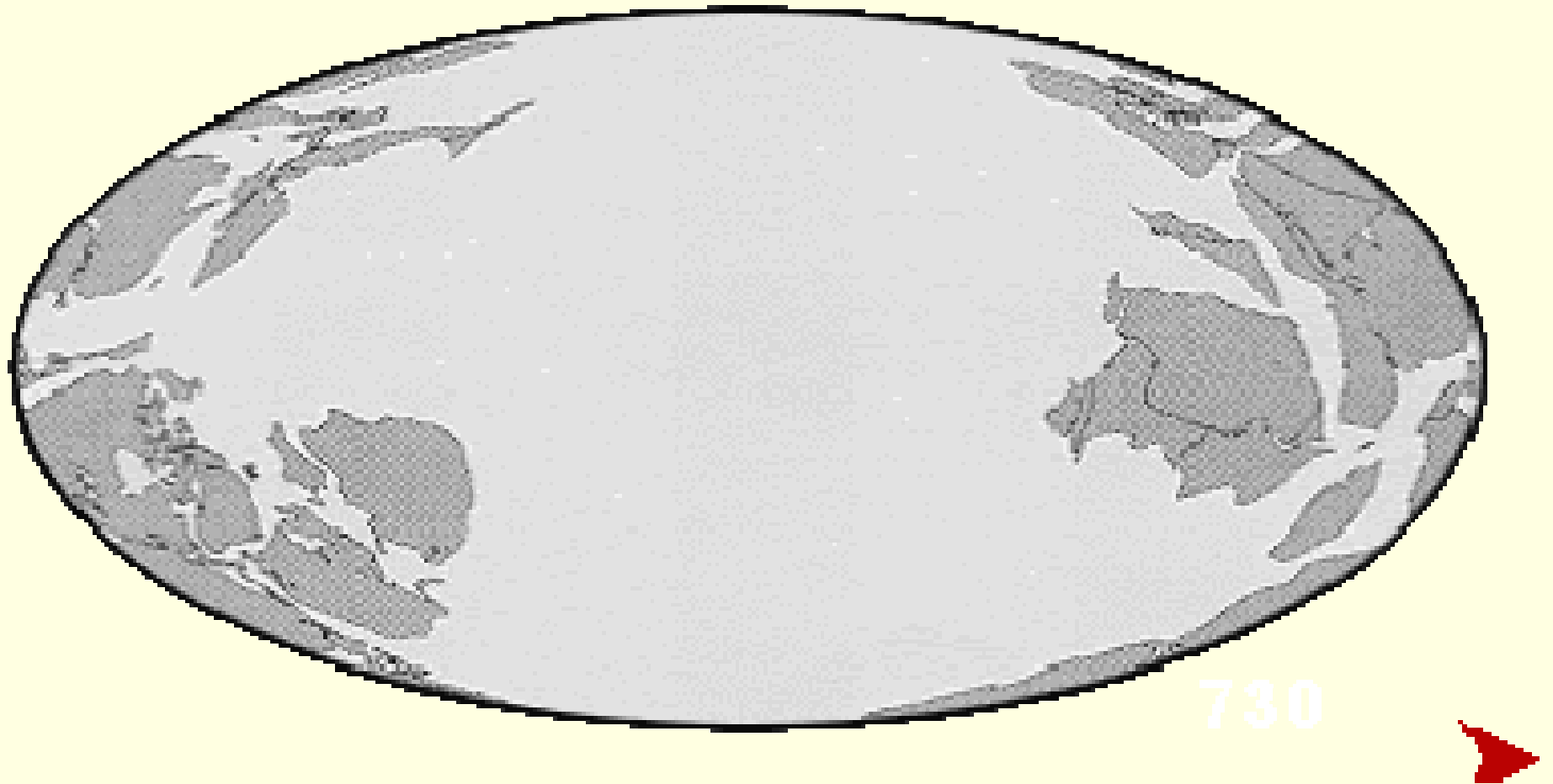


(Core Pressure -mountain of 4000 cars piled up)



Continental Drift

Continental Drift



Continental Drift

Alfred Wegener -1912

- large “supercontinent” (Pangaea) existed and then split into pieces
- fossil & glacial deposit evidence

Wegener not able to provide
MECHANISM for his theory

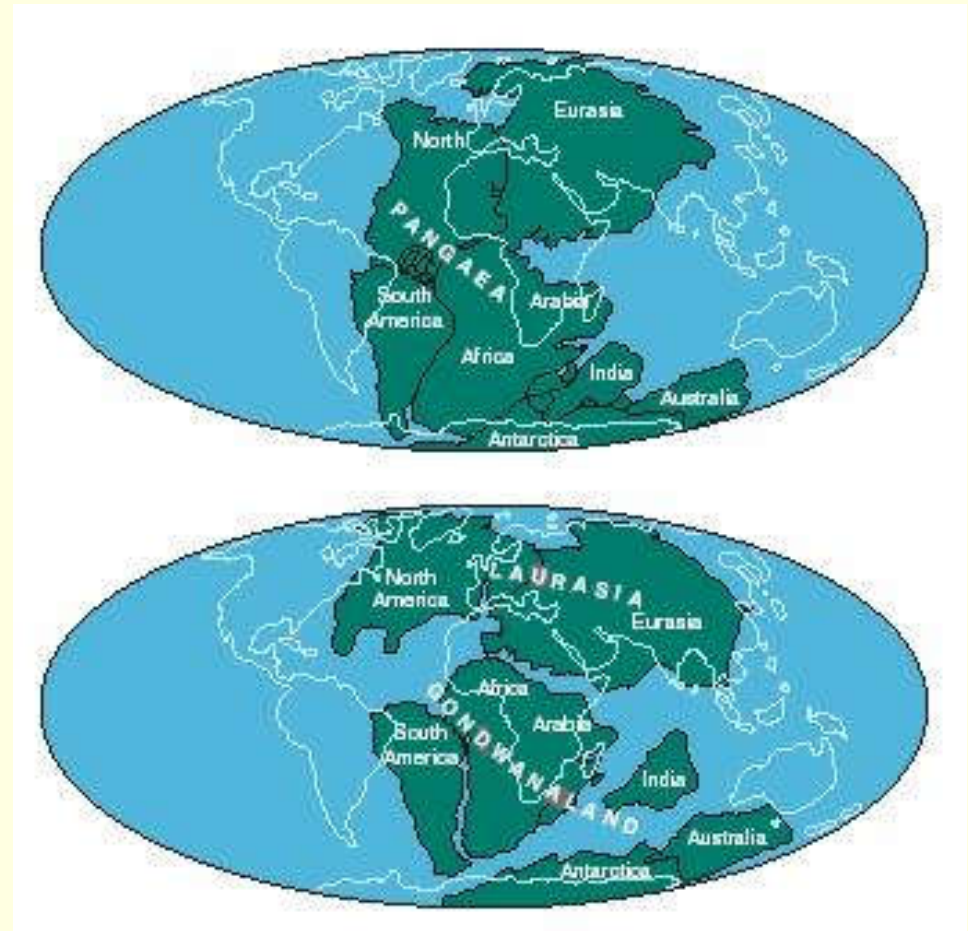
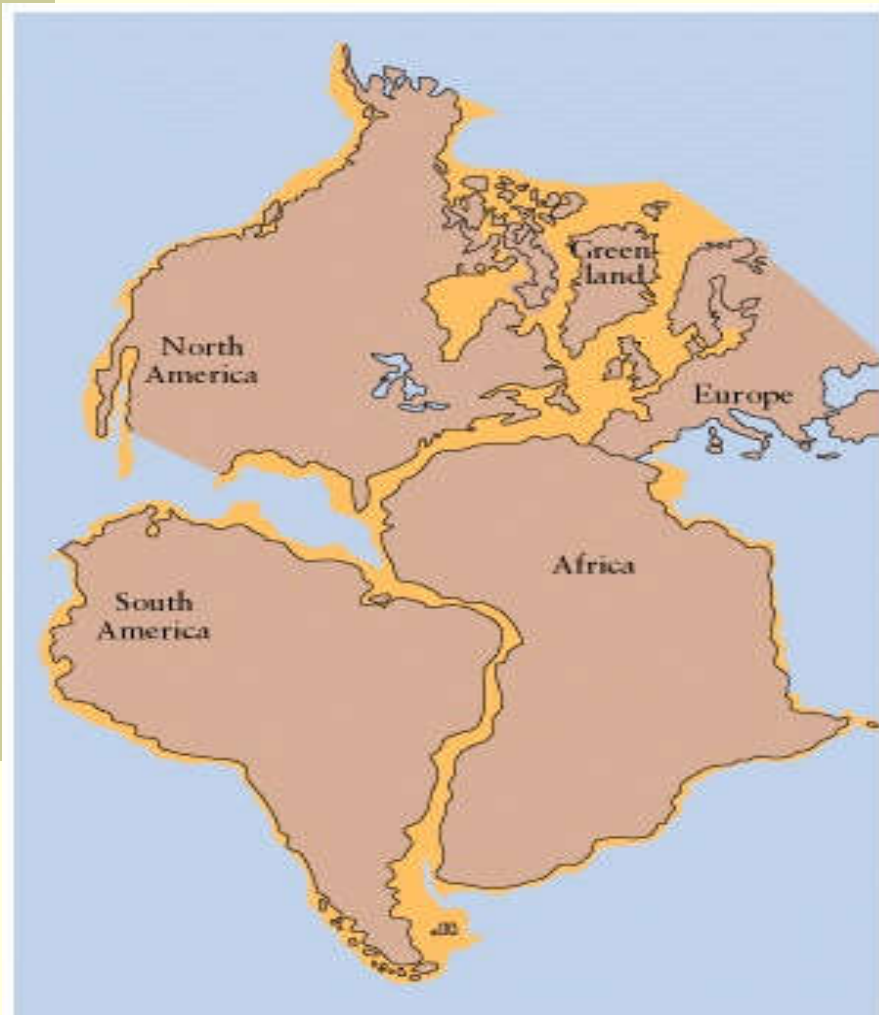
Major mechanism later found in the
OCEANS



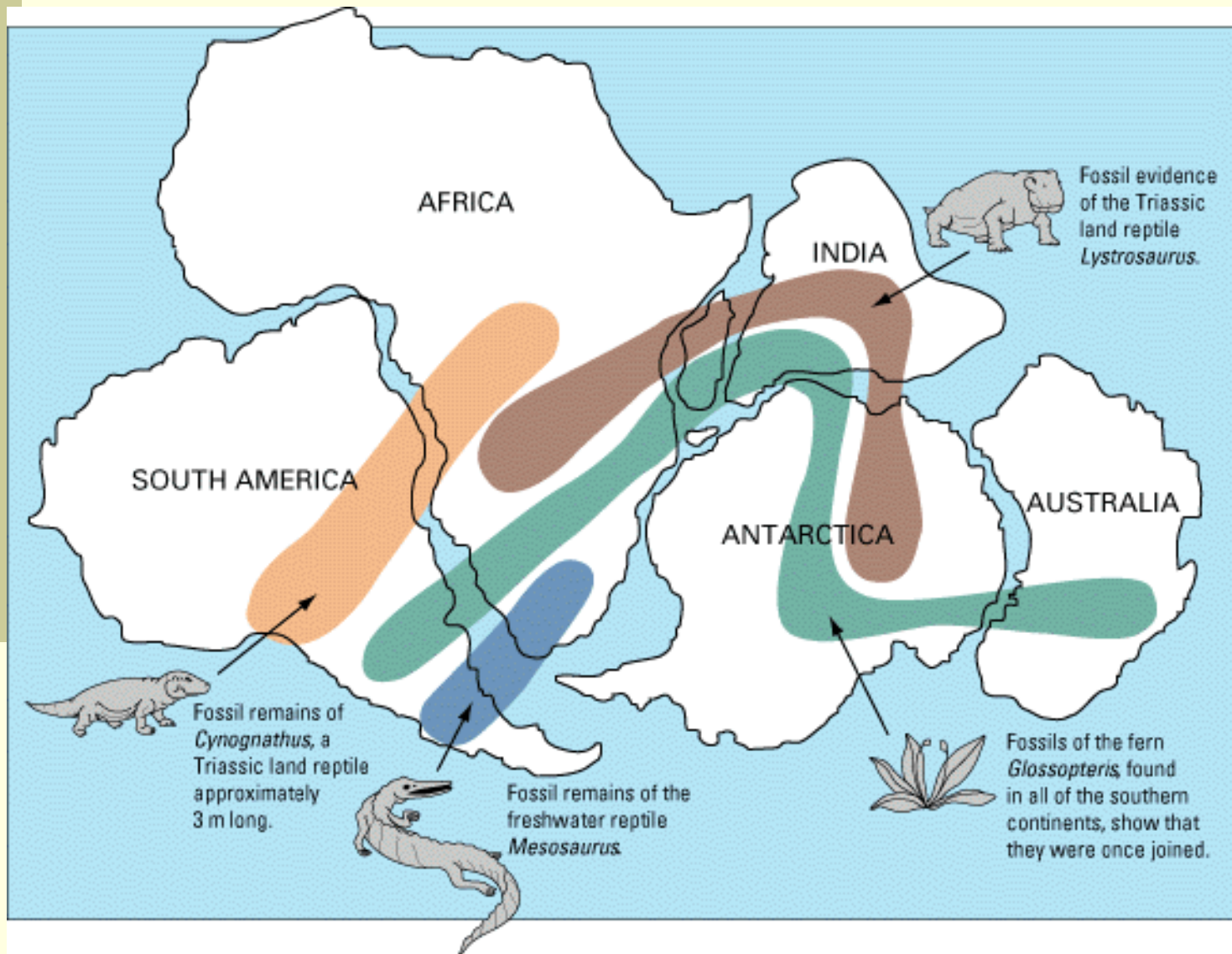
Evidence for Continental Drift

- There is a noticeable jigsaw fit between many continents - for example, between the East Coast of South America and the West Coast of Africa, which suggests that the continents were once assembled together.
- A number of identical fossils have been found distributed across the southern continents. Fossils of the Mesosaurus dating back 280 million years ago are found in South America and Africa Plant Fossils, such as Glossopteris (a tree) have been found in South America, Africa, India and Australia.

Evidence for Continental Drift



Evidence for Continental Drift



Evidence for Continental Drift

- A number of continents show evidence of matching geological sequences with rocks of similar age, type, formation and structure occurring in different countries
- A number of climatic anomalies have been discovered which suggest that continents must once have been in a different position and therefore have experienced a different climate. Coal which only forms under wet / warm conditions have been found beneath the Antarctica ice cap and there is evidence of glaciation in Brazil

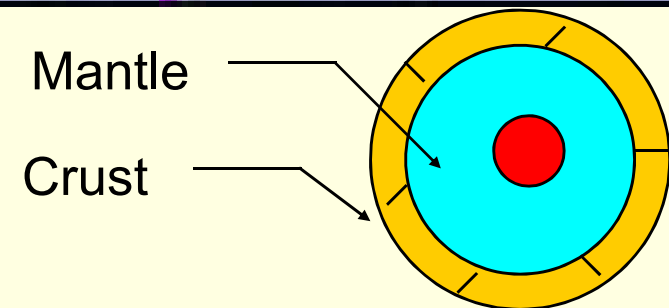
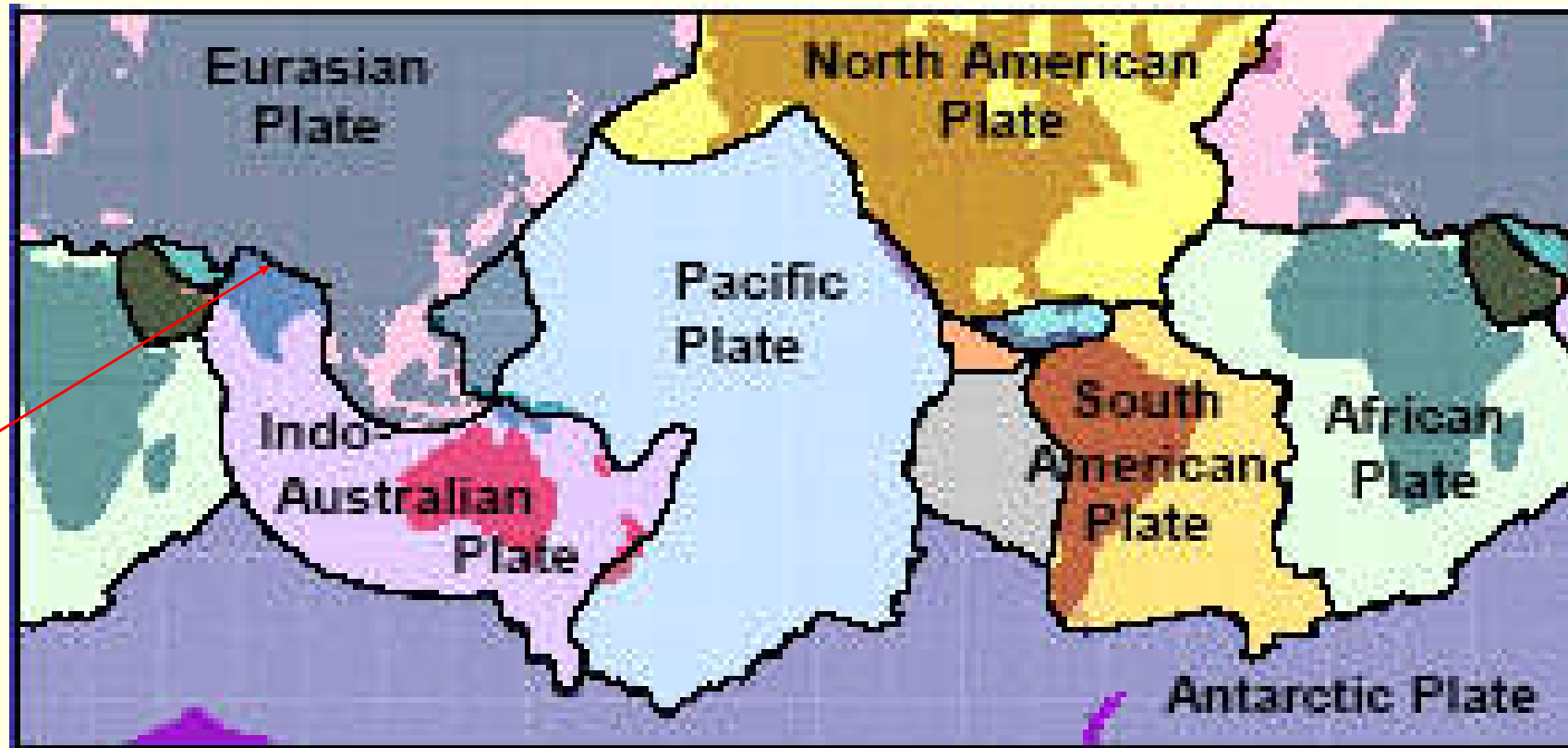
Continental Drift

1. That the continents were once joined.
Therefore, they must have moved apart over time.
2. Contracting Earth theory was not consistent with the facts.
3. Wegener proposed a mechanism for continental drift: pushing of the continents by gravitational forces that derived from the sun and the moon (similar to tides).

:

Plate Tectonics

Major Tectonic Plates on the Earth Surface



HOW EARTHQUAKE GENERATES ?

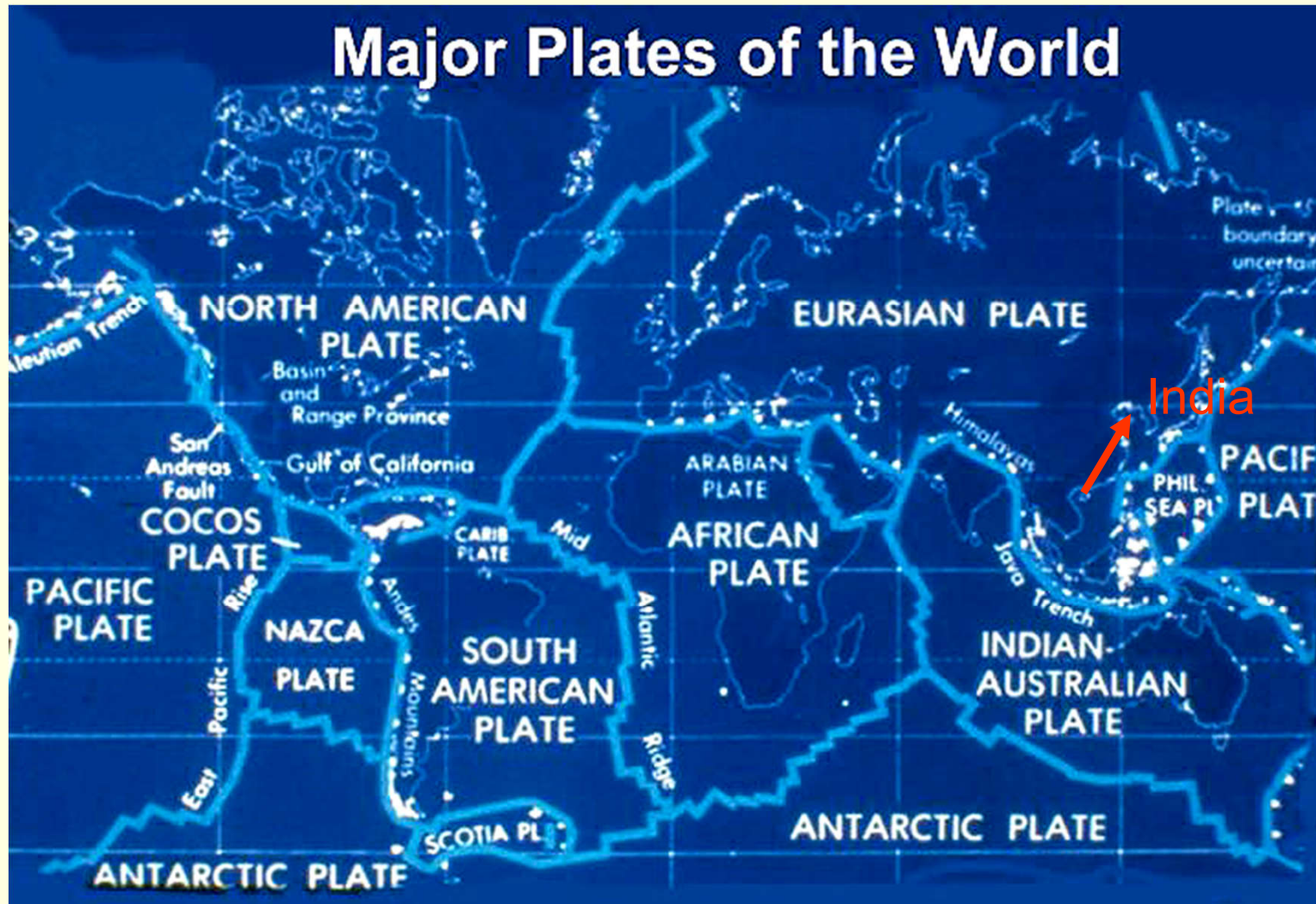
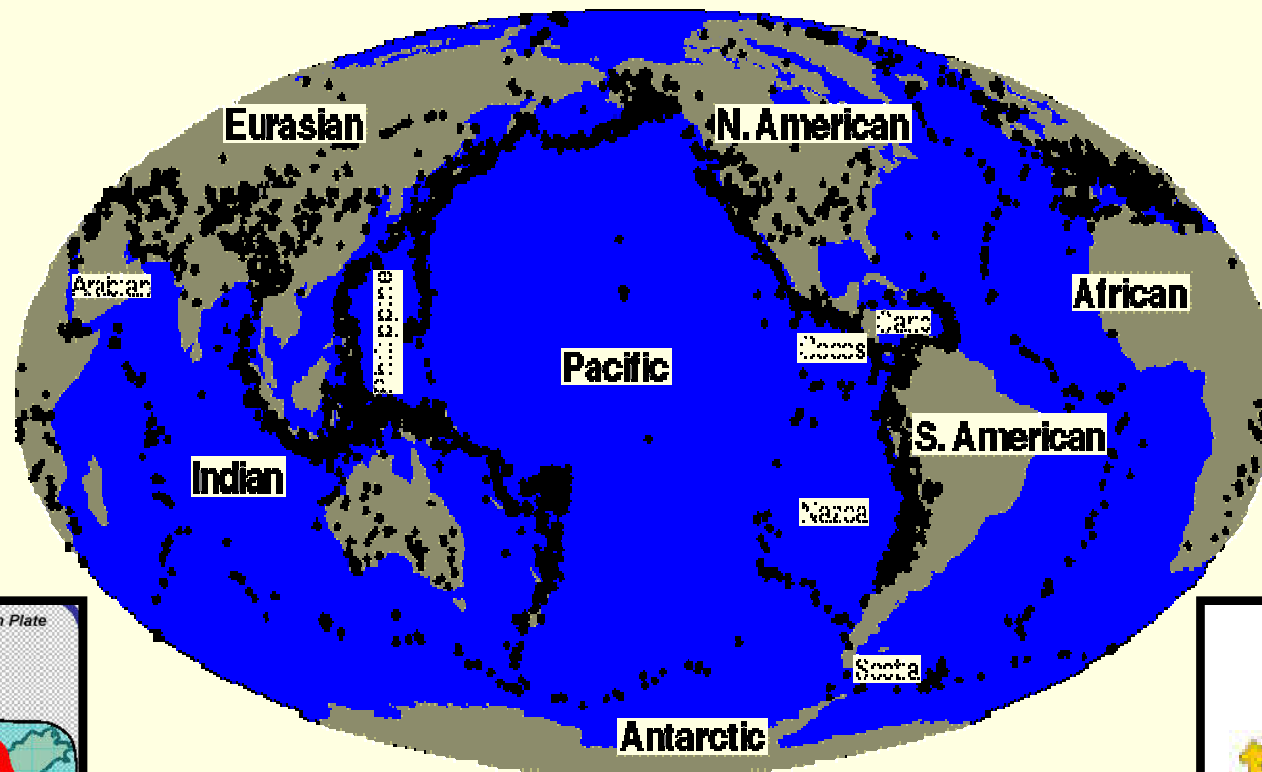
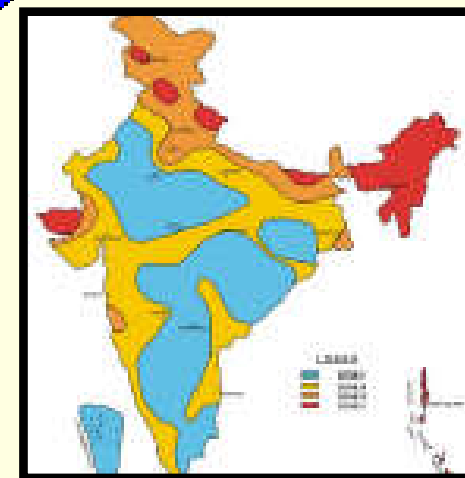
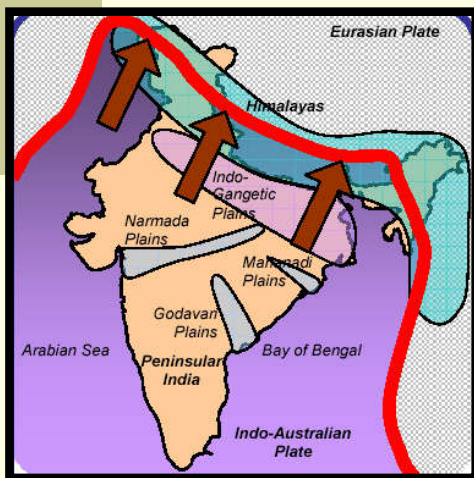


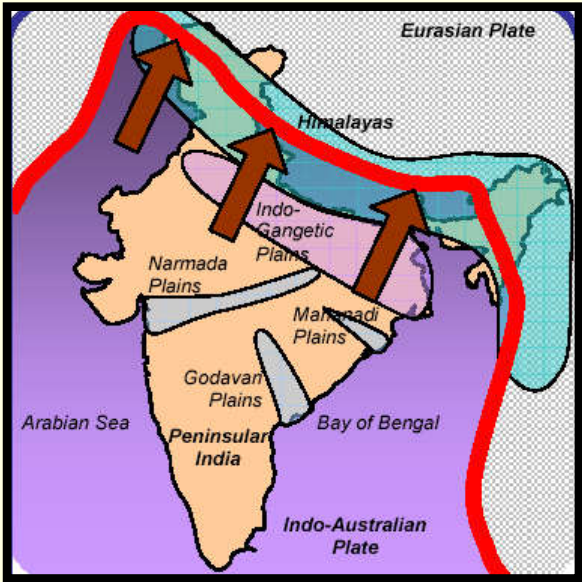
Plate Tectonics – Epicenters of recent earthquakes of moderate magnitude



**8 to 10 cm
Every year**



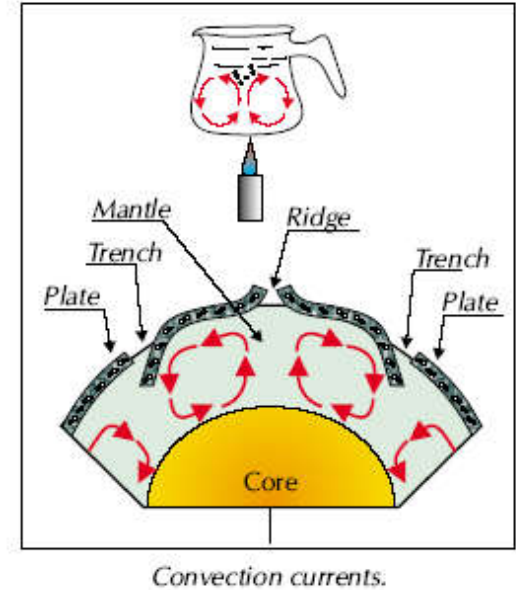
1000



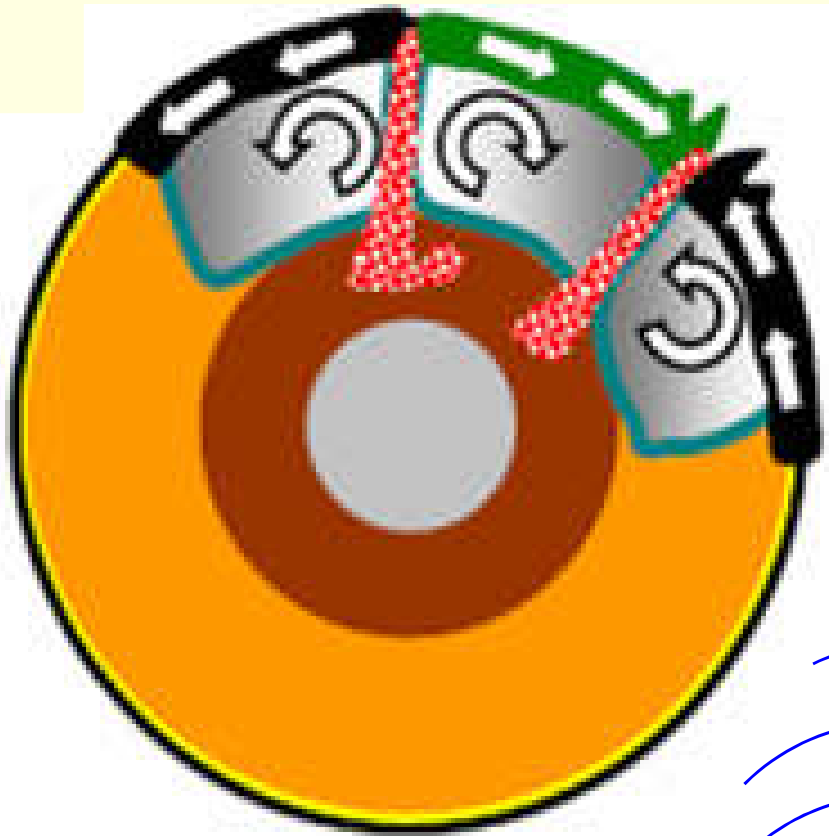
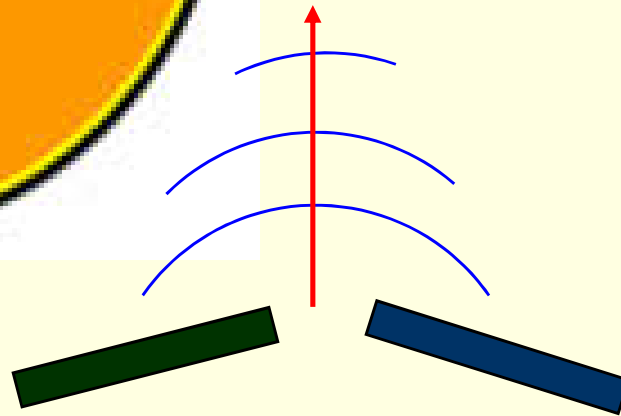
8 to 10 cm
Every year

Why Plates Move?

Some scientists have suggested that convection currents flowing in the mantle may cause the plates to move. Because of the great heat in the mantle, parts of the mantle may flow like a very thick liquid. Compare the diagram of the coffee pot to the diagram of the mantle. A plate might move above a huge convection current like an object riding along on a giant conveyor belt.



Warmer material to Raise



What drives Earth processes?

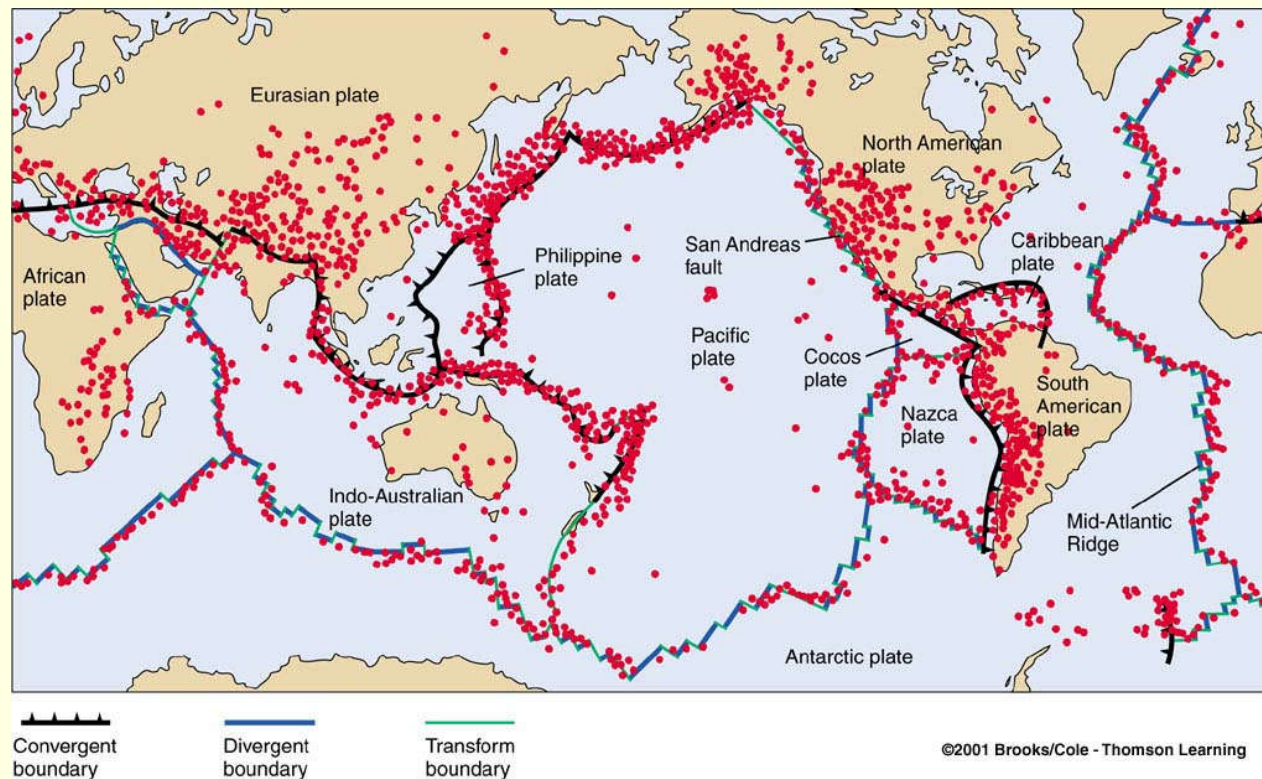
Gravity and density differences

External (e.g. hydrologic cycle, erosion)

Internal (e.g. mantle convection)

Where Do Earthquakes Occur and How Often?

- 95% of all earthquakes occur along the plate boundaries
- most of these result from convergent margin activity
- remaining 5% occur in interiors of plates and on spreading ridge centers
- more than 150,000 quakes strong enough to be felt are recorded each year



Properties of Plate Tectonic Theory

1. Continental crust is less dense, or lighter, than Oceanic crust so it doesn't sink. It is never destroyed and is considered permanent.
2. Oceanic crust is heavier so it can sink below Continental crust. It is constantly being formed and destroyed at ocean ridges and trenches.
3. Continental crust can carry on beyond the edges of the land and finally end far below the sea. This explains why the edges of all the continents don't have deep trenches right up against their coastlines.

Properties of Plate Tectonic Theory

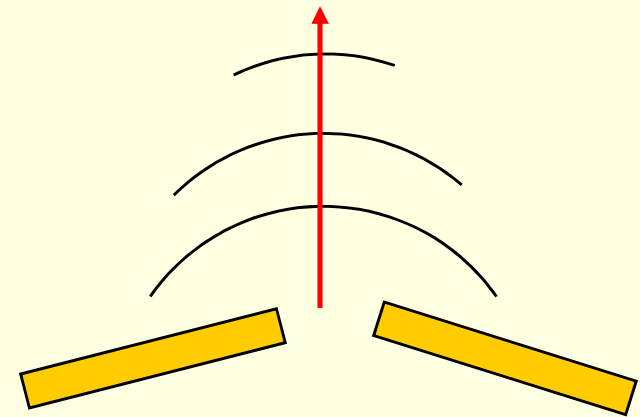
4. Plates can never overlap. This means that they must either collide and both be pushed up to form mountains, or one of the plates must be pushed down into the mantle and be destroyed.
5. There can never be gaps between plates, so if two plates move apart, as in the middle of the Atlantic, new rock will be formed to fill the space.

Properties of Plate Tectonic Theory

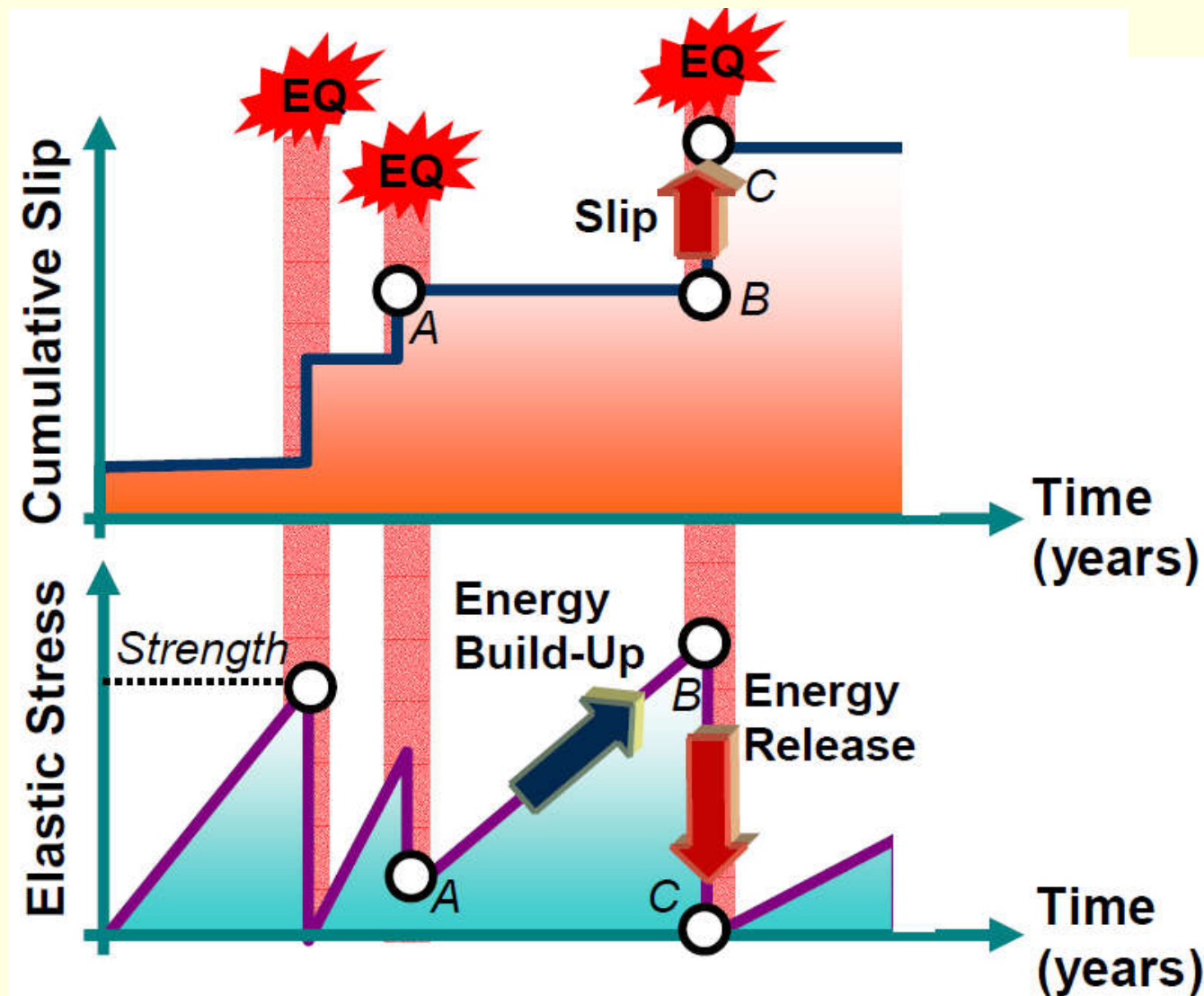
6. Earth is not getting bigger or smaller, so the amount of new crust being formed must be the same as the amount being destroyed.
7. Plate movement is very slow. Nobody could 'see' the continents moving. When the plates make a sudden movement, it is called an Earthquake, and it is the only time we are directly aware of the plates moving.

Plate Tectonic Theory

- Surface is made of 12 major plates – constantly drifting over semi molten mass
- Plates collide - stresses will develop
- Strain energy due to deformation ➤ Resilience, Energy is released
- Released in the form of waves



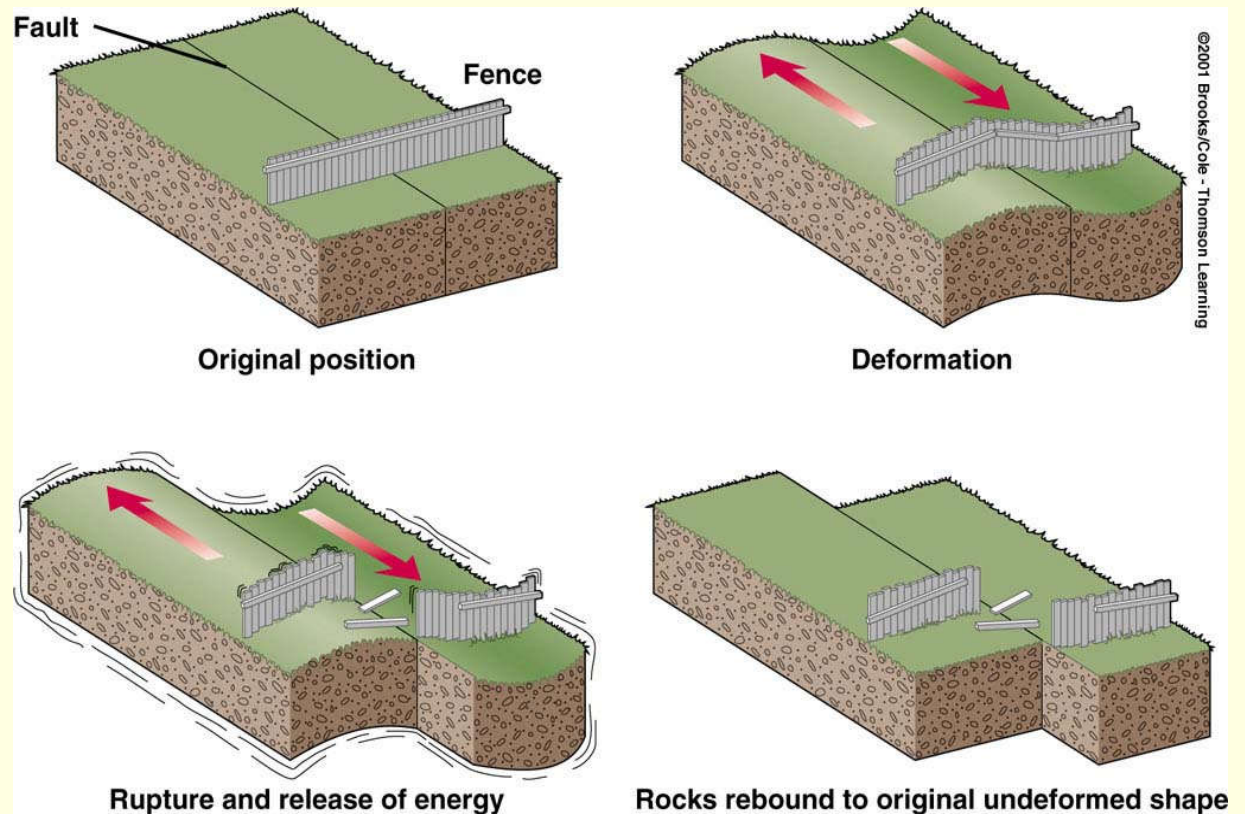
Elastic Rebound Theory



What is **Elastic Rebound Theory**?

Explains how energy is stored in rocks

- Rocks bend until the strength of the rock is exceeded
- Rupture occurs and the rocks quickly rebound to an undeformed shape
- Energy is released in waves that radiate outward from the fault



Earthquakes

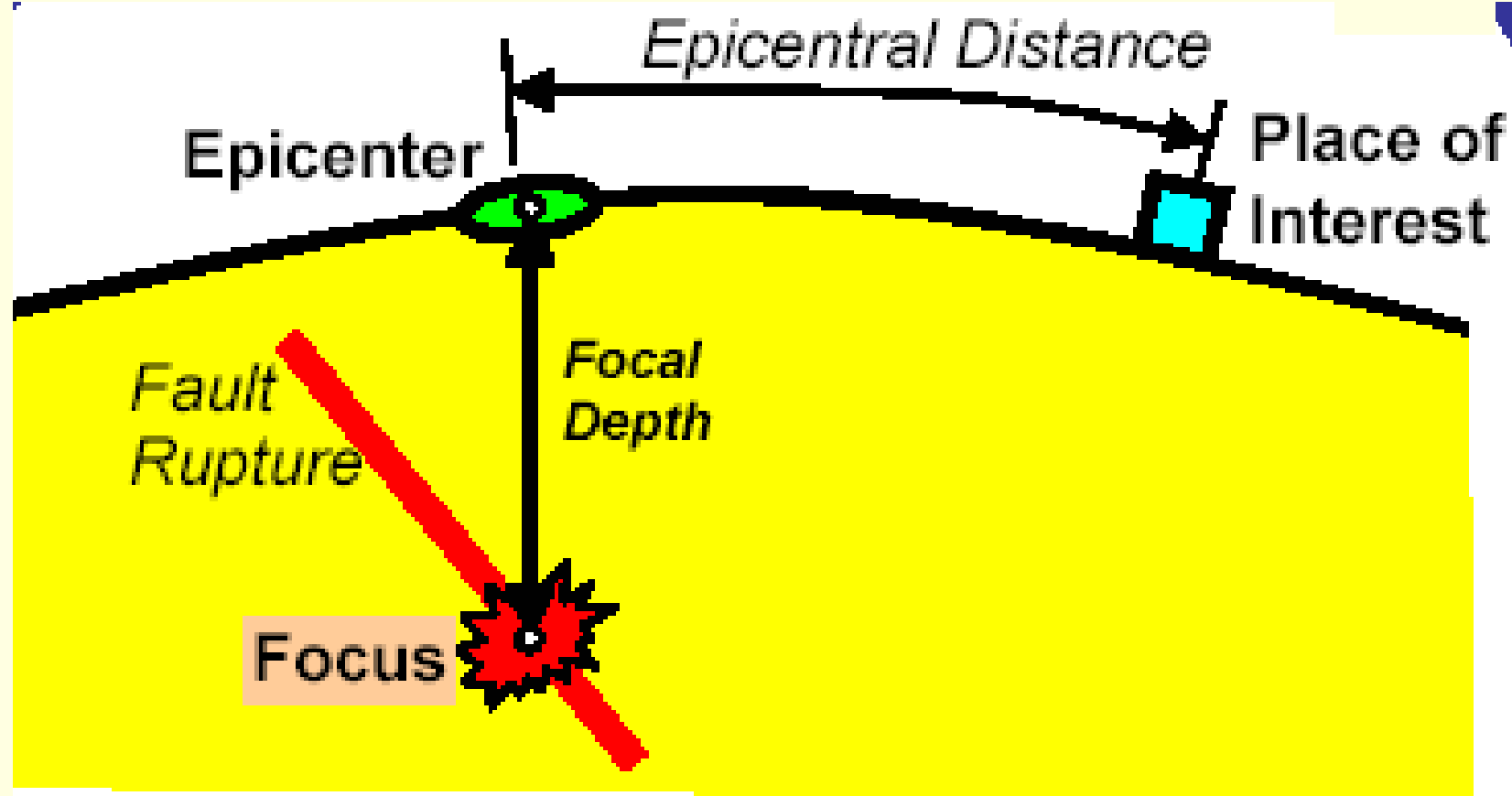
Vibrations of the earth surface caused by waves originating from a source of disturbance in the earth mass is called Earthquake.

Earthquake may be caused by volcanic eruption or by strain building process inside the earth mass.

UNPREDICTABLE

Earthquake Shaking



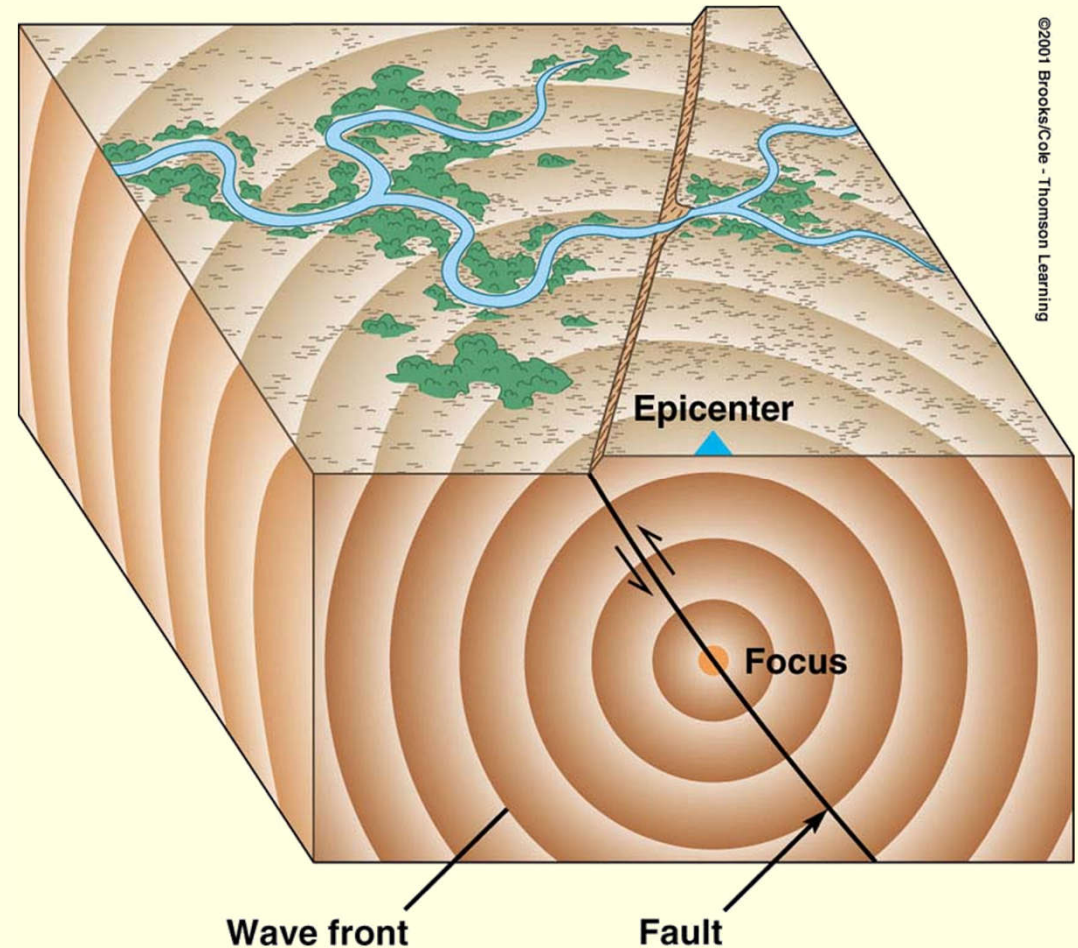


Earthquake may be caused by volcanic eruption or by strain building process inside the earth mass.

UNPREDICTABLE

The **Focus** and **Epicenter** of an Earthquake

- The point within Earth where faulting begins is the focus, or hypocenter
- The point directly above the focus on the surface is the epicenter

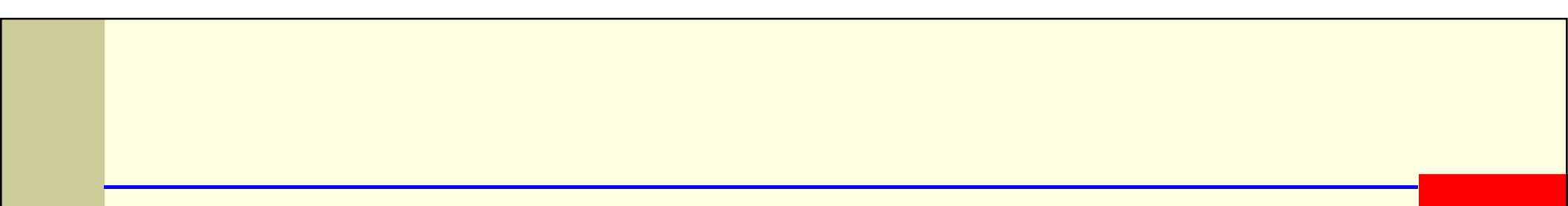


FOCUS or HYPOCENTER: Location from where earthquake originates. It may be a point, line or a plane. It will be deep below the earth surface.

EPICENTER: Projection of focus on the surface of earth. It is a point which is closest to point of release of energy.

FOCAL DEPTH: Distance between focus and epicenter. The closer the focal depth, more damaging is the earthquake.

EPICENTRAL DISTANCE: Distance between point of interest and epicenter.



Faults



Minor faults in Bhuj Earthquake (2001)

Fault movement



Chichi Earthquake, Taiwan, 1999



SHIH KONG DAM



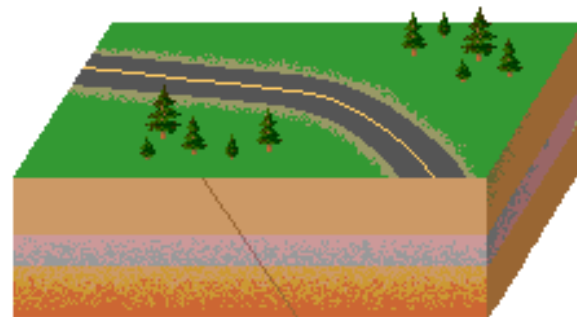
Surface Fault Rupture



Faults

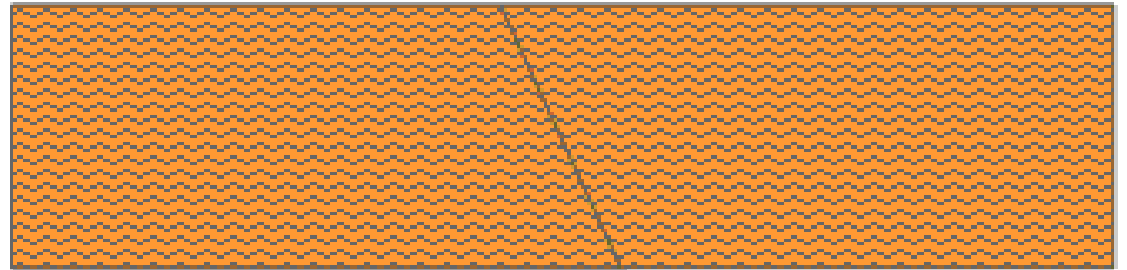


- Normal Fault
- Reverse Fault
- Strike Slip Fault



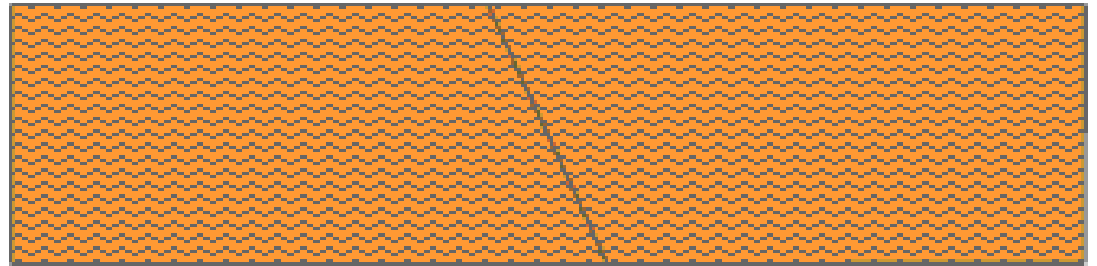


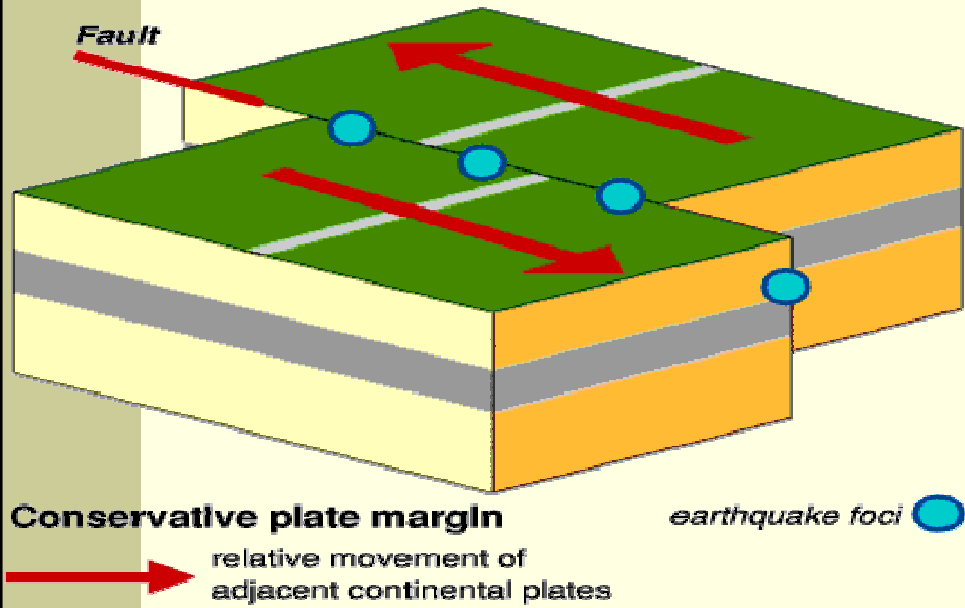
Normal Fault





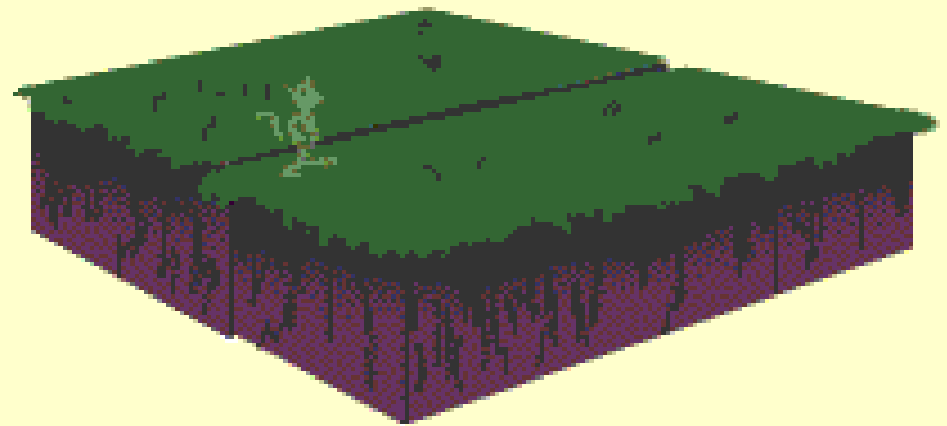
Reverse Fault

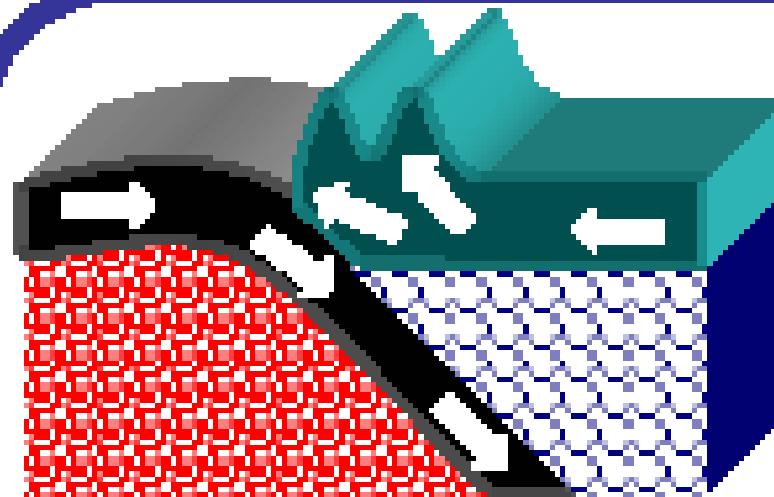




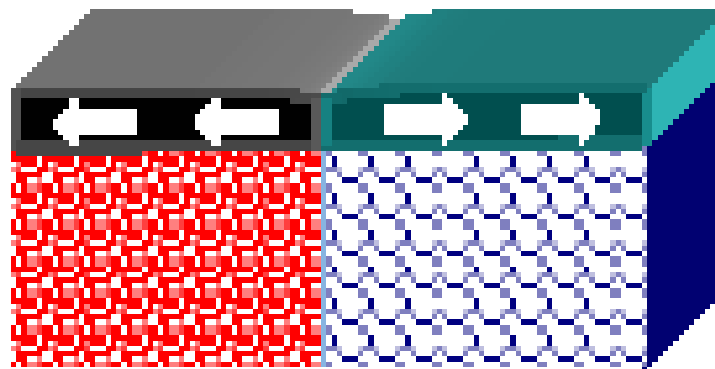
Strike Slip Fault

Two plates slide laterally

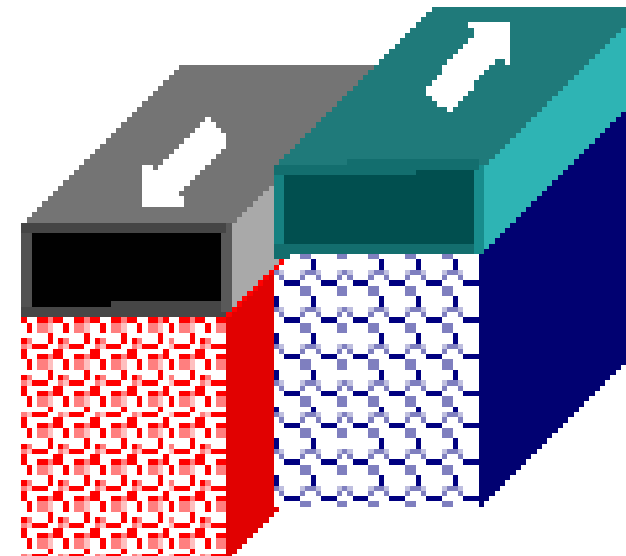




Convergent Boundary

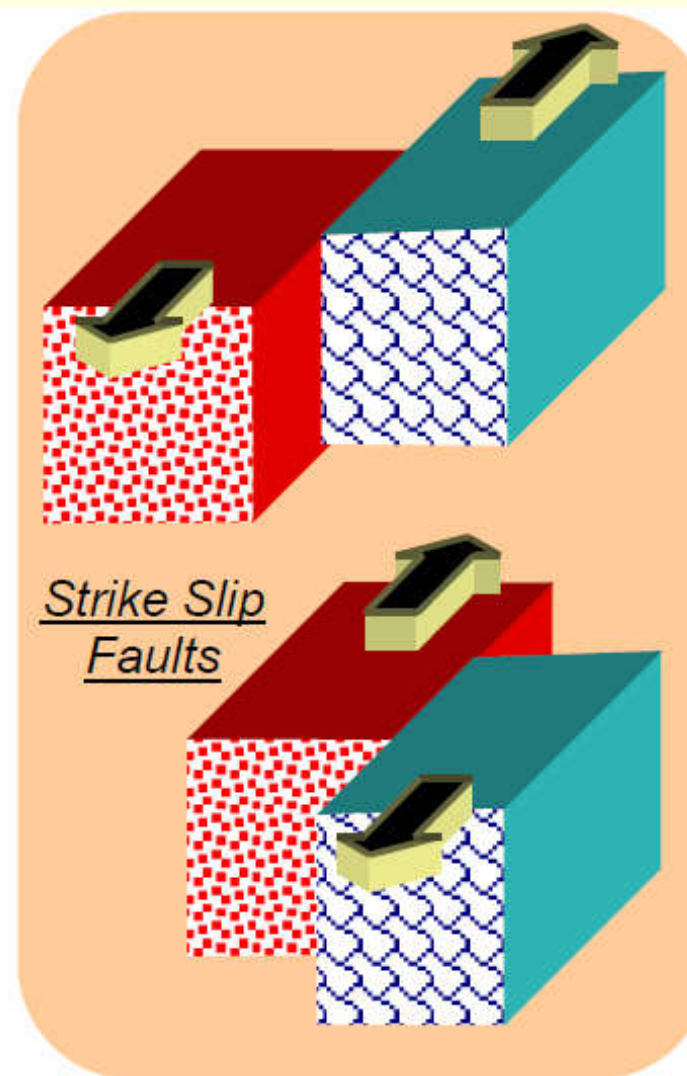
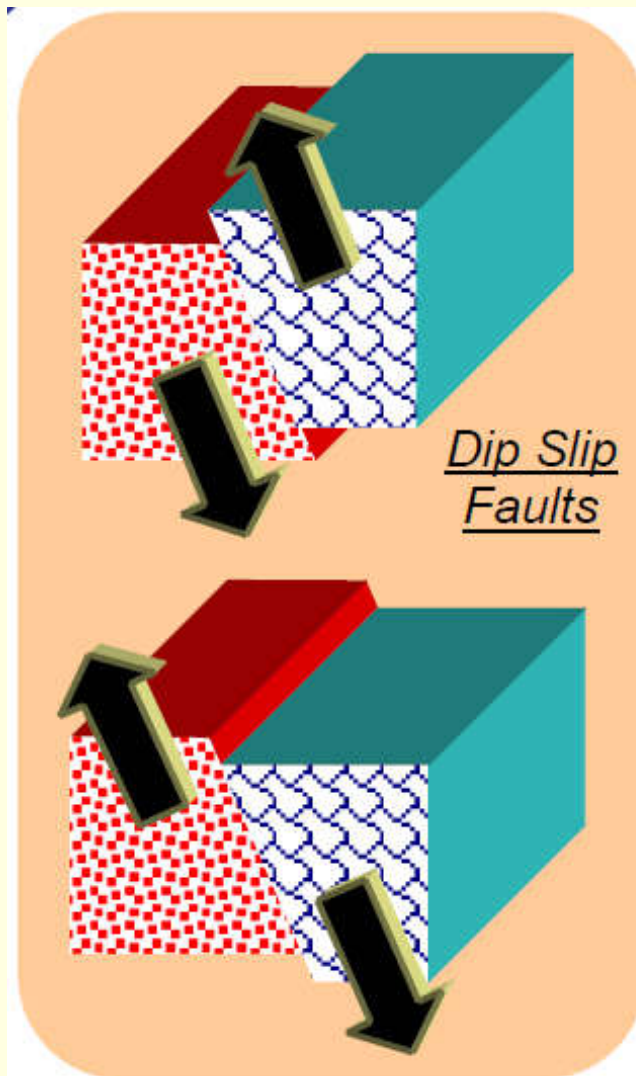


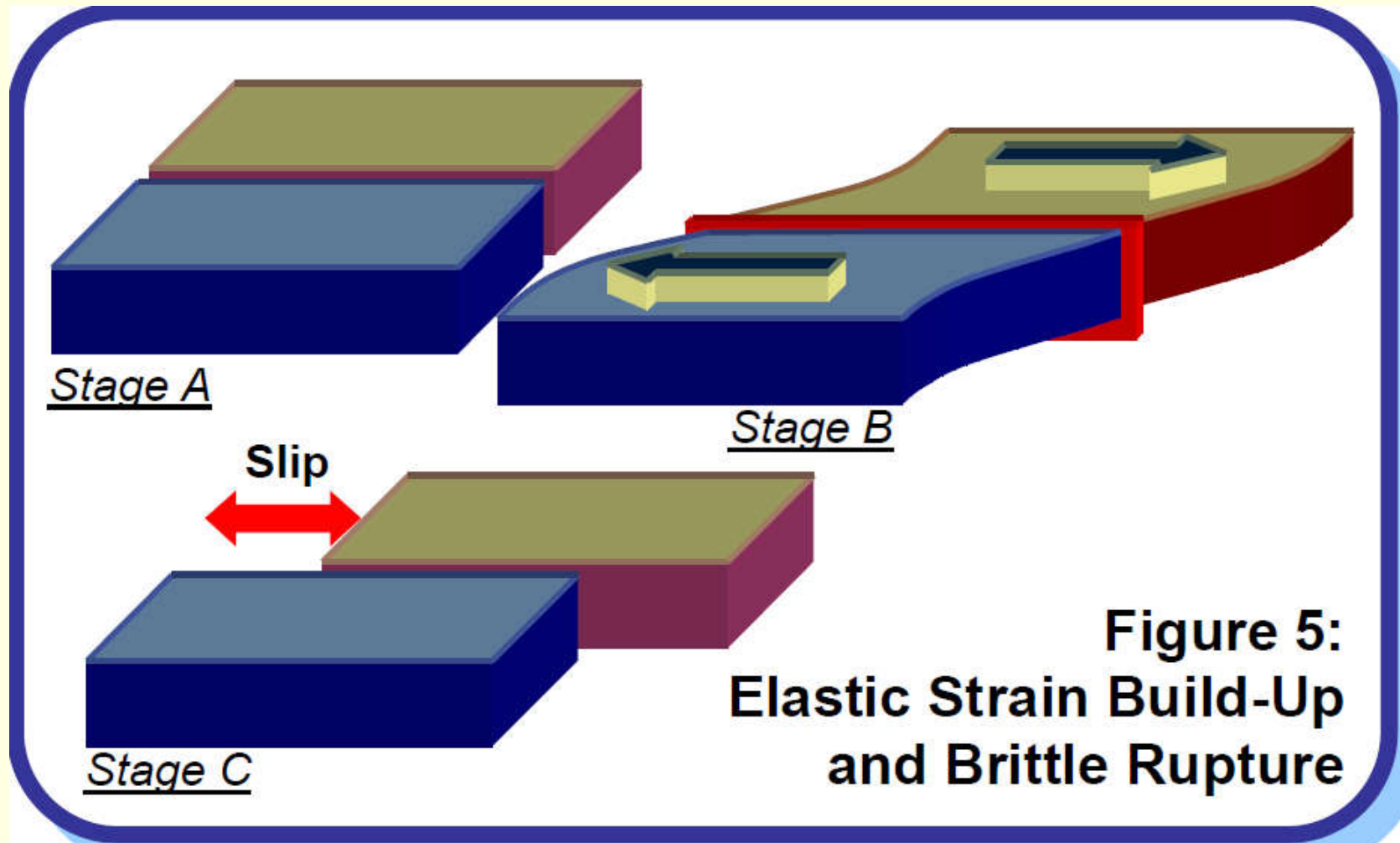
Divergent Boundary

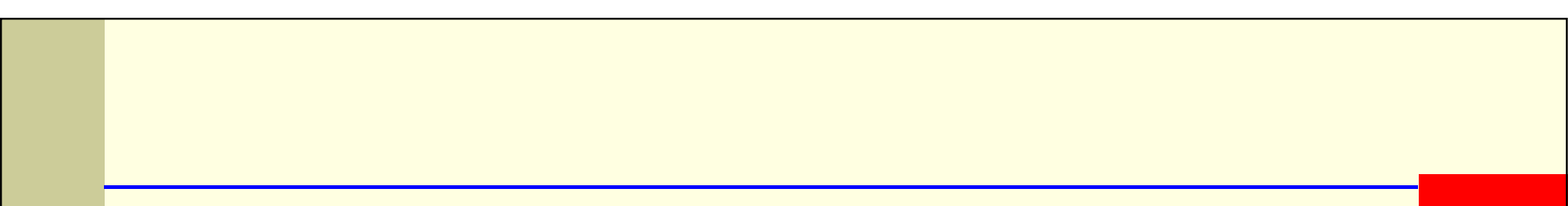


Transform Boundary

**Figure 4: Types of
Inter-Plate Boundaries**

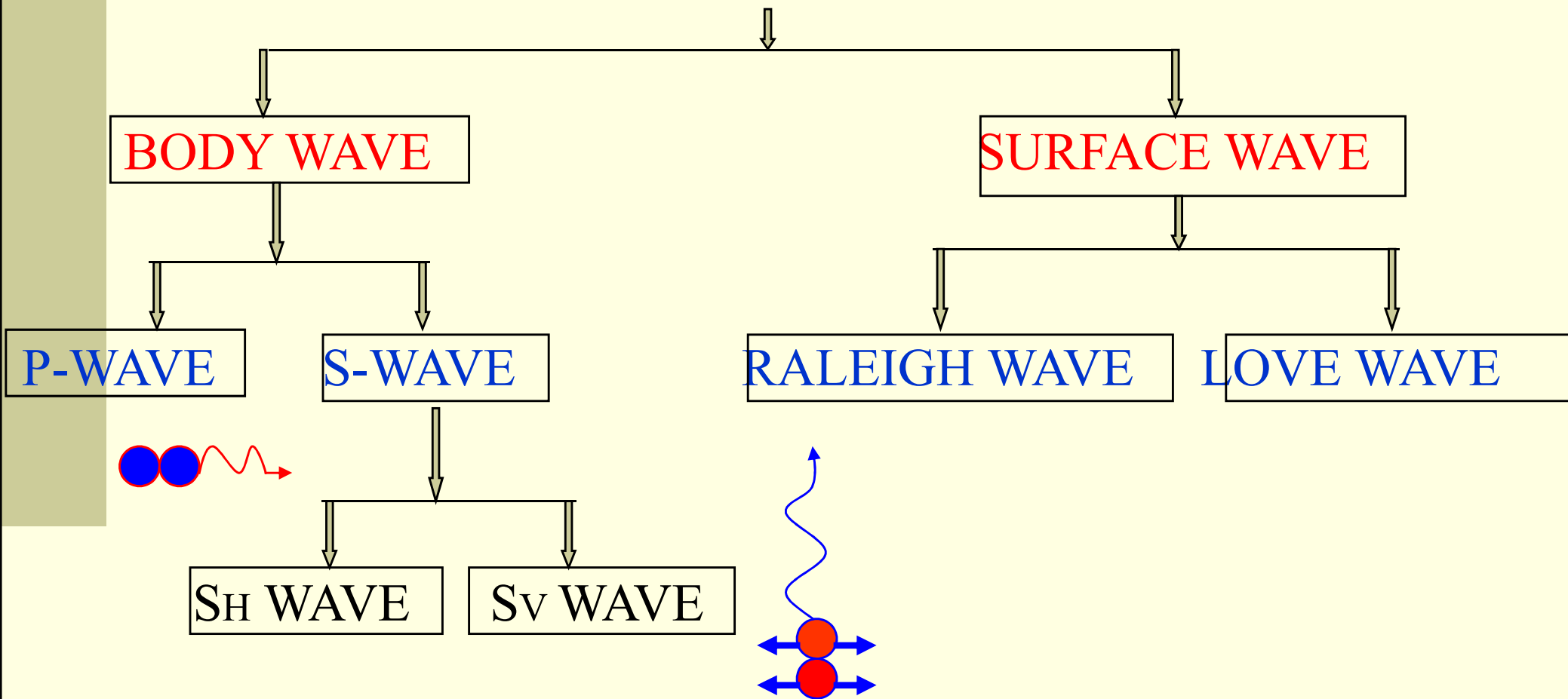






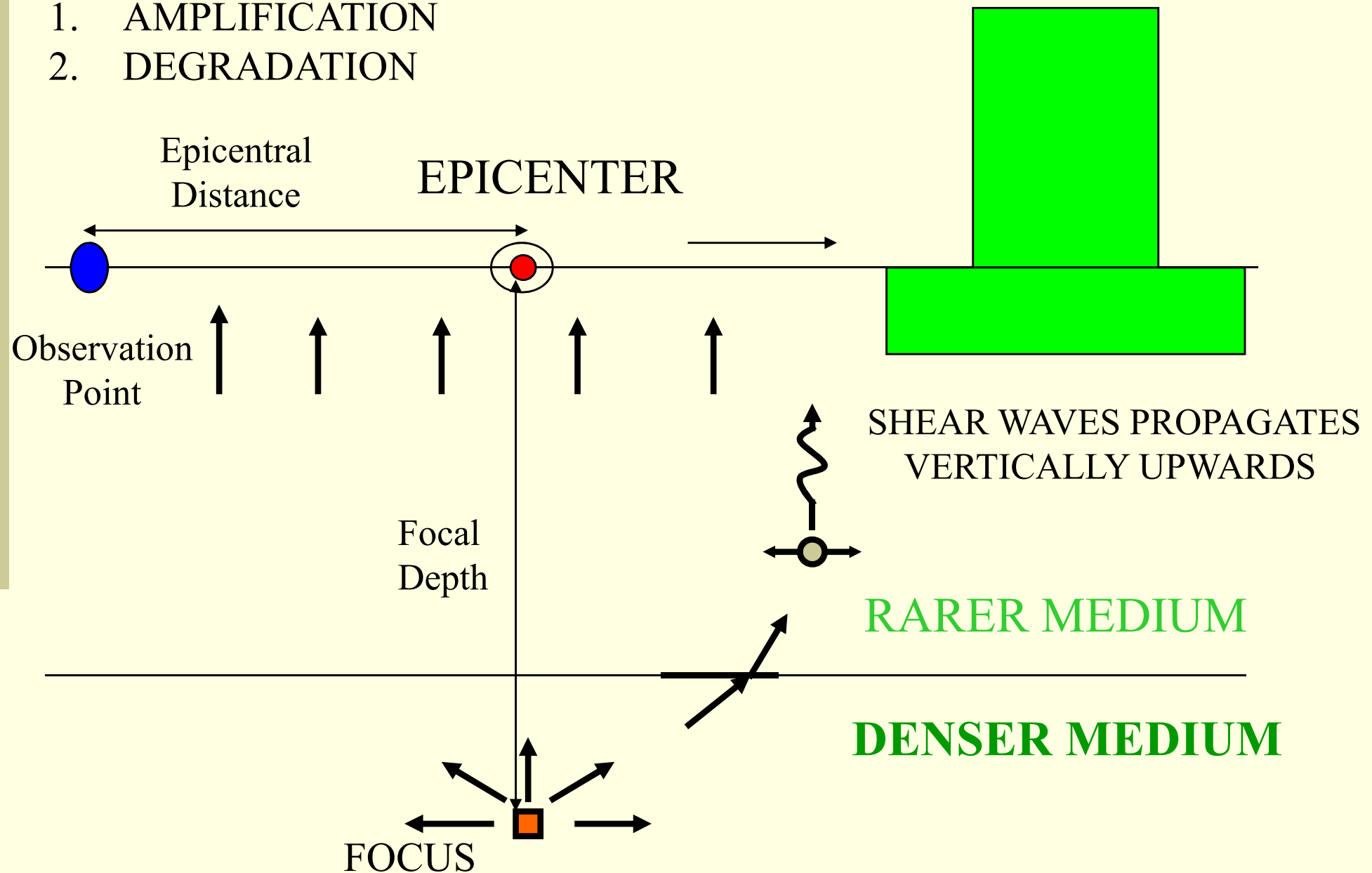
Seismic Waves

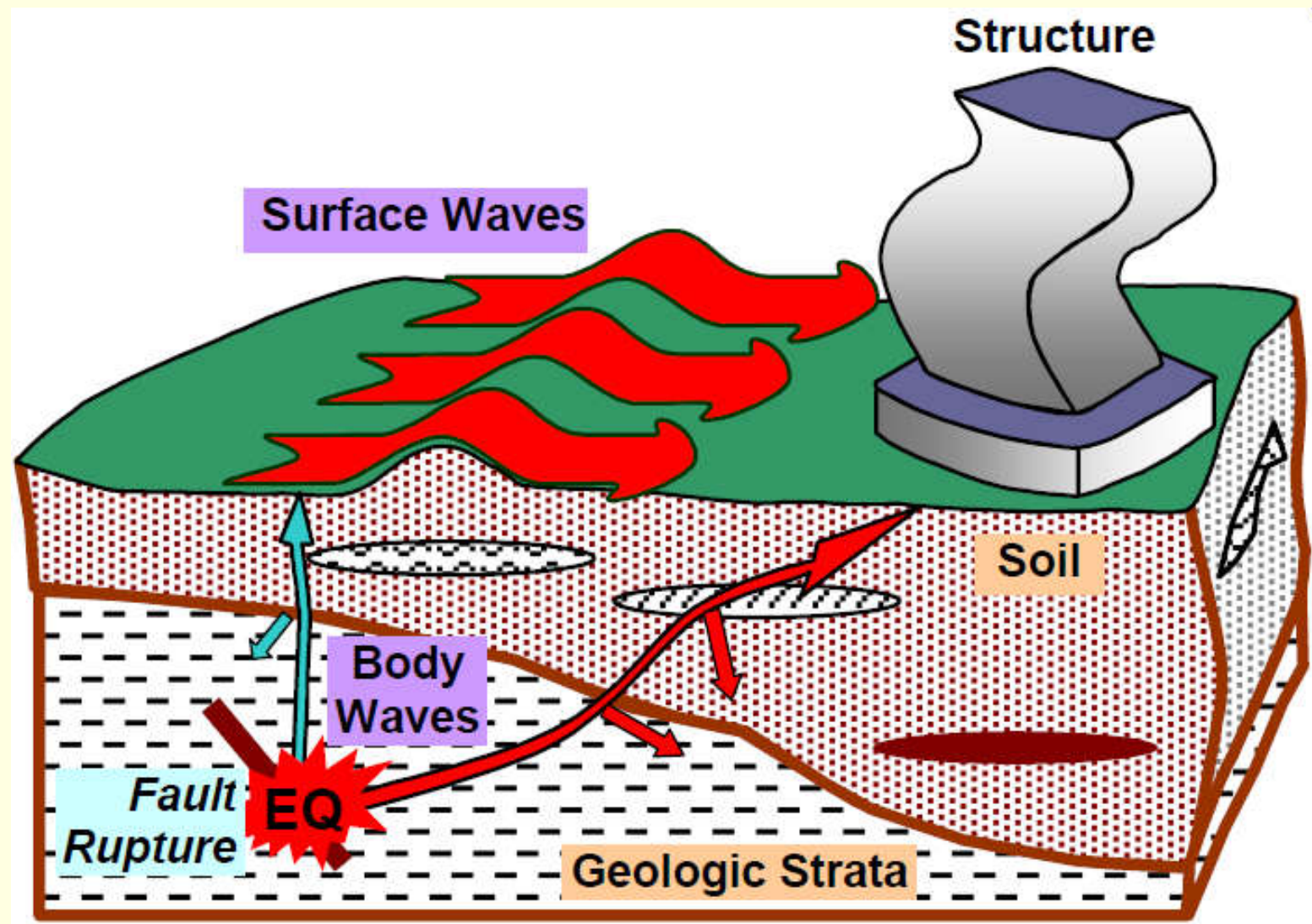
WAVE PROPAGATION IN ELASTIC HALF SPACE



Wave Propagation during Earthquake

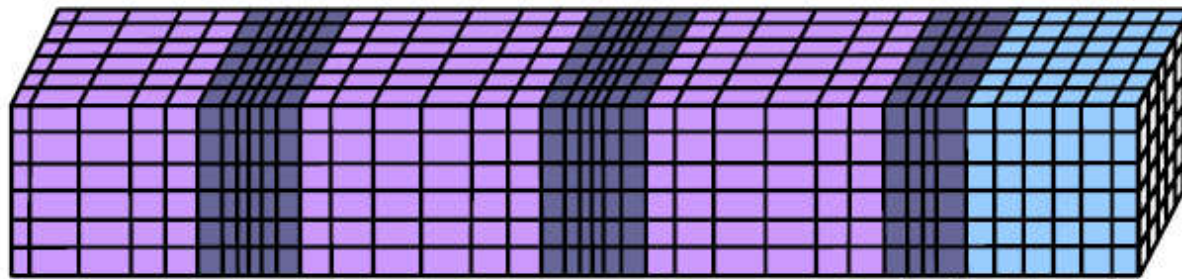
1. AMPLIFICATION
2. DEGRADATION





P-Waves

Push and pull

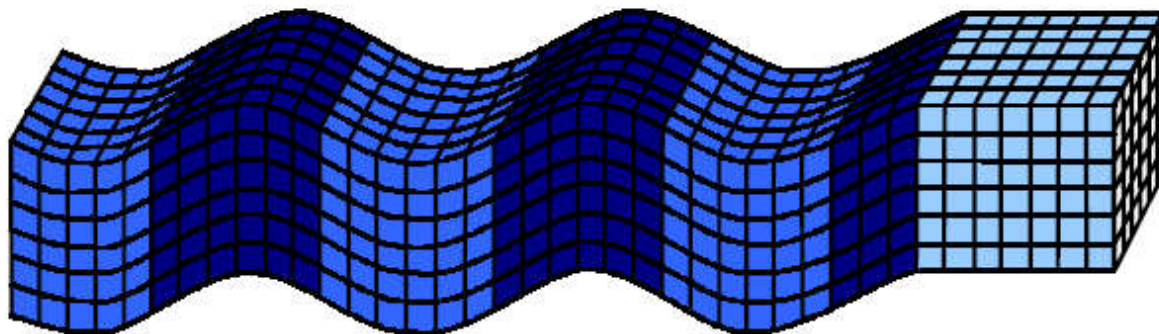


Extension

Compression

S-Waves

Up and down

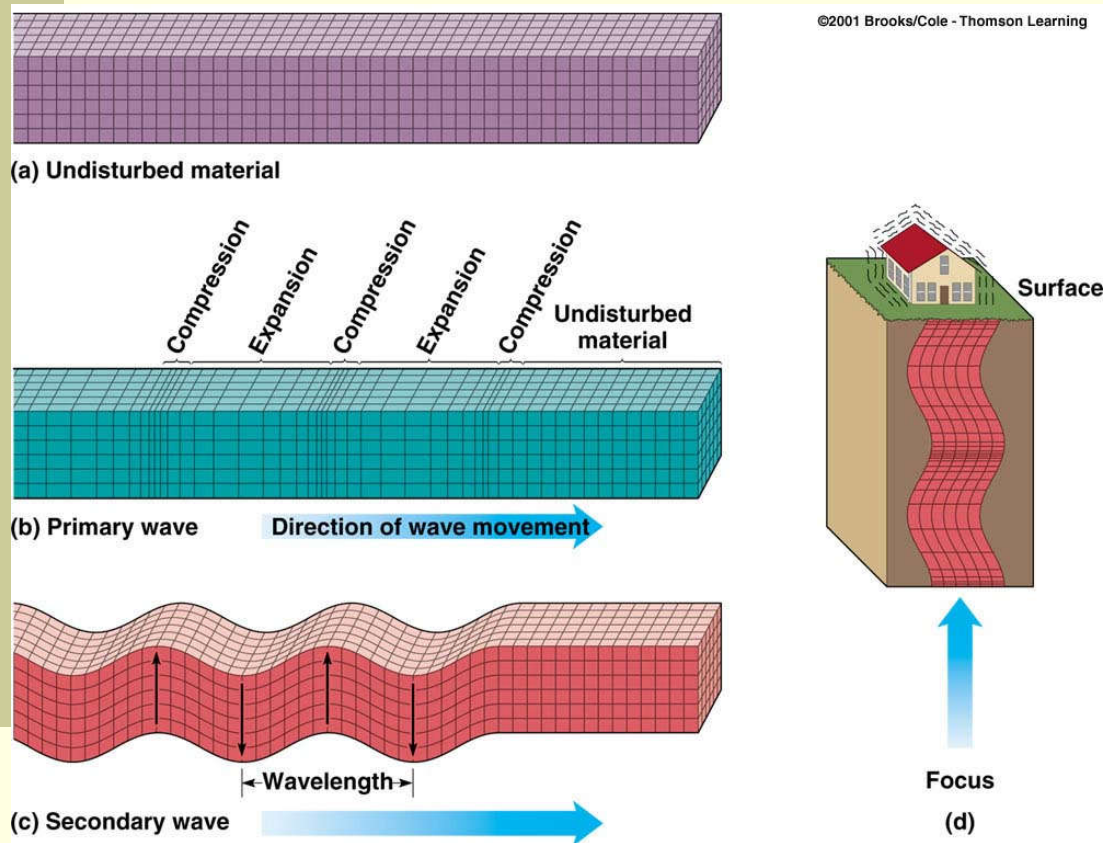


Side to side

Direction of
Energy Transmission

Body Waves

Body Waves: P and S waves



P or primary waves

- fastest waves
- travel through solids, liquids, or gases
- compressional wave, material movement is in the same direction as wave movement

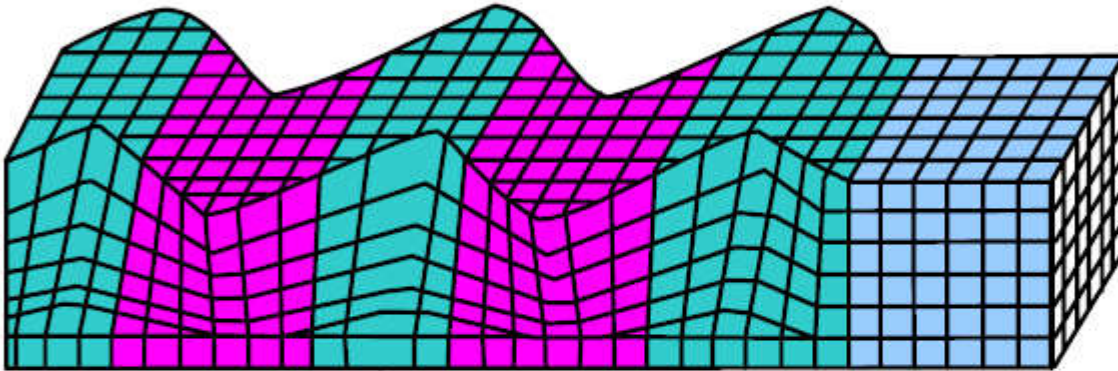
S or secondary waves

- slower than P waves
- travel through solids only
- shear waves - move material perpendicular to wave movement

Direction of
Energy Transmission

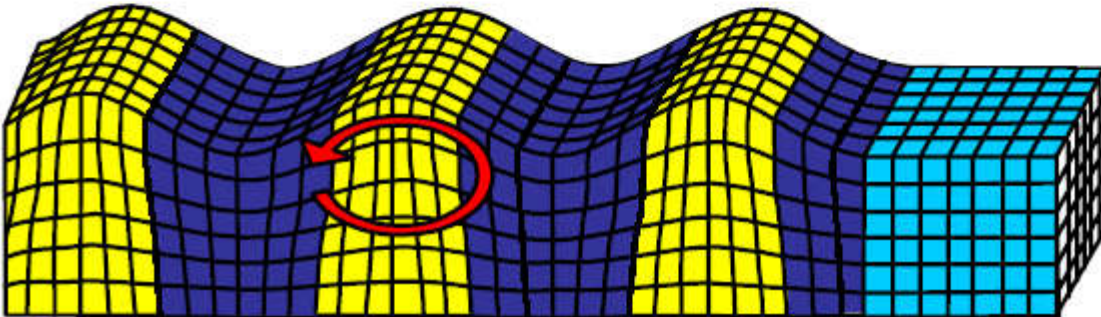
Love Waves

Sideways in horizontal plane



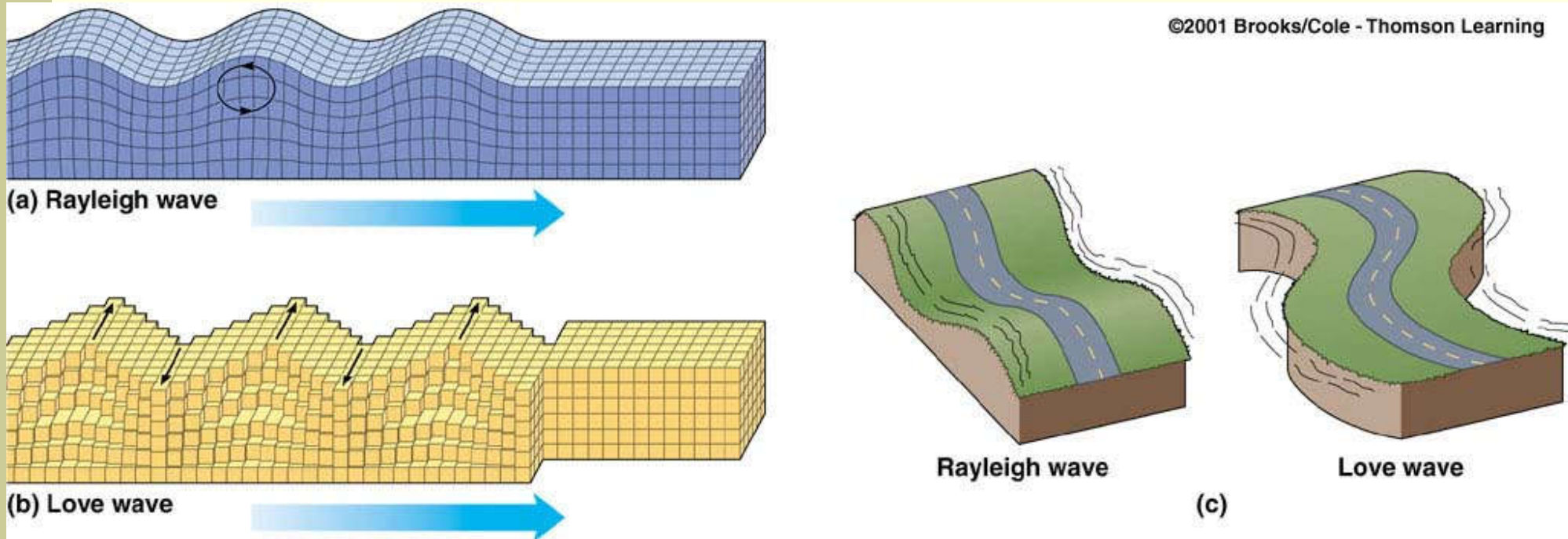
Rayleigh Waves

Elliptic in vertical plane



Surface Waves

Surface Waves: R and L waves

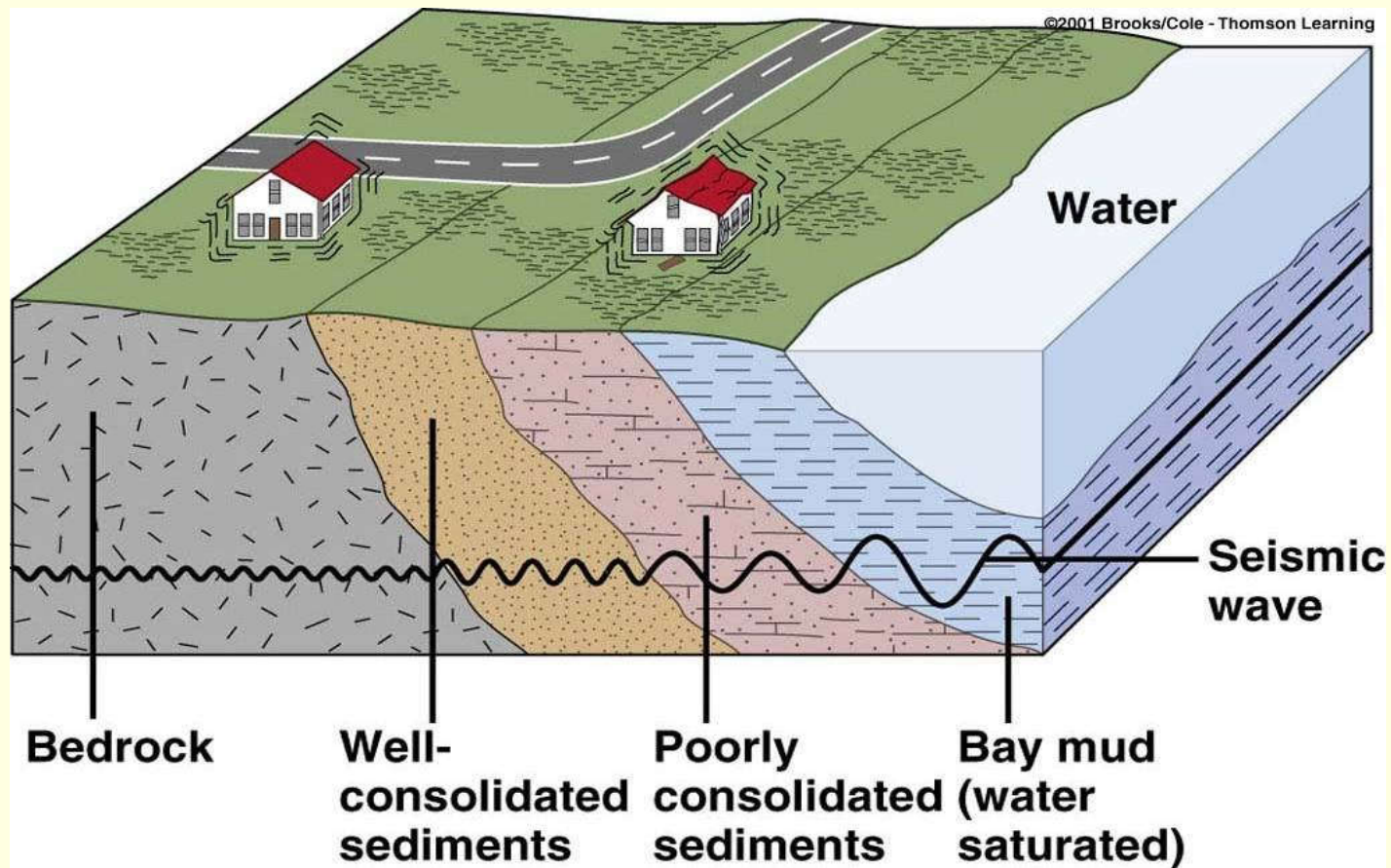


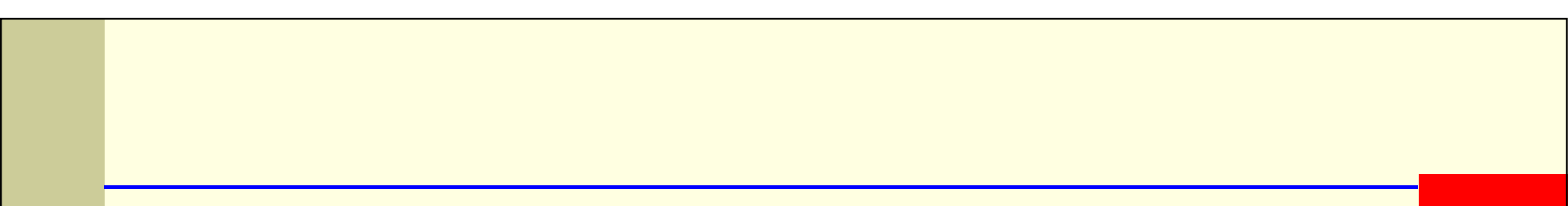
- **Surface Waves**
 - Travel just below or along the ground's surface
 - Slower than body waves; rolling and side-to-side movement
 - Especially damaging to buildings



What are the Destructive Effects of Earthquakes?

Ground Shaking amplitude, duration, and damage increases in poorly consolidated rocks





Seismic Instruments

Seismic Measuring Instruments

Two types of Instruments

1. Seismographs - Instrument
Seismogram - Record

Sensitive, less accurate

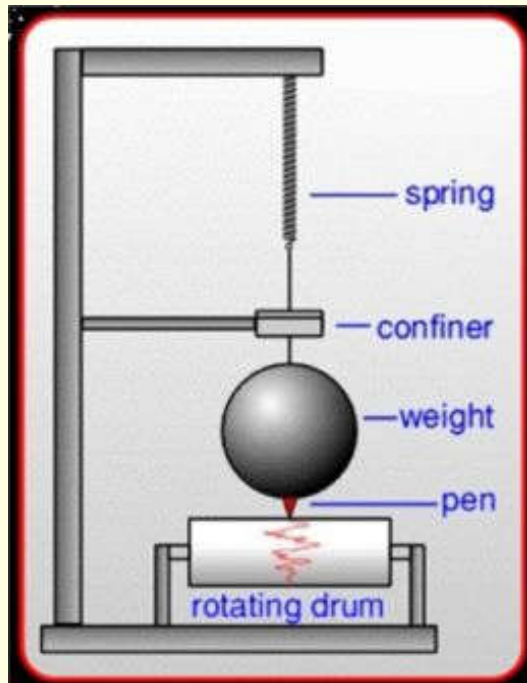
2. Accelerographs - Instrument
Accelerogram – Record

Accurate, less sensitive

Seismograph

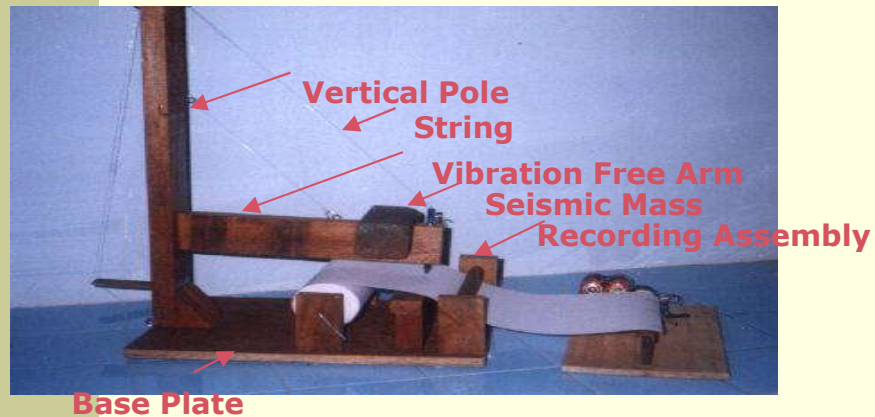


Charles
Richter



Cheng Heng's Earthquake
weathercock

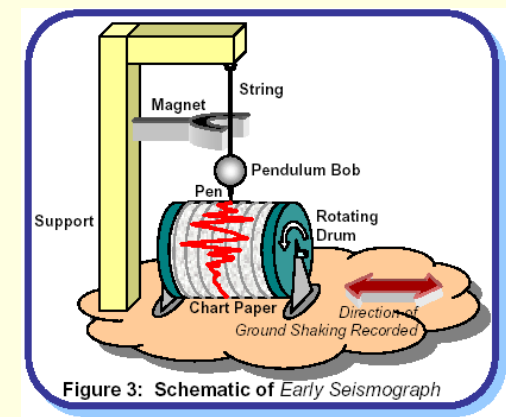
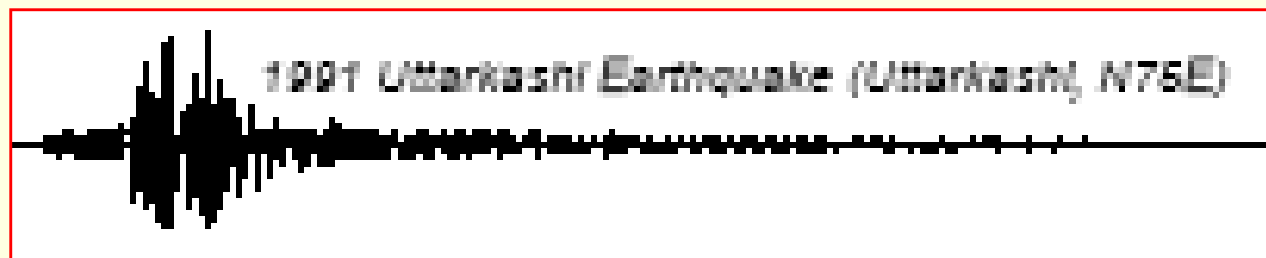
Seismic Measuring Instruments



Seismographs - Instrument
Seismogram - Record

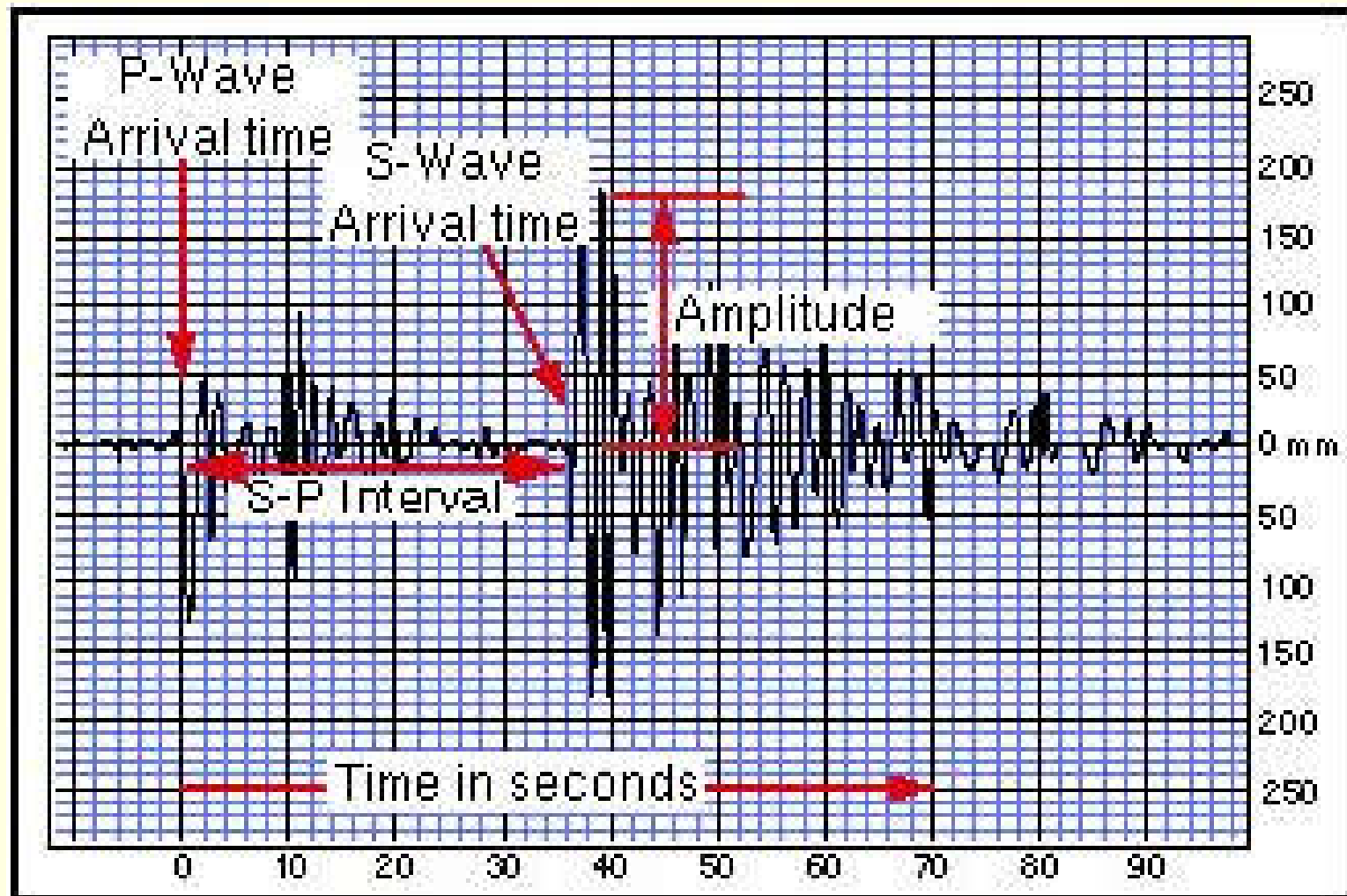


Strong Motion Accelerographs - Instrument
Accelerogram - Record

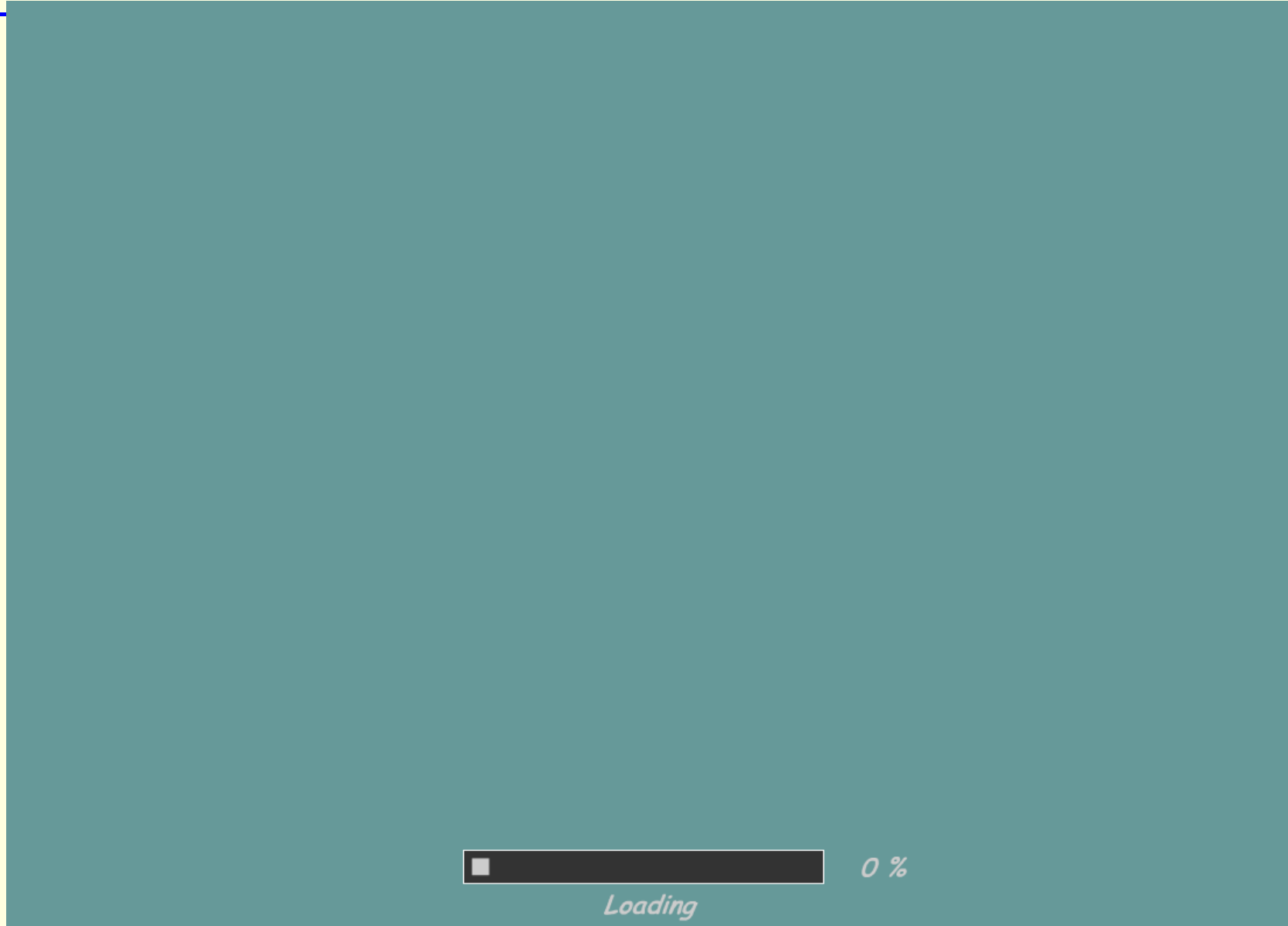


Seismogram

Unpredictable ? ? ? ? ?

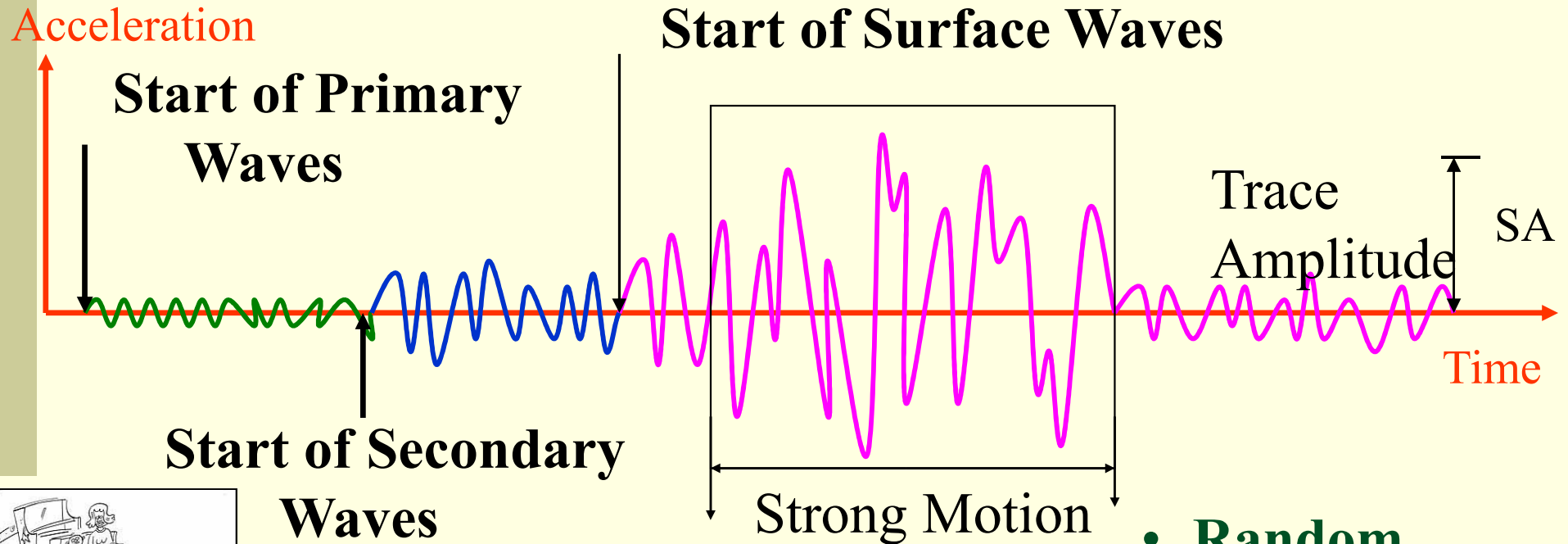


Typical Seismograph

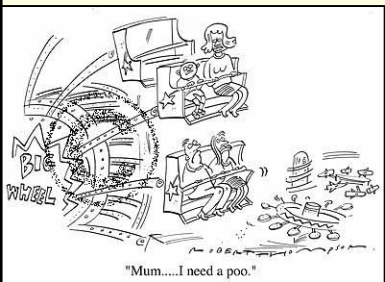


Typical Seismogram

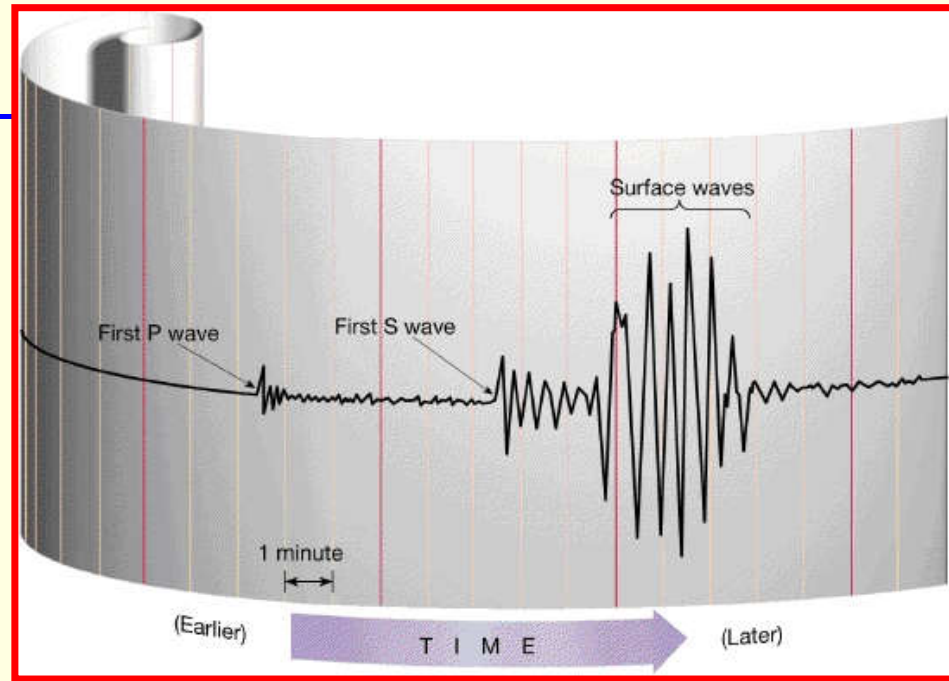
- PGA
- Predominant Frequency
- Duration of Strong Motion



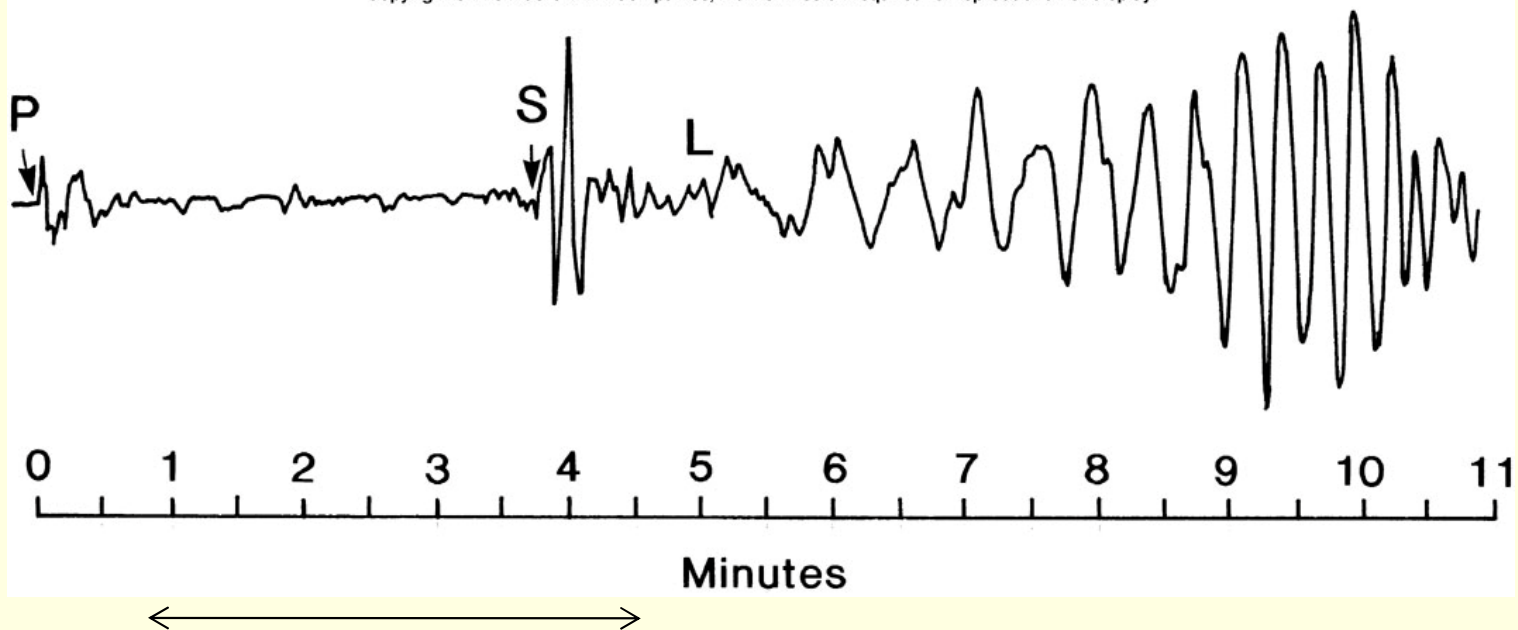
- Random
- Time Dependent
- Cyclic



Seismogram Printout

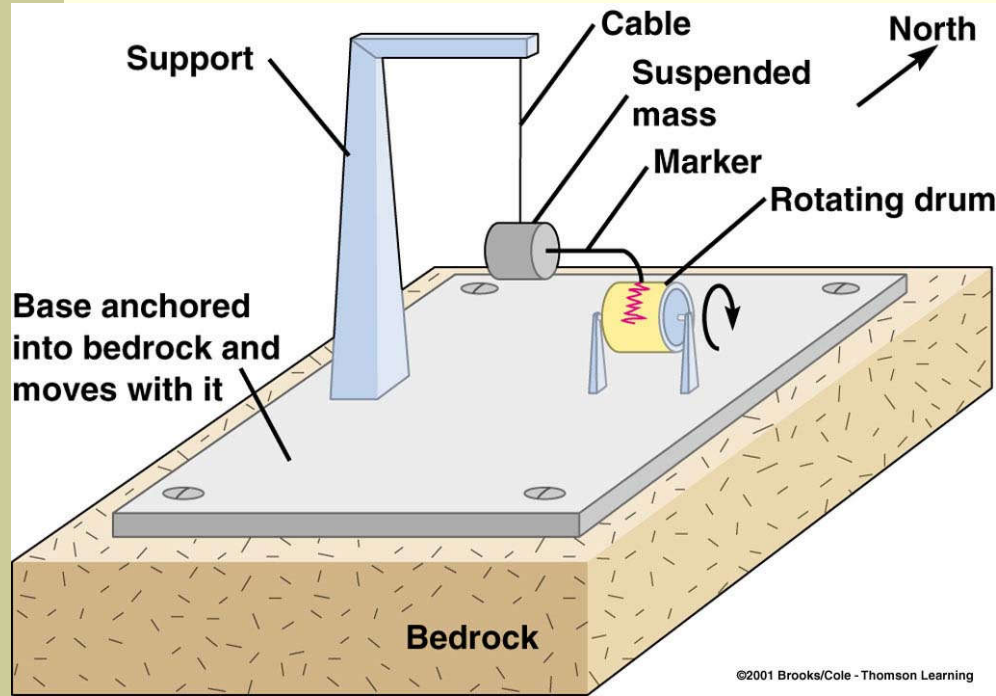


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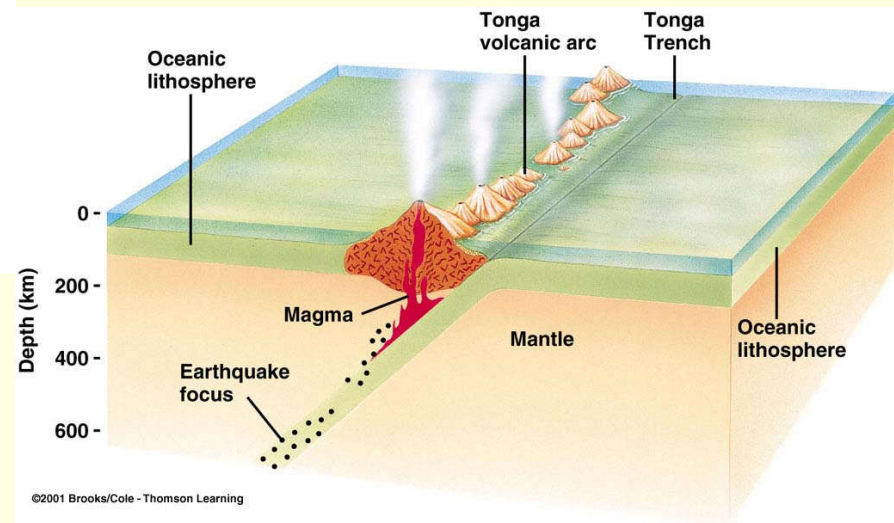


Typical result from a Seismograph





Seismographs record earthquake events



At convergent boundaries,
focal depth increases along
a dipping seismic zone
called a **Benioff zone**



Accelerogram Records

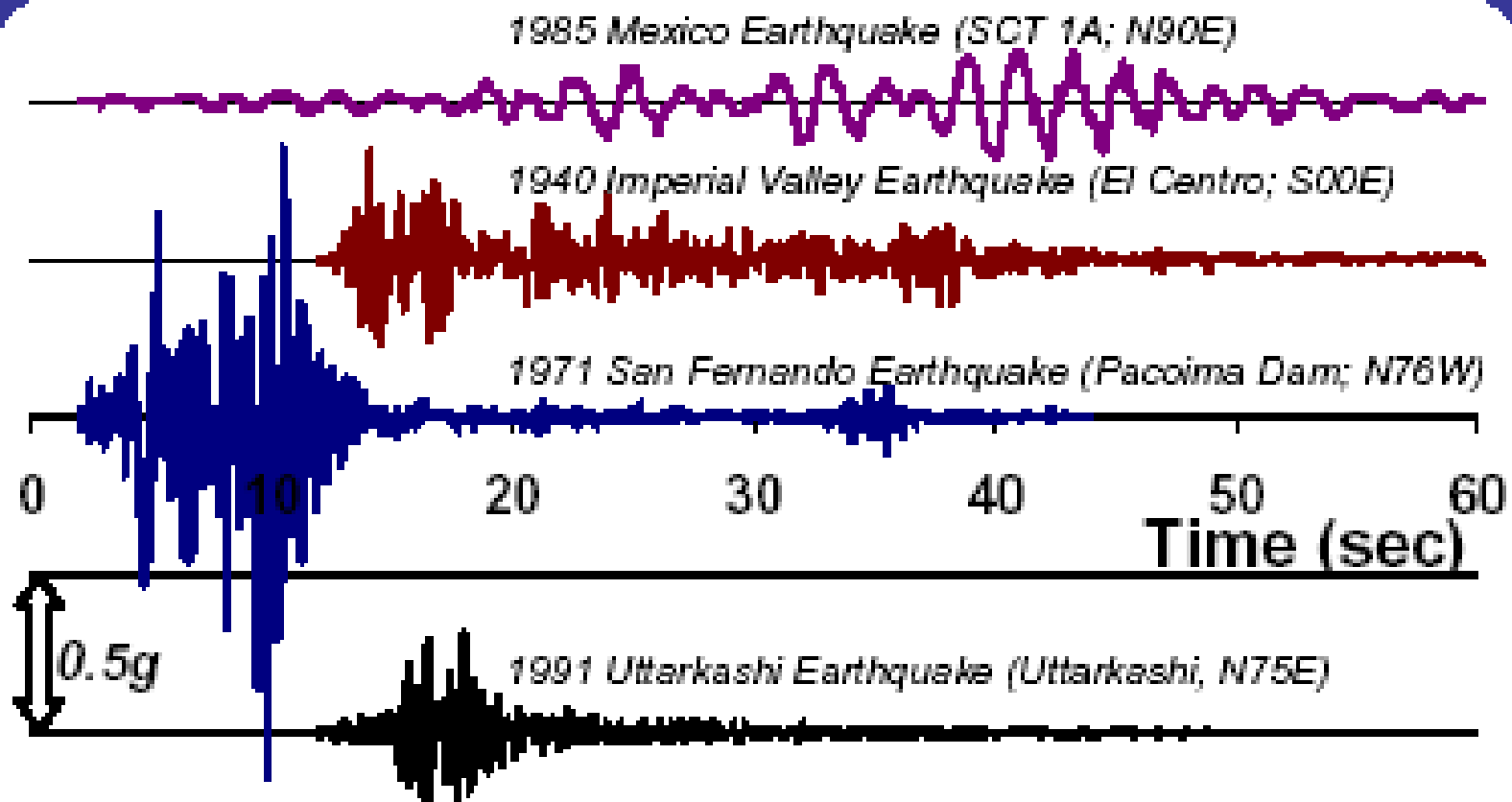


Figure 4::
Some typical recorded accelerograms

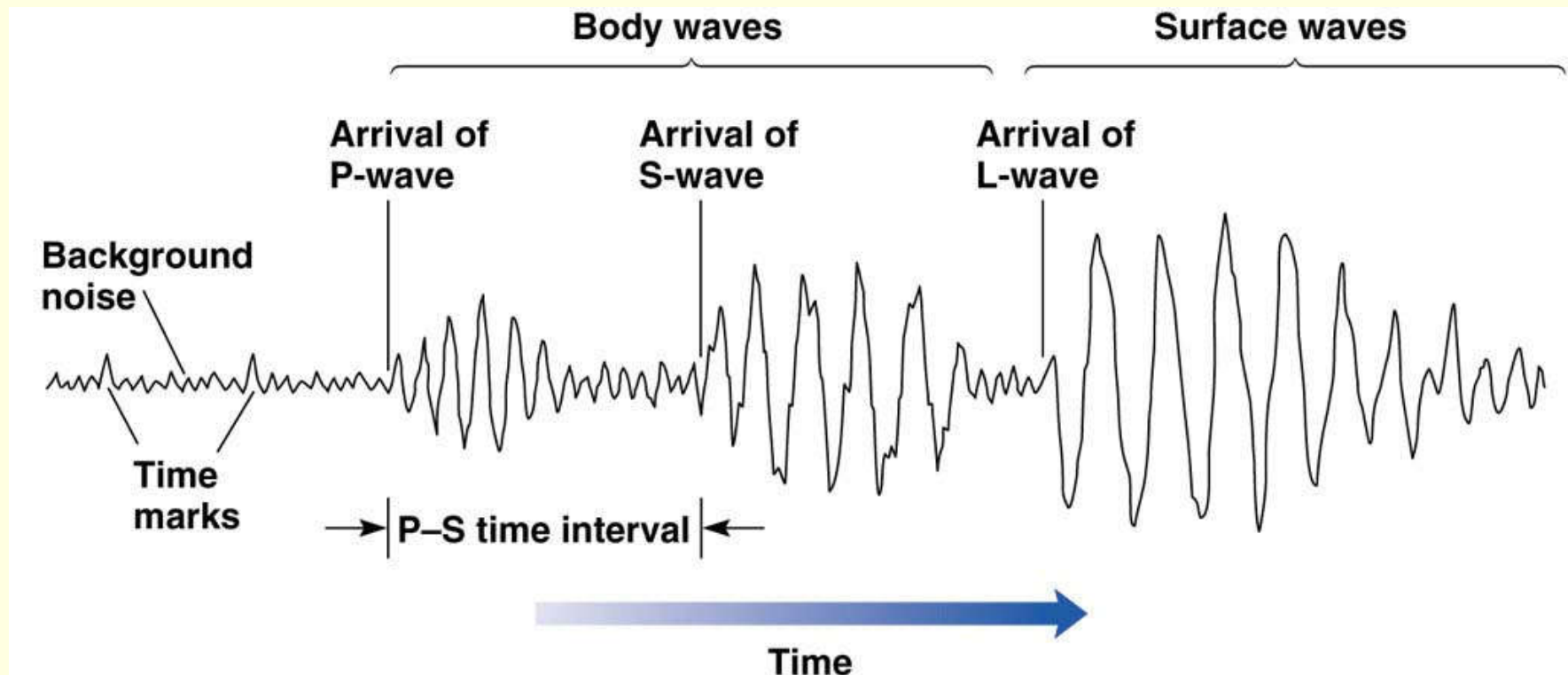
How is an Earthquake's Epicenter Located?

Seismic wave behavior

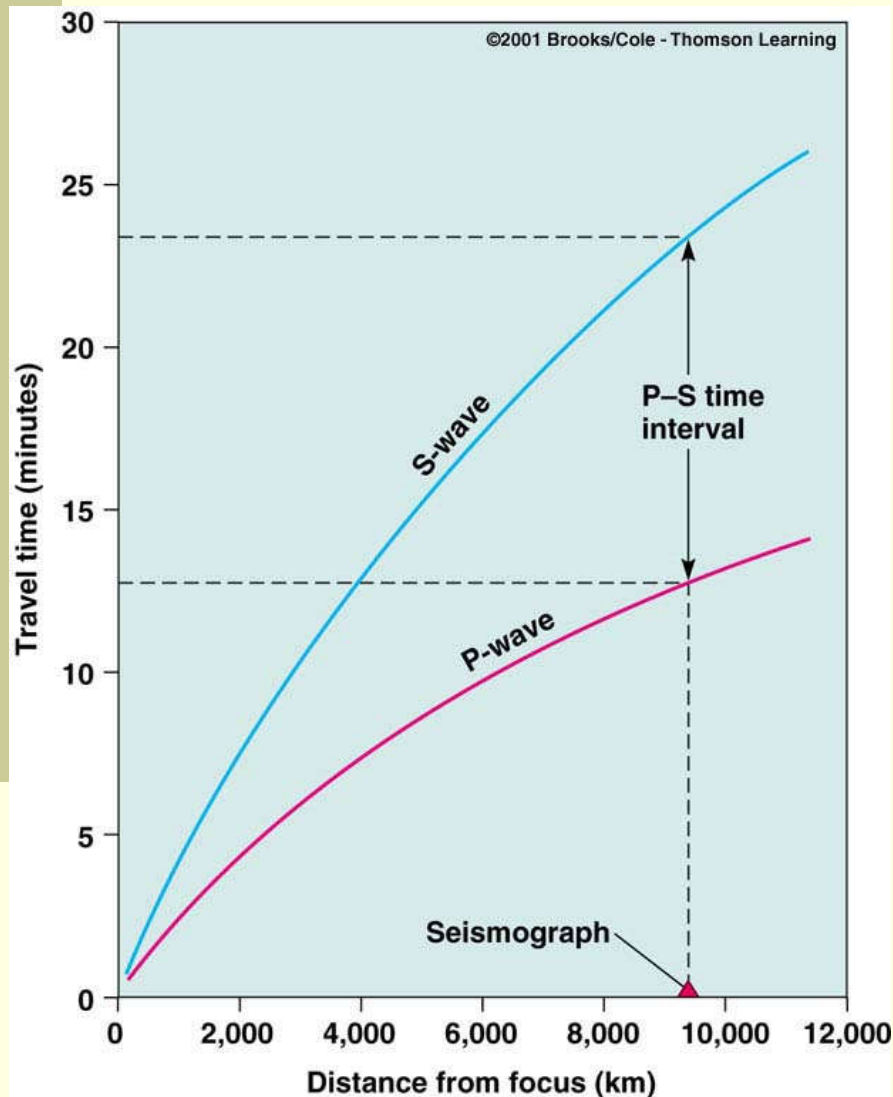
P waves arrive first, then S waves, then L and R

Average speeds for all these waves is known

After an earthquake, the difference in arrival times at a seismograph station can be used to calculate the distance to the epicenter.

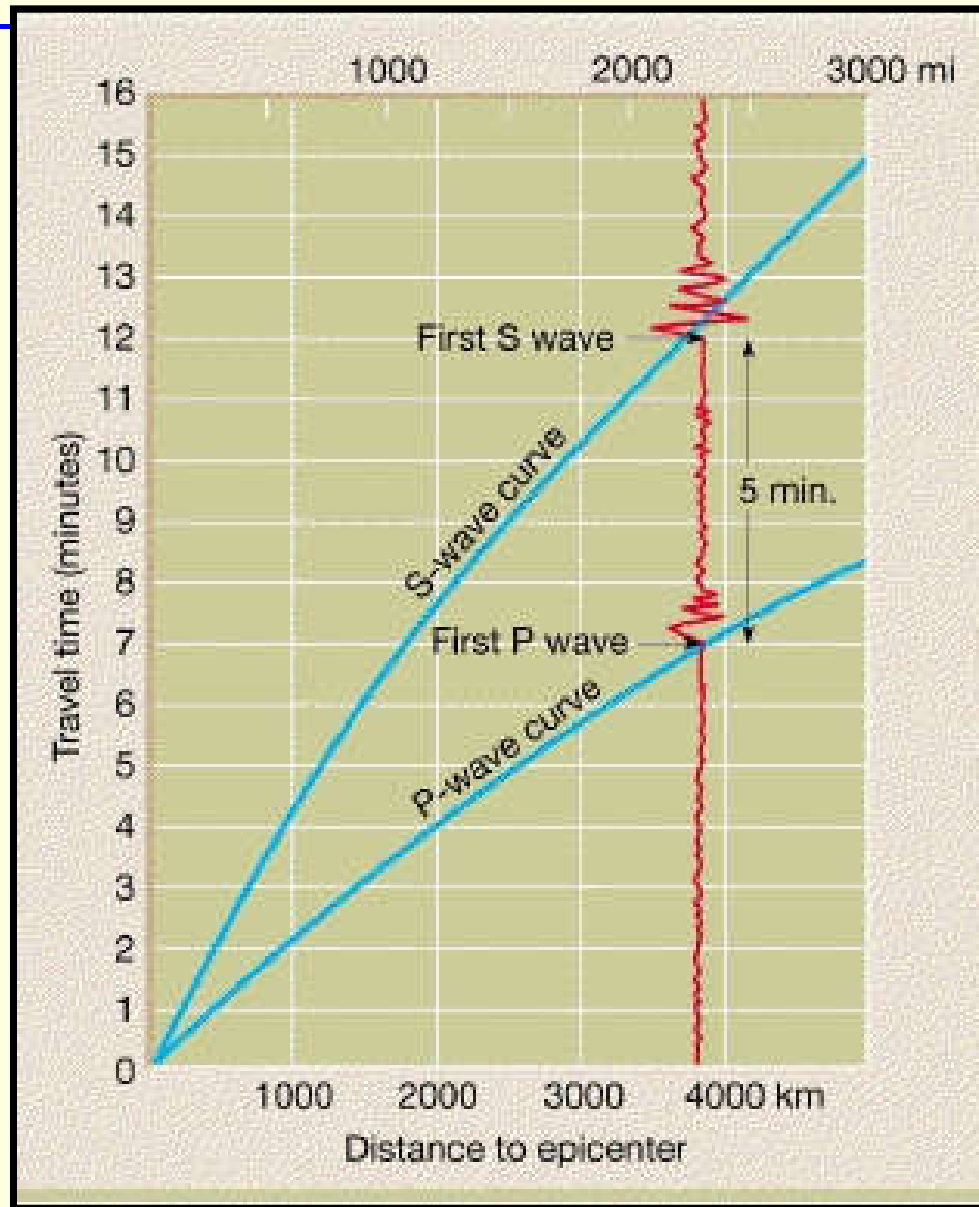


How is an Earthquake's Epicenter Located?



Time-distance graph showing the average travel times for P- and S-waves.

The farther away a seismograph is from the focus of an earthquake, the longer the interval between the arrivals of the P- and S- waves



Time-Travel Curve

Δ_{S-P} 11 mnts - 8600 km

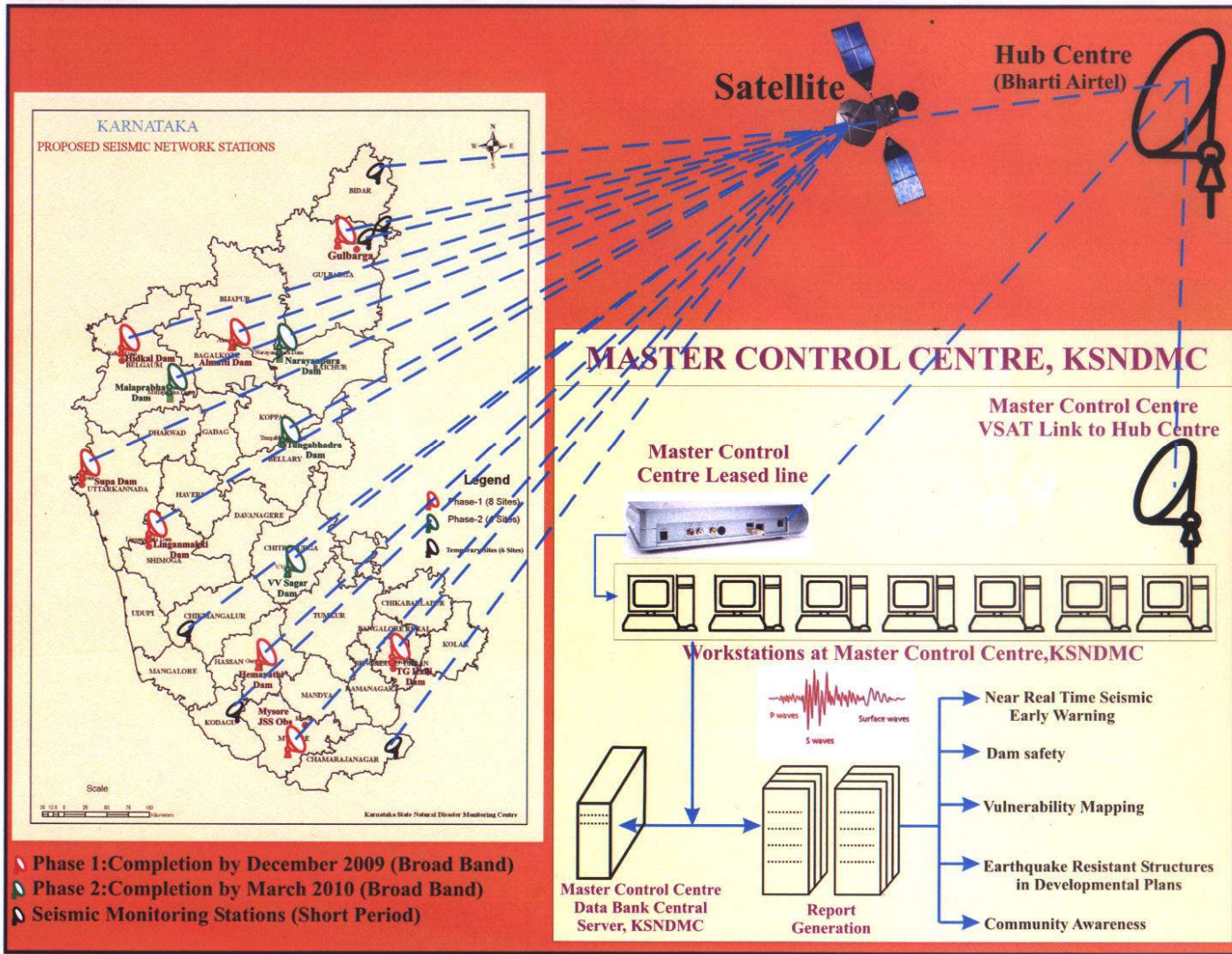
Δ_{S-P} 8 mnts - 5600 km

Δ_{S-P} 3 mnts - 1500 km

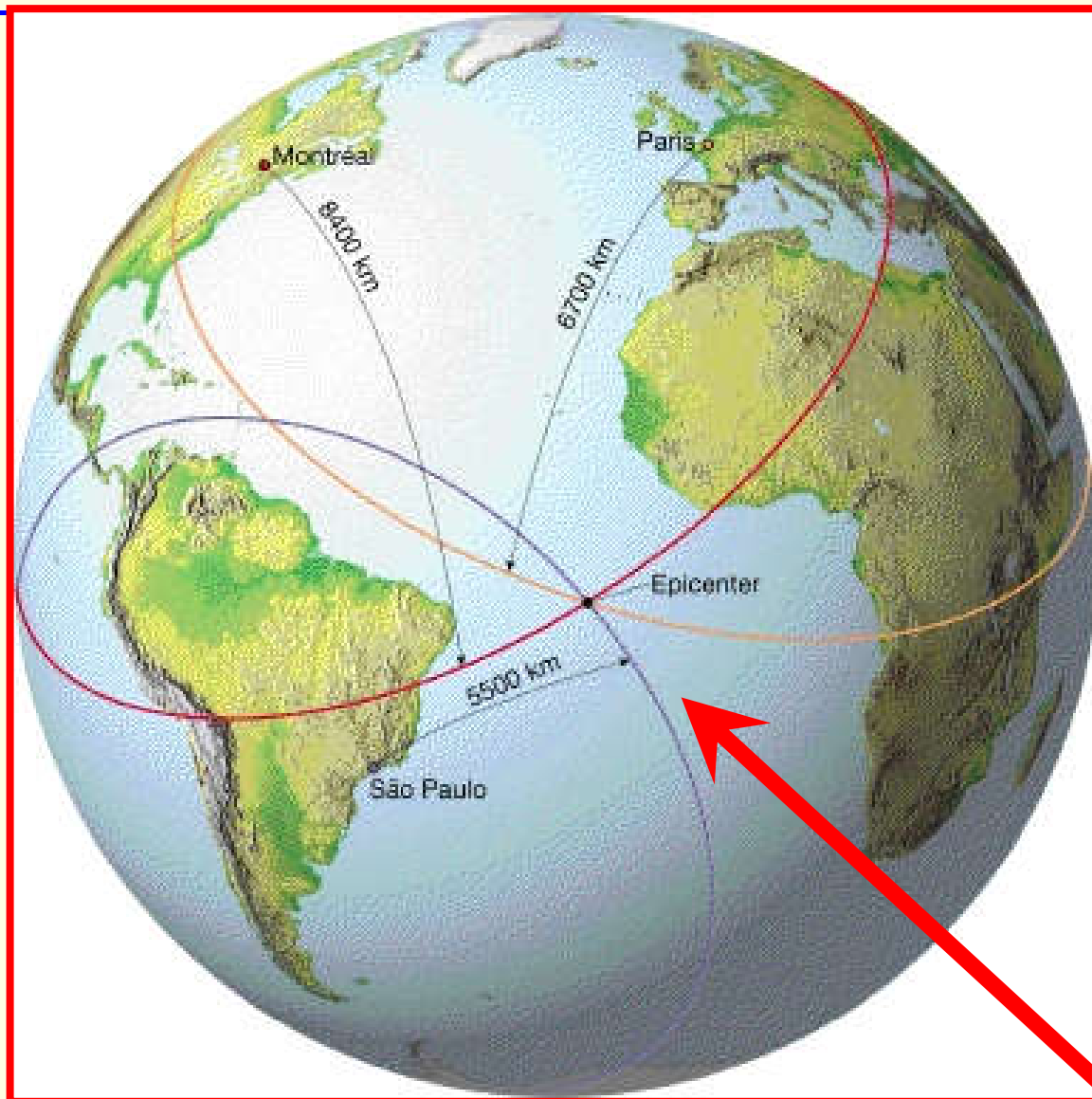
Annexure-I

A Typical Schematic Diagram of the Proposed VSAT Enabled Near Real Time Seismic Monitoring Network for KARNATAKA

KSNDMC

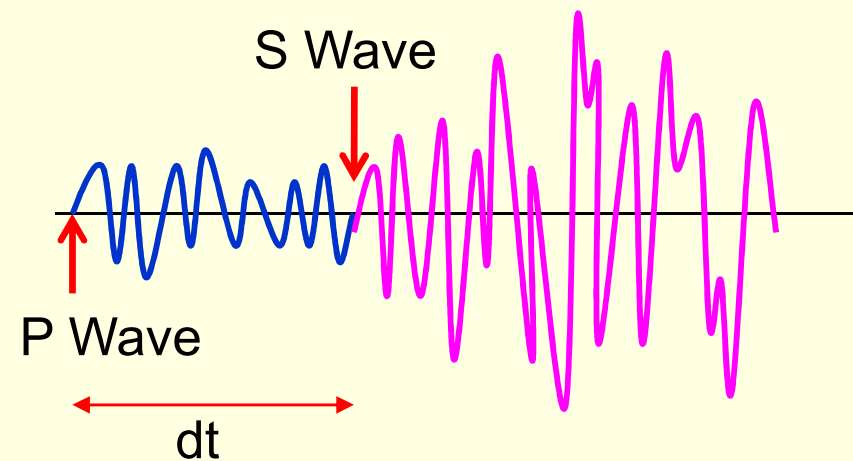


Karnataka State Natural Disaster Management Center



**Triangulation
of 3 stations
to locate
earthquake
epicenter**

Epicenter Location



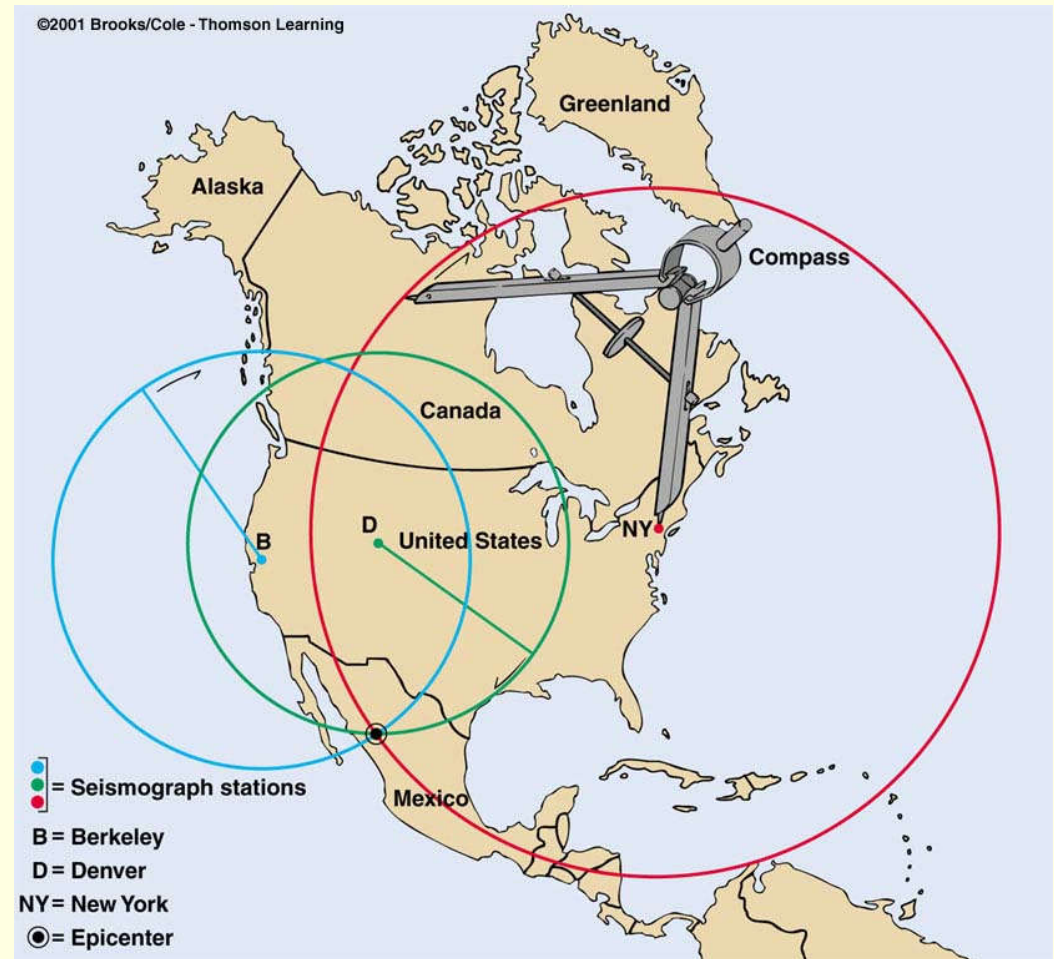
Epicenter

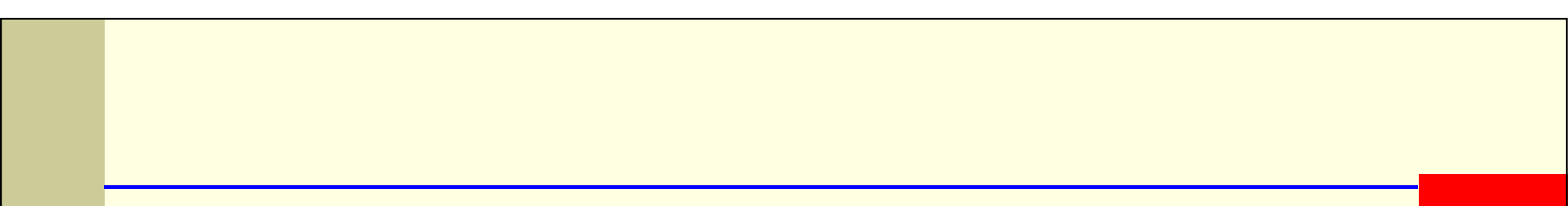
$$D = \frac{dt}{\left(\frac{1}{V_s} - \frac{1}{V_p} \right)}$$

$$V_p = 4.8 \text{ km/s}, V_s = 3 \text{ km/s}$$

How is an Earthquake's Epicenter Located?

- Three seismograph stations are needed to locate the epicenter of an earthquake
- A circle where the radius equals the distance to the epicenter is drawn
- The intersection of the circles locates the epicenter





Magnitude & Intensity

Earthquake Magnitude

How can we compare the sizes of earthquakes?

What is Richter magnitude?

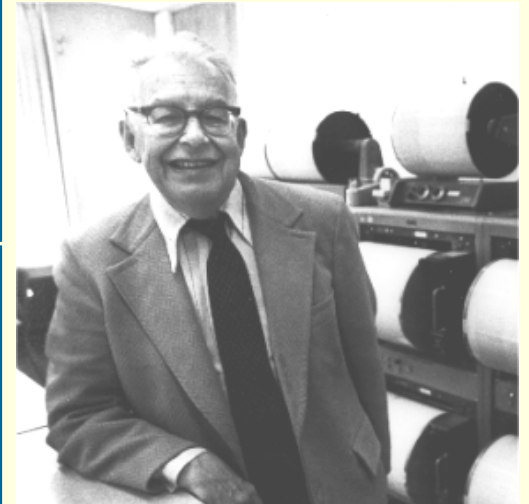
How does magnitude relate to the energy released by an earthquake?

When rocks shift suddenly along a fault, they generate waves. These waves shake the ground, producing earthquakes. Seismographs record the wave amplitudes, which are used to calculate the earthquake magnitude and the energy released by the rupture.

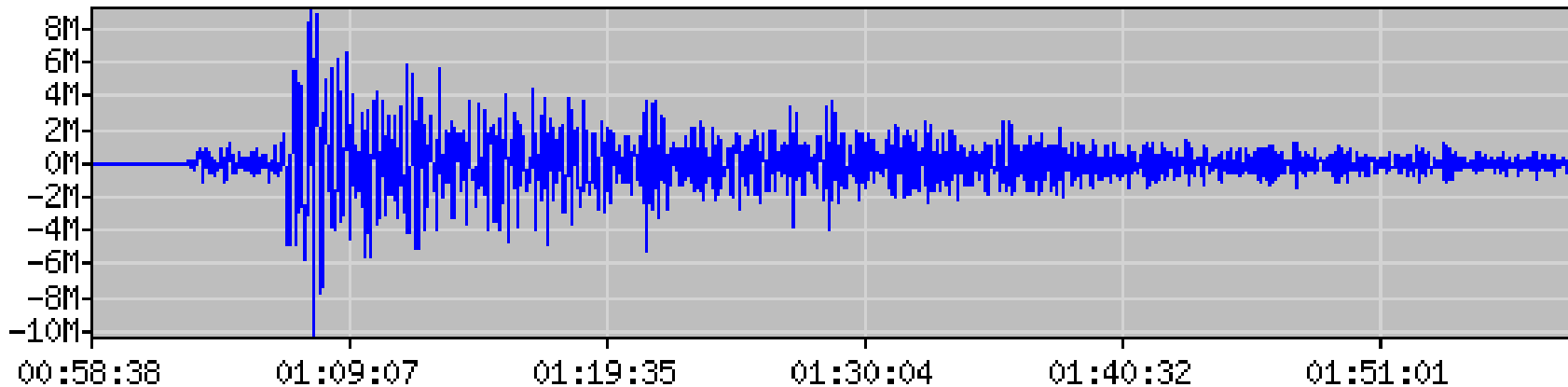
The **intensity** of shaking is one way to assess the size of an earthquake. A value is assigned based on damage reports and personal interviews of people who experienced the quake. The intensity depends on location; in general, the closer the observer to the earthquake, the higher the intensity. Intensity values assist in seismic hazard and historical earthquake analysis.

In 1935 **Charles Richter** developed a method to compare the sizes of California earthquakes based on waves recorded by seismographs. In his method, a single **magnitude** is assigned based on maximum wave amplitudes.

Modern seismologists have modified his method and now analyze a large section of the waves recorded on a seismograph to calculate a **seismic moment**. The seismic moment is then converted to **moment magnitude**, which is the standard size reported by the U.S. Geological Survey.



:COCO.II.00.BHZ Filter:none samp/sec=20 dec factor=126



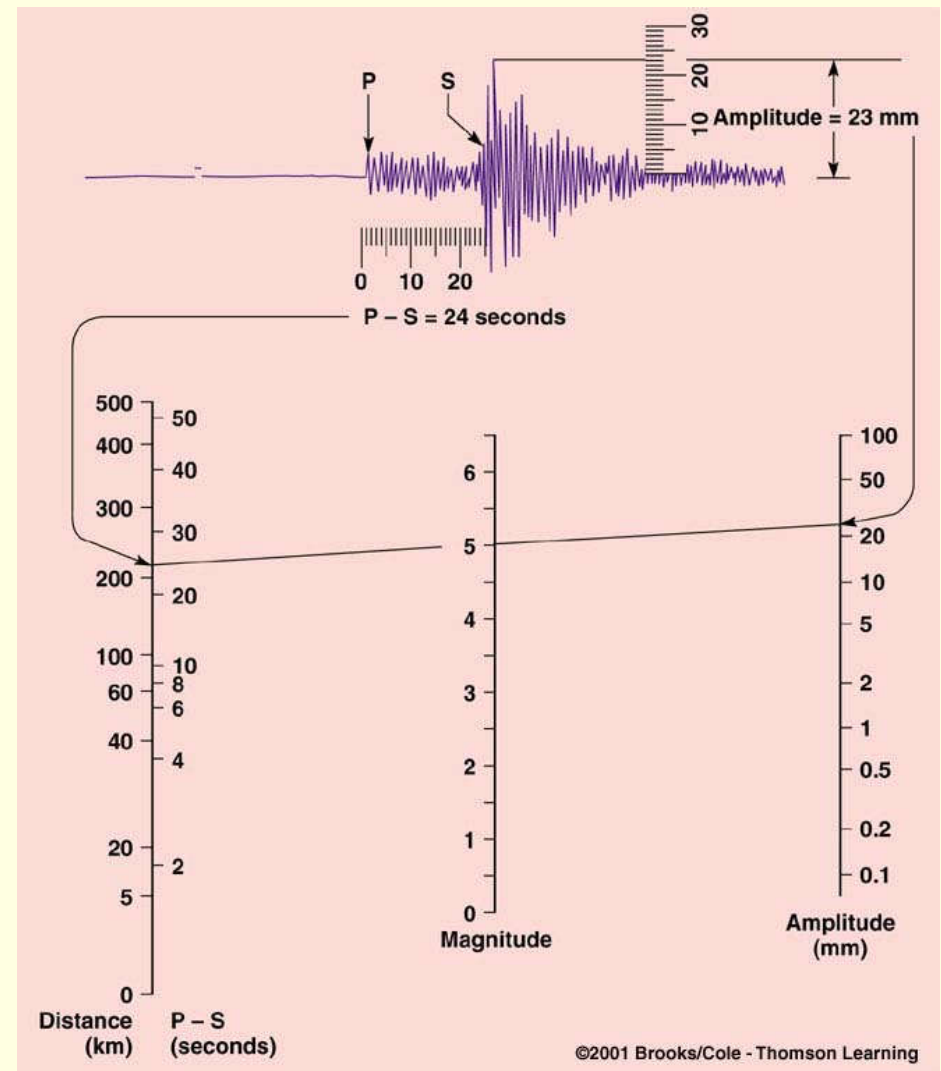
A seismogram is a graph of wave amplitude Vs. Time. In old seismographs, a pen drew the recording on a piece of paper. In new seismographs, the signal is recorded digitally.

This is a seismogram of the magnitude-9.1 Sumatra-Andaman Islands earthquake that occurred on December 26, 2004. The recording seismograph is located on the Cocos Islands in the Indian Ocean.

How the Magnitude of an Earthquake Measured?

Magnitude

- Richter scale measures **total amount of energy** released by an earthquake; independent of intensity
- Amplitude of the largest wave produced by an event is corrected for distance and assigned a value on an open-ended logarithmic scale



MAGNITUDE

A number – RICHTER Scale $M = \log_{10} A$

Energy released at focus

$$\log_{10} E = 11.4 + 1.5 M$$

$$\log_{10} E = 4.8 + 1.5 M_s$$

Each increase in M > the energy by 32 times

Strength of earthquake (Atom bomb – 5.0)

Measure of strain energy released at hypocenter.

Determined by seismographs

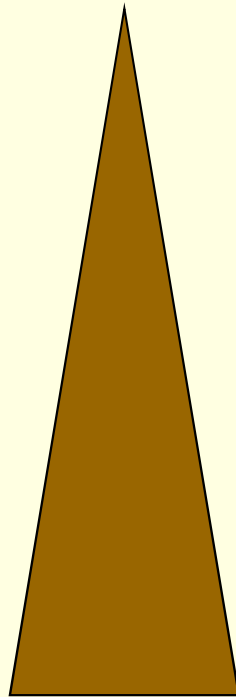
It is independent of place

How does the amplitude of a magnitude-8 earthquake compare to the amplitude of smaller events?

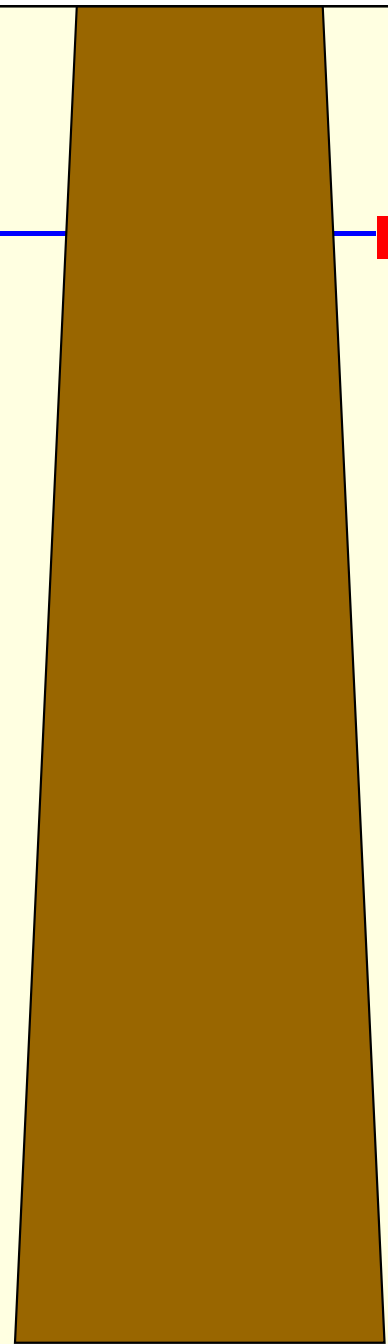
If we likened earthquakes to hills and mountain peaks, each peak is 10 times the height of the previous one.



Mag. 6



Mag. 7 = 10× larger than Mag 6



**Mag. 8 = 10× larger than Mag. 7
= 100× larger than Mag. 6**

Moment Magnitude Scale (MMS or Mw) is more popular presently.

The magnitude is based on seismic moment of the earthquake

Seismic Moment $M_o = m_u \times \text{Length} \times \text{width} \times \text{slip}$

m_u = modulus of rigidity N/m²

$M_w = 2/3 (\log_{10} M_o - 9.1)$ is better for bigger earthquakes.

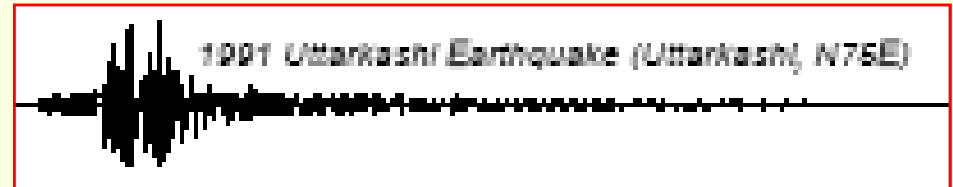
It is equal to the rigidity of the Earth multiplied by the average amount of slip on the fault and the size of the area that slipped

INTENSITY: Is not Quantitative- **Mercalli's scale**

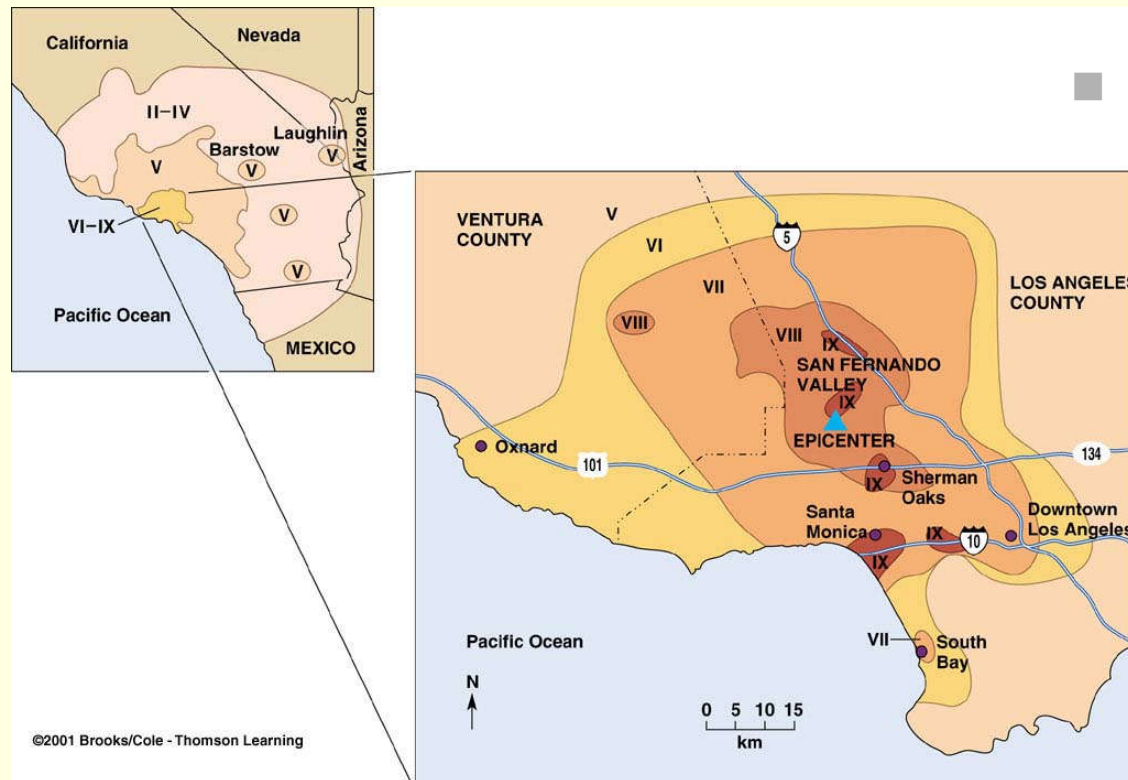
Measure of damaging effect of earthquake at a site

Depends on

- **Local soil conditions,**
- **Type & Quality of structures,**
- **Epicentral distance etc.**
- **Focal Depth**



How are the Size and Strength of an Earthquake Measured?



Intensity

- **subjective** measure of the kind of damage done and people's reactions to it
- isoseismal lines identify areas of equal intensity

- **Modified Mercalli Intensity Map**
 - 1994 Northridge, CA earthquake, magnitude 6.7



**Fire Crackers
of different
magnitudes of
blasts**



Intensity of Earthquake – Modified Mercalli's Scale

I	Insignificant	Only detected by instruments
II	Very Light	Felt by sensitive persons, Oscillation of hanging objects
III	Light	Small vibratory motion
IV	Moderate	Felt inside building, Noise produced by moving objects
V	Slightly Strong	Felt by most persons, some panic, minor damages
VI	Strong	Damage to non seismic resistant structures
VII	Very Strong	People panic, serious damage to poor construction
VIII	Destructive	Serious damage to structures in general
IX	Ruinous	Serious damage to well built structures, almost total destruction of non-seismic resistant structures
X	Disastrous	Only seismic resistant structures remain standing
XI	Extremely Disastrous	General Panic, almost total destruction, ground cracks & opens
XII	Catastrophic	Total destruction

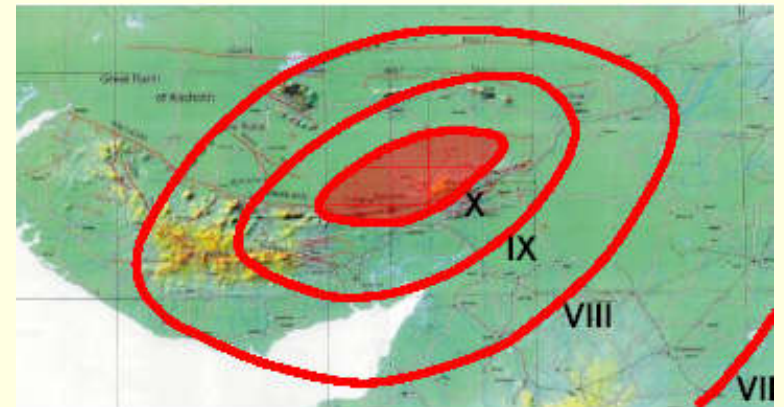
Intensity VIII - Destruction of Buildings

- (a) Fright and panic. Also, persons driving motorcars are disturbed. Here and there branches of trees break off. Even heavy furniture moves and partly overturns. Hanging lamps are damaged in part.
- (b) Most buildings of Type C suffer damage of Grade 2, and few of Grade 3. Most buildings of Type B suffer damage of Grade 3, and most buildings of Type A suffer damage of Grade 4. Occasional breaking of pipe seams occurs. Memorials and monuments move and twist. Tombstones overturn. Stonewalls collapse.
- (c) Small landslips occur in hollows and on banked roads on steep slopes; cracks develop in ground up to widths of several centimeters. Water in lakes becomes turbid. New reservoirs come into existence. Dry wells refill and existing wells become dry. In many cases, changes in flow and level of water are observed.

Note:

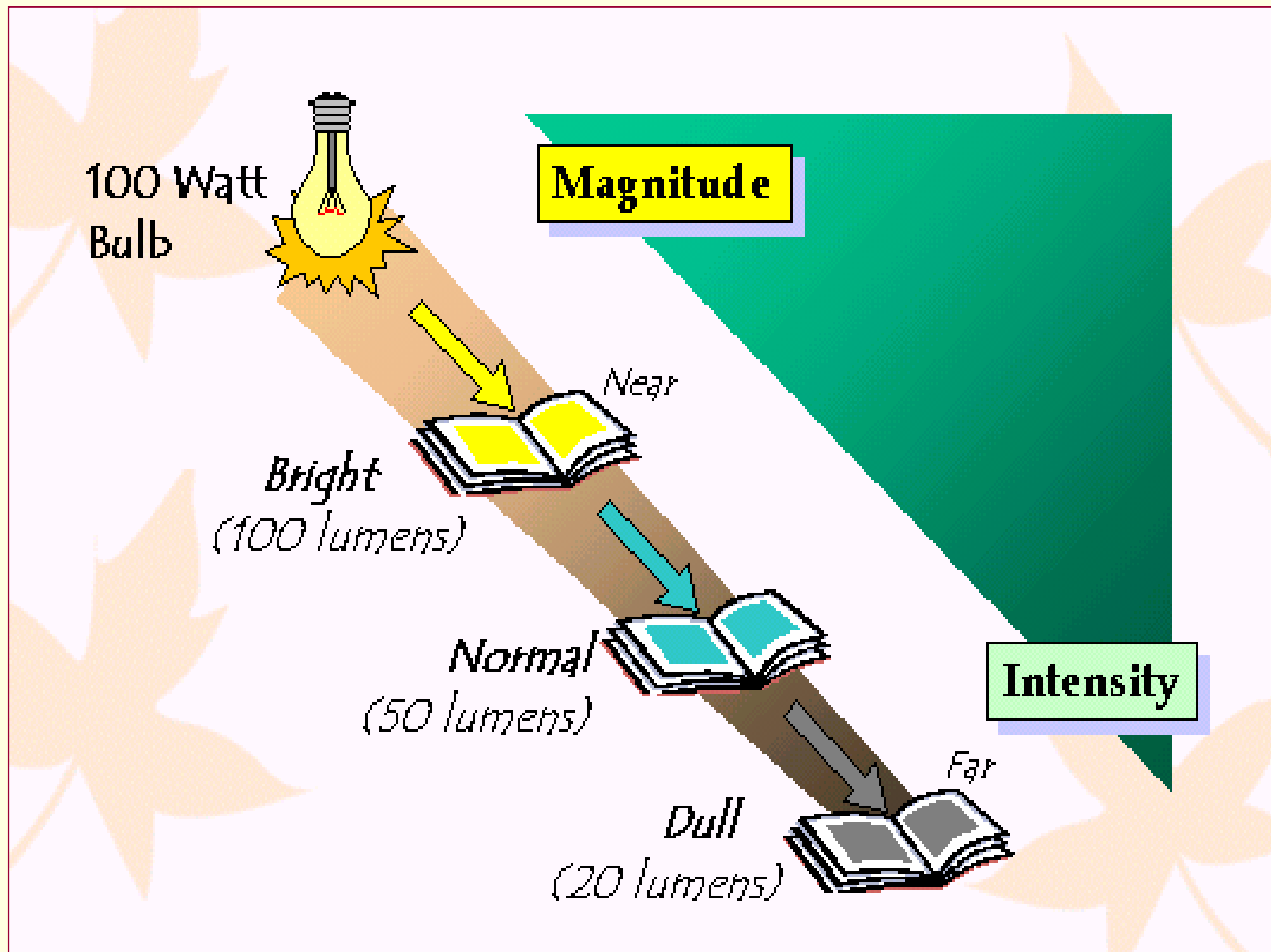
- *Type A structures* - rural constructions; *Type B* - ordinary masonry constructions; *Type C* - Well-built structures
- *Single, Few* - about 5%; *Many* - about 50%; *Most* - about 75%
- *Grade 1 Damage* - Slight damage; *Grade 2* - Moderate damage; *Grade 3* - Heavy damage; *Grade 4* - Destruction; *Grade 5* - Total damage

Intensity of Earthquake Modified Mercalli's Scale

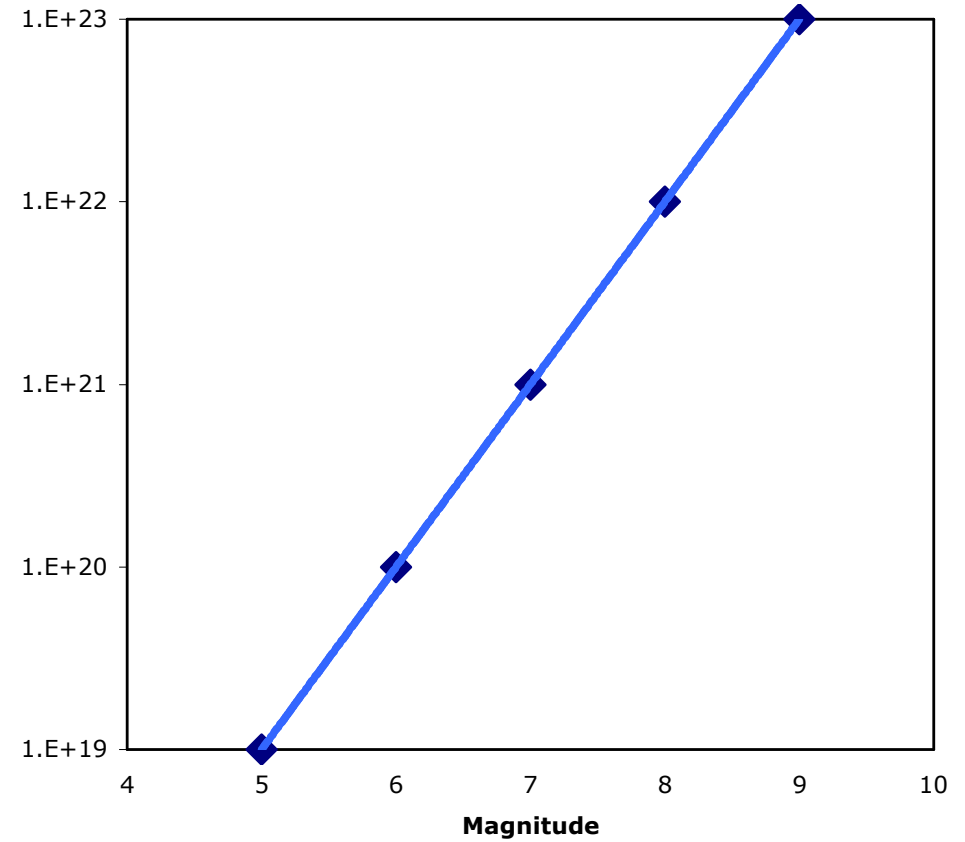
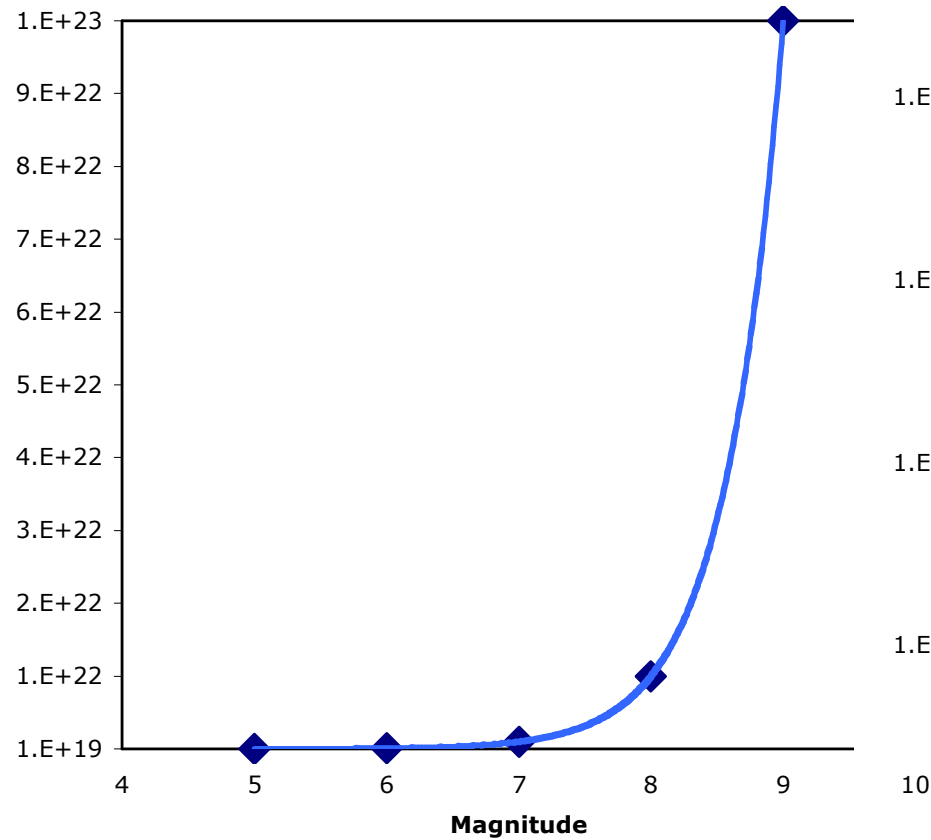


Isoseismal Map of Bhuj (2001)

Magnitude & Intensity



Energy released and Magnitude Relationships



Energy (Log)

Causes for Earthquake

- Tectonic earthquake
- Volcanic earthquake
- Rock fall or collapse of cavity
- Microseism
- Explosion (Controlled blast)
- Reservoir induced earthquake
- Mining induced earthquake
- Cultural Noise (Industry, Traffic etc.)

EARTHQUAKE DAMAGE

1. An earthquake does not cause death or injury by itself.
2. People are hurt by falling plaster and collapsing walls or falling of heavy objects.
3. Collapsing buildings and vibrations can cause short circuits and electric fires.
4. Lighted gas or stoves may also cause fires.
5. All this leads to panic and confusion.
6. With some precautions it is possible to avoid such confusion.

Classification of Earthquakes

Based on Focal Depth

Based on magnitude

Based on origin

Based on location

Based on Epicentral distance

Based on Focal Depth

Shallow Focus earthquakes
(<70 km)

Intermediate focus earthquakes
(70 to 300 km)

Deep focus earthquakes
(> 300 km)

Based on magnitude

Micro earthquakes ($M < 3$)

Intermediate earthquakes (M 3 to 5)

Moderate earthquakes (M 5 to 6)

Strong earthquakes (M 6 to 7)

Major earthquakes (M 7 to 8)

Great earthquakes ($M > 8$)

Based on origin

Tectonic earthquakes

Plutonic earthquakes

Explosions

Collapse earthquakes

Volcanic earthquakes

Reservoir induced earthquakes

Based on location

Inter-plate earthquakes

Convergent boundaries

Divergent boundaries

Transform plane boundaries

Intra-plate earthquakes

Dip slip fault

Strike slip fault

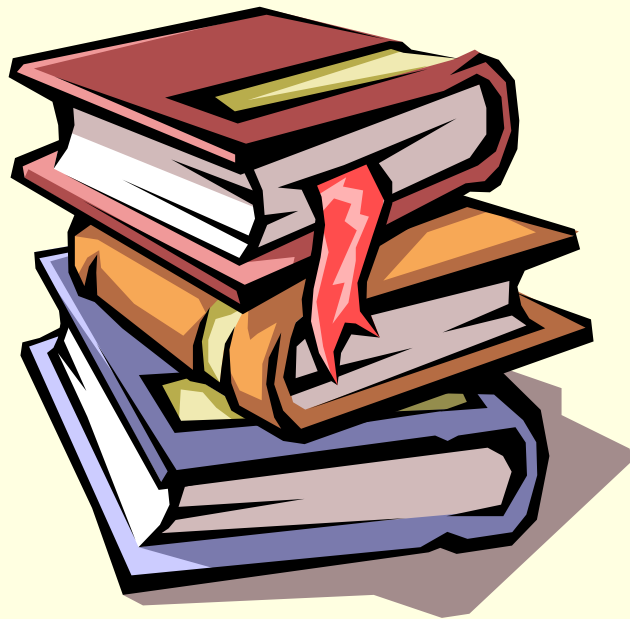
Based on Epicentral distance

Local shock (4 km range)

Near shock (4 to 10 km range)

Distant shock (10 to 20 km range)

Telescopic shock (> 20 km range)



Thank you